Quantum Series

ENGINEERING

QUANTUM *Series*

Semester - 7 Common to All Branches

Renewable Energy Resources

- Topic-wise coverage of entire syllabus in Question-Answer form.
- Short Questions (2 Marks)

Session 2021-22 Odd Semester

Includes solution of following AKTU Question Papers

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CONTENTS

KOE 074 : RENEWABLE ENERGY RESOURCES

UNIT-1 : INTRODUCTION & SOLAR CELLS **(1–1 M to 1–25 M)**

Introduction: Various non-conventional energy resources-Introduction, availability, classification, relative merits and demerits.

Solar Cells: Theory of solar cells, Solar cell materials, solar cell array, solar cell power plant, limitations.

UNIT-2 : SOLAR THERMAL ENERGY **(2–1 M to 2–34 M)**

Solar Thermal Energy: Solar radiation, flat plate collectors and their materials, applications and performance, focussing of collectors and their materials, applications and performance; solar thermal power plants, thermal energy storage for solar heating and cooling, limitations.

UNIT-3 : GEOTHERMAL ENERGY & FUEL CELLS **(3–1 M to 3–26 M)**

Geothermal Energy: Resources of geothermal energy, thermodynamics of geo-thermal energy conversion-electrical conversion, non-electrical conversion, environmental considerations.

Magneto-hydrodynamics (MHD): Principle of working of MHD Power plant, performance and limitations.

Fuel Cells: Principle of working of various types of fuel cells and their working, performance and limitations.

UNIT-4 : THERMOELECTRICAL **(4–1 M to 4–30 M)**

Thermo-electrical and thermionic Conversions: Principle of working, performance and limitations.

Wind Energy: Wind power and its sources, site selection, criterion, momentum theory, classification of rotors, concentrations and augments, wind characteristics, performance and limitations of energy conversion systems.

UNIT-5 : BIO-MASS & OTEC **(5–1 M to 5–21 M)**

Bio-mass:Availability of bio-mass and its conversion theory.

Ocean Thermal Energy Conversion (OTEC): Availability, theory and working principle, performance and limitations.

Wave and Tidal Wave: Principle of working, performance and limitations.

Waste Recycling Plants.

SHORT QUESTIONS **(SQ–1 M to SQ–25 M)**

SOLVED PAPERS (2015-16 TO 2018-19) **(SP–1 M to SP–12 M)**

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UNIT

Introduction and Solar Cells

CONTENTS

Part-1	:	Various Non Conventional Energy Resources	**1–2M to 1–9M**
Part-2	:	Availability of Non Conventional Energy Resources	**1–9M to 1–11M**
Part-3	:	Classification of Non Conventional Energy Resources	**1–11M to 1–12M**
Part-4	:	Relative Merits and Demerits	**1–12M to 1–13M**
Part-5	:	Theory of Solar Cells, Solar Cell Materials	**1–13M to 1–19M**
Part-6	:	Solar Cell Array	**1–19M to 1–20M**
Part-7	:	Solar Cell Power Plant	**1–20M to 1–22M**
Part-8	:	Limitations	**1–22M to 1–24M**

PART-1

Various Non Conventional Energy Resources.

Questions-Answers

Long Answer Type and Medium Answer Type Questions

Que 1.1. What are the conventional and non conventional energy sources ? Describe the fossil fuel as a conventional energy source.

Answer

A. Conventional (Non Renewable) Energy Sources : These are the sources of energy which are exhaustible *i.e.*, cannot be replaced if once they are used.

Example : Coal, Petroleum products, Natural gas etc.

B. Non Conventional (Renewable) Energy Sources : These are the sources of energy which are inexhaustible *i.e.*, can be used to produce energy again and again.

Example : Sun, Water, Animal dung, Agro-waste etc.

C. Fossil Fuel as a Conventional Energy Source : Some of the fossil fuels are discussed below :

a. Coal Energy :

1. Coal is a conventional energy source.
2. It is formed due to degradation of trees and plants buried under layers of silt.
3. It is composed of mainly carbon and hydrocarbons.
4. Coal is found in Jharkhand, U.P., M.P., Bihar etc. in India.
5. Uses of coal :
   - i. Coal is used to generate electricity. Power plants use coal for heating the water to generate steam, which runs the turbines to generate electricity.
   - ii. Various industries use heat obtained from coal in making plastics, tar, synthetic fiber, etc.
   - iii. Coal is heated in furnace to make coke, which is used to melt iron for making steel.
6. Environmental problems :

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i. Due to combustion of coal, carbon dioxide is produced which is responsible for causing global warming.

ii. Coal also produces sulphur dioxide which is a cause for acid rain.

b. Natural Gas :

1. Natural gas formed by decomposition of dead animals and plants buried under the earth.
2. It is mainly composed of methane (CH_4) with small amount of propane and ethane.
3. Natural gas is the cleanest fossil fuel.
4. **Uses of natural gas :**
   i. It is used as a domestic and industrial fuel.
   ii. It is also used in thermal power plants for generating electricity.
5. **Advantages :**
   i. Natural gas has a high calorific value and it burns without any smoke.
   ii. It can be easily transported through pipelines.

Que 1.2. **Give a brief review of various sources of renewable energy.**

OR

Discuss renewable forms of energy. Highlight their merits and demerits.

Answer

Various Sources of Renewable Energy :

A. Solar Energy :

1. Solar energy is a clean, cheap and abundantly available renewable energy and it is also the most important of the non conventional sources of energy because it is non-polluting and, therefore helps in decreasing the green house effect.
2. Solar energy can be used as :
   i. By direct conversion to a fuel by photosynthesis.
   ii. By direct conversion to electricity by photovoltaic.
   iii. By conversion to electricity via thermo-electric power system.
3. The sun releases the enormous amount of energy due to continuous fusion reaction taking place inside the sun.
4. The sun sends out the energy in the form of radiations at the rate of 3.7×10^{20} MW.

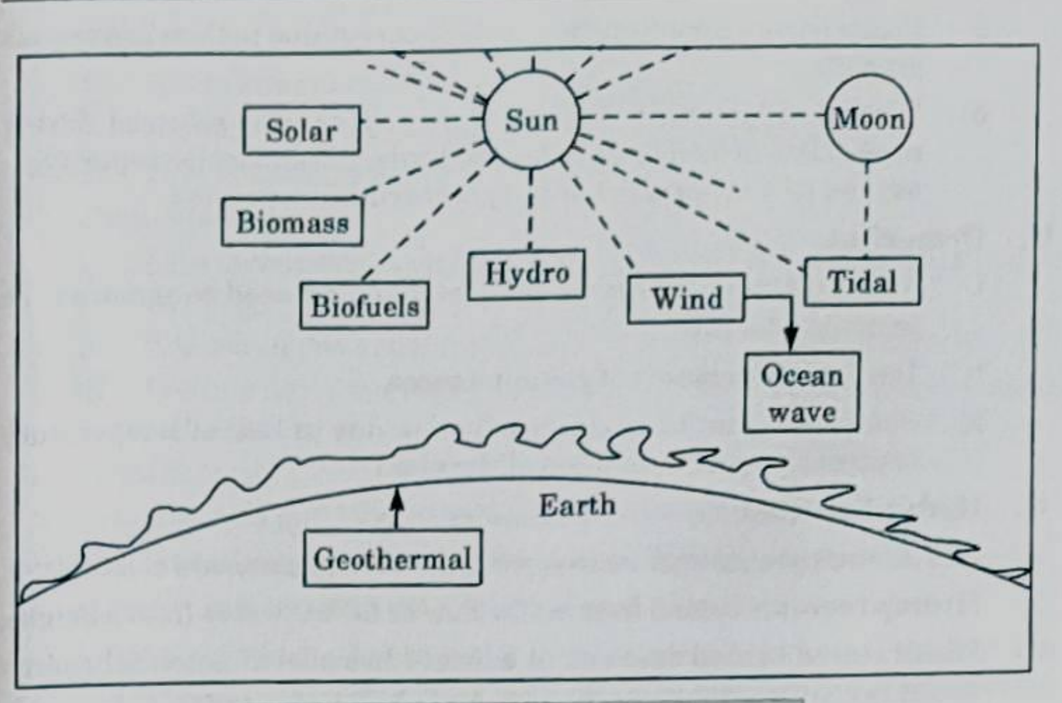

Fig. 1.2.1. Renewable sources of energy.

5. However, the energy intercepted by the earth is about 1.85×10^{11} MW.
6. This energy available is several times more than all the energy produced and consumed in the world.

a. Merits of Solar Energy :

i. Noiseless operation.

ii. Occupies less space on floor as there is no need of storage vessels.

iii. Cheaper initial cost and no need of containers to store the fuel.

b. Demerits :

i. Solar equipments fail to work in nights, cloudy days or rainy season.

ii. Large space is required for the collection of solar energy at a useful rate.

B. Biomass :

1. Green plants trap solar energy through the process of photosynthesis and convert it into organic matter. This organic matter is known as biomass.
2. Wood, charcoal, agricultural waste produces the bio-energy after burning, and cow dung, garbage are anaerobically decomposed to obtain the energy.
4. Dried animal dung or cattle dung cakes are used directly as fuels in rural area but it produces smoke and has low efficiency of burning.

a. Merits :

i. Cost of obtaining bio-energy through plantations is lesser than cost of obtaining energy from fossil fuels.

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ii. Plants ensure a continuous supply of energy due to their continuous growth.

iii. Growth of biomass consumes more CO_2 than is released during combustion of biomass besides producing the atmosphere–purifying oxygen as a by–product of the photosynthesis process.

b. Demerits :

i. Accumulation of water in the pipe line and need to remove it at periodic intervals.

ii. Inefficient designing of gas appliances.

iii. Seasonal variation in gas production due to lack of temperature control and poor insulation of the plant.

C. Hydro Energy :

1. It is a renewable energy source, which is used to generate electricity.
2. Hydropower is obtained from water flow or falling water from a height.
3. Water stored behind dam and at a height has a lot of potential energy which is converted into mechanical and electrical energy.
4. The water is released gradually and is allowed to fall under the gravitational force and drive which rotate hydraulic turbines.
5. The generators attached with turbine produce the electricity.

a. Advantage :

i. Hydropower does not pollute the water or the air during operation and no waste products are formed.

b. Limitation :

i. The generation of electricity by hydroelectric power plants results in pollution and ecological disturbance like flooding situation and adverse effects on flora and fauna.

D. Wind Energy :

1. Wind energy is a renewable source of non polluting energy and it has tremendous potential which if harnessed, can easily satisfy the energy demands of a country.
2. Estimates reveal that 2 % of the total solar energy falling on earth is converted to kinetic energy in the atmosphere.
3. 30 % of this kinetic energy occurs in the lowest 1000 m of elevation *i.e.*, wind in the lowest kilometer has maximum kinetic energy which can be converted into mechanical energy which in turn can be utilized to generate electricity or to perform some other useful work.
4. Since, the energy possessed by wind is by virtue of its motion, so the device used to extract its energy should be capable of slowing down the wind.

a. Merits :

i. Abundance availability for no price.

ii. Useful at remote places also for electricity generation.

iii. Non polluting and eco-friendly.

b. Demerits :

i. Less favourable in city locations as the wind is available at higher locations.

ii. It is unreliable and intermittent. It is not available regularly.

iii. Present day wind energy systems are a source of immense noise pollution.

E. Tidal Energy :

1. Gravitational pull by sun and moon result in the tides.
2. This type of energy can be harnessed by constructing the tidal barrage.
3. Energy can be harnessed from high as well as from low tides.
4. During high tides, the sea water flows into the reservoir of the barrage and operates the turbine which in turn produces electricity, by rotating generators.
5. During low tides, the water stored in the reservoir flows into the sea and again operates the turbine.
6. In this way the energy can be harnessed from high and low tides.

a. Merits :

i. Very less area is required because they are on bays.

ii. It is free from pollution as it does not use any fuel.

iii. It does not produce any unhealthy waste like gases, ash, refuse, etc.

b. Demerits :

i. Tidal power plants can be developed only if natural sites are available.

ii. The capital cost of the plant is high.

iii. Utilization of tidal energy on small scale has not yet proved economical.

F. Ocean Thermal Energy :

1. OTEC *i.e.*, ocean thermal energy conversion plants convert the heat of the ocean into electrical energy, with the help of temperature difference.
2. The large temperature difference between warm surface sea water (28 - 30 °C) and cold deep sea water (5 - 12 °C) is used to generate electricity, with the help of ocean thermal energy conversion system.

a. Merits :

i. The OTEC process exploits the temperature difference between the warm surface and cold bottom water to produce electricity. In India, the conditions for operation of OTEC plant are favourable because of this temperature different that remains constant throughout the year.

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ii. Inexpensive transmission of electricity is possible provided the OTEC plant is less than 30 km from the sea shore.

b. Demerits :

i. At greater distance (from plant to sea shore) the transmission cost of electricity is increased.

ii. OTEC technology is in infant stage.

iii. OTEC technology is costly and difficult.

G. Wave Energy :

1. The motion of the sea surface in the form of wind waves forms a source of energy.
2. Floating propellers are placed in shallow waters, near the shores and due to motion of the waves, the propellers also get the motion and this kinetic energy can be used to drive turbines.
3. This is cheap, clean and inexhaustible source of energy.

a. Merits :

i. Ocean wave energy source is renewable and free of cost; hence its importance will increase with time.

ii. Waves are continuous. They come and go between 6 second.

iii. Collector size of wave machines is comparatively smaller than solar devices.

b. Demerits :

i. Corrosion of materials used in plant.

ii. Marine growth of algae in the plant.

iii. Obstruction to ships.

H. Geothermal Energy :

1. The energy harnessed from the hot rocks present inside the earth is called as geothermal energy.
2. There is an increase in the temperature of the earth with increasing depth below the surface.
3. The fission of radioactive material naturally occurring in the rocks increases the temperature of the earth as we move down from the earth's surface.
4. Hot molten rocks called 'magma' are present in the core of the earth. This causes sometimes volcanic action.
5. This hot steam is used to operate turbines to generate electricity.
6. Artificially it can also be harnessed with the help of pipes by drilling the hot rocks, which make the hot water to gush out through pipes which turns the turbine of the generator to produce electricity.

a. Merits :

i. It is cheap and clean source of energy.

ii. Geothermal plants require little land area.

b. Limitations :

i. Air pollution results in case of release of gases like H_2S, NH_3 present in the steam waste.

ii. Noise pollution results from the drilling operations.

K. Hydrogen Energy :

1. Hydrogen is considered as an alternative future source of energy.
2. It is a non conventional energy resource.
3. Hydrogen energy has a tremendous potential because it can be produced from water which is available in abundance in nature.
4. In sun's core, hydrogen atoms combine to form helium atom which is known as fusion reaction.
5. It gives the radiant energy which sustains the life on the earth.
6. Hydrogen can be separated from water by means of electrical energy.
7. It can also be obtained from fossil fuels.

a. Advantages :

i. Hydrogen energy has very high energy content.

ii. Its burning is non-polluting.

b. Disadvantages :

i. Highly flammable.

ii. It is more expensive.

Que 1.3. Discuss the main features of various types of renewable and non-renewable energy sources. Also explain the importance of non-conventional energy sources in the context of global warming.

AKTU 2016-17, 2018-19; Marks 10

Answer

A. Types of Renewable Energy Sources : Refer Q. 1.2, Page 1–3M, Unit-1.

B. Types of Non-renewable Energy Sources : Refer Q. 1.1, Page 1–2M, Unit-2.

C. Importance of Non-conventional Energy Sources in the Context of Global Warming :

1. Global warming is the phenomenon of rise in temperature of environment due to the rise in concentration of the various gases like CH_4, CO_2, aerosols, NO_x, etc which is caused by the burning of conventional fossil fuels in industries.
2. Non-conventional energy resources like solar energy, wind energy, ocean thermal energy, etc are eco-friendly resources and do not produce any kind of pollutants or harmful gases like CO_2, CH_4, SO_x, NO_x, etc.

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3. Thus, we can say that non-conventional energy resources are playing an important role in context of global warming by not producing harmful gases which increases the temperature of environment.

PART-2

Availability of Non Conventional Energy Resources.

Questions-Answers

Long Answer Type and Medium Answer Type Questions

Que 1.4. Discuss the prospects of non conventional energy sources in India.

Answer

Prospects of Non Conventional Energy Sources in India :

1. India is heavily dependent on fossil sources of energy for most of its demand.
2. This has necessitated the country to start aggressively pursuing alternative energy sources – solar, wind, biofuels, and small hydro and more.
3. India is the 4th largest country with regard to installed power generation capacity in the field of renewable energy sources.
4. Wind, hydro, biomass and solar are main renewable energy sources.
5. India has tremendous potentialities to harness the much-needed energy from renewable sources.
6. The country has an estimated renewable energy potential of around 90000 MW from commercially exploitable sources; wind, 48500 MW; small hydro, 15000 MW and biomass/bio-energy, 25000 MW.
7. In addition, India has the potential to generate 35 MW per square km using solar photovoltaic and solar thermal energy.
8. India is among top 5 destinations worldwide for solar energy development.
9. Government has launched Jawaharlal Nehru National Solar Mission which aims to generate 20 GW by 2022.
10. Wind energy is the fastest growing renewable energy sector.
11. Wind energy, with an installed capacity of nearly 15 GW, accounts for the bulk of installed renewable energy capacity in India making it fifth

top country in the world, after USA, China, Germany, and Spain in terms of installed capacity.

12. Coastal areas of Gujarat, Tamil Nadu, Andhra Pradesh, as well as vast areas of Maharashtra and Madhya Pradesh provide a good potential for its development.
13. Globally, India is in the fourth position in generating power through biomass and has the potential to become a world leader in the utilisation of biomass.
14. Biomass power projects with an aggregate capacity of 773.3 MW through over 100 projects have been installed in the country.
15. For the last 15 years, biomass power has become an industry attracting annual investment of over 1000 billion, generating more than 9 billion unit of electricity per year.
16. More than 540 million tonnes of crop and plantation residues are produced every year in India and a large portion is either wasted, or used inefficiently.
17. By using these surplus agricultural residues, by conservative estimates more than 16000 MW of grid quality power could be generated through biomass.
18. With numerous rivers and their tributaries in India, small hydro RE (< 25 MW) presents an excellent opportunity with an estimated potential of 15000 MW with only 17 percent of this sector exploited so far.
19. Over 674 projects aggregating to about 2558.92 MW generating capacity has been set up in the country as on 31.12.2009. Most of the potential is in Himalayan states as river-based projects and in other states on irrigation canals.

Que 1.5. What is MNRE ? What are the mission and function of it ?

OR

Write short note on MNRE. **AKTU 2015-16, Marks 04**

Answer

A. MNRE :

1. The Ministry of New and Renewable Energy (MNRE) is the nodal ministry of the government of India for all matters relating to new and renewable energy.
2. The broad aim of the ministry is to develop and deploy new and renewable energy for supplementing the energy requirements of the country.
3. Creation CASE and ministry :

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a. Commission for Additional Sources of Energy (CASE) in 1981.
b. Department of Non Conventional Energy sources (DNES) in 1982.
c. Ministry of Non Conventional Energy Sources (MNES) in 1992.
d. Ministry of Non Conventional Energy Sources (MNES) renamed as ministry of New and Renewable Energy (MNRE) in 2006.

4. The role of new and renewable energy has been assuming increasing significance in recent times with the growing concern for the country's energy security.
5. Energy self-sufficiency was identified as the major driver for new and renewable energy in the country in the wake of the two oil shocks of the 1970s.
6. The sudden increase in the price of oil, uncertainties associated with its supply and the adverse impact on the balance of payments position led to the establishment of the commission for additional sources of energy in the department of science and technology in March 1981.
7. The commission was charged with the responsibility of formulating policies and their implementation, programmes for development of new and renewable energy apart from coordinating and intensifying *R & D* in the sector.

B. MISSION :

1. The Jawaharlal Nehru national solar mission was launched on the 11th January, 2010 by the Prime Minister.
2. The mission has set the ambitious target of deploying 20,000 MW of grid connected solar power by 2022 is aimed at reducing the cost of solar power generation in the country through :
   i. Long term policy,
   ii. Large scale deployment goals,
   iii. Aggressive R & D, and
   iv. Domestic production of critical raw materials, components and products, as a result to achieve grid tariff parity by 2022.
3. Mission will create an enabling policy framework to achieve this objective and make India a global leader in solar energy.

PART-3

Classification of Non Conventional Energy Resources.

Questions-Answers

Long Answer Type and Medium Answer Type Questions

Que 1.6. What are the conventional and non-conventional energy sources ? Write short notes on classification of energy resources.

AKTU 2015-16, Marks 10

Answer

A. Conventional and Non Conventional Energy Resources : Refer Q. 1.1, Page 1–2M, Unit-1.

B. Classification :

i. Primary Energy Resources :

1. These resources are obtained from the environment.
   Example : Fossil fuels, solar energy, hydro energy and tidal energy.
2. These resources can further be classified as :

i. a Conventional Energy Sources :
Example : Thermal power.

b. Non Conventional Energy Sources :
Example : Wind energy, geothermal, ocean energy, solar energy and tidal energy.

ii. a. Renewable : These sources are being continuously produced in nature and are inexhaustible.
Example : Wood, wind energy, biomass, biogas, solar energy etc.

b. Non Renewable : These are finite and exhaustible.
Example : Coal, petroleum etc.

C. Secondary Energy Resources : These resources do not occur in nature but are derived from primary energy resources.
Example : Electrical energy from coal burning, H_2 obtained from hydrolysis of H_2O.

PART-4

Relative Merits and Demerits.

Questions-Answers

Long Answer Type and Medium Answer Type Questions

Que 1.7. What are the advantages and limitations of non conventional or renewable energy resources ?

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Answer

A. Advantages :

1. Renewable energy is an indigenous resource available in considerable quantities to all developing nations and capable, of having a significant local, regional or national economic impact. The use of renewable energy could help to conserve foreign exchange and generate local employment if conservation technologies are designed, manufactured, assembled and installed locally.
2. Several renewable resources are financially and economically competitive for certain applications, such as in remote locations, where the costs of transmitting electrical power or transporting conventional fuels are high, or in those well endowed with biomass, hydro or geothermal resources.
3. Since conversion technology tends to be flexible and modular, it can usually be rapid deployed. Other advantages of modular over very large individual unit include easy in adding new capacity, less risk in comparison with 'lumpy' investments, lower interest on borrowed capital because of shorter lead times and reduced transmission and distribution costs for dispersed rural locations.
4. Rapid scientific and technological advantages are expected to expand the economic range of renewable energy applications over the next 8-10 years, making it imperative for international decision makers and planners to keep awareness of these developments.

B. Limitations :

1. Inadequate documentation and evaluation of past experience, a scarcity of validated field performance data and a lack of clear priorities for future work.
2. Weak or non-existent institutions and policies to finance and commercialize renewable energy systems.
3. Technical and economic uncertainties in many renewable energy systems, high economic and financial costs for some systems in comparison with conventional supply option and energy efficiency measures.
4. Skeptical attitudes towards renewable energy systems on the part of the energy planners and a lack of qualified personnel to design, manufacture, market, operate and maintain such systems.
5. Inadequate donor coordination in renewable energy assistance activities, with little or no information exchange on successful and unsuccessful projects.

PART-5

Theory of Solar Cells, Solar Cell Materials.

Questions-Answers

Long Answer Type and Medium Answer Type Questions

Que 1.8. Write short note on solar cells and its materials.

Answer

A. Solar Cells :

1. Photovoltaic energy is the conversion of sunlight into electrical energy through a photovoltaic cell, commonly called a solar cell.
2. Solar cells are the solid state electronic device used to convert the electromagnetic energy of solar radiation directly into direct current electricity. This conversion takes place inside the cell.
3. When sunlight strikes the solar cell, electrons are knocked loose. They move towards the treated front surface. An electron unbalance is created between the front and back. When the two surfaces are joined by a connector, like a wire, an electric current flows between the negative and positive sides.
4. These individual solar cells are arranged together in an array.

B. Solar Cell Materials :

1. The solar cell is made of different material and silicon and silicon is one used for nearly 90 % applications.
2. The choice of material depends on the band energy gap, efficiency and cost.
3. The maximum efficiency of solar cell is achieved with the band gap energy of 1.12 eV-2.3 eV.
4. The various materials like aluminum silicon, Si (1.12eV), Aluminium antimonide, AlSb (1.27 eV), Cadmium telluride, CdTe (1.5 eV), Zink telluride, ZnTe (2.1 eV), Cadmium sulphide, CdS (2.42 eV) etc. are the materials suitable for solar cell.
5. The smaller the energy gap, the large number of photon of solar spectrum will be useful to produce the required energy for electrons to jump the forbidden band gap.

Que 1.9. Write short note on :

A. Principle of solar photovoltaic, and
B. Photovoltaic effect.

Answer

A. Principle of Solar Photovoltaic :

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1. It is a field of solar energy utilization by which solar radiation is converted into electrical energy using a device called photovoltaic cell or solar cell.
2. A solar cell is made up of a semiconductor material like silicon (Si) or gallium arsenide GaAs.
3. In semiconductors, atoms carry four electrons in the outer valence orbit, some of which can be dislodged to move freely in the materials, if extra energy is supplied.
4. Then, a semiconductor attains the property to conduct the current. This is the basic principle on which the solar cell works and generates power.

B. Photovoltaic Effect :

1. When a solar cell is illuminated, electron-hole pairs are generated and the electric current I is obtained.
2. I is the difference between the solar light generated current I_L and the diode dark current I_j.
3. Mathematically :

$$I = I_L - I_j = I_L - I_o\left[\exp\left(\frac{eV}{KT}\right) - 1\right]$$

Where,
I_o = Saturation current,
e = Electronic charge,
T = Absolute temperature, and
K = Boltzmann's constant.
$= 1.38 \times 10^{-23}$ J/K.

Que 1.10. Explain the mechanism of photoconduction in a PV cell. **AKTU 2016-17, 2018-19; Marks 10**

Answer

1. In p-n junction after the photons are absorbed, the free electrons of the n-side will tend to flow to the p-side, and the holes of the p-side will tend to flow to the n-side to compensate for their respective deficiencies.
2. This diffusion will create an electric field E_F from the n-region to the p-region.
3. This field will increase until it reaches equilibrium of V_e, the sum of the diffusion potentials for holes and electrons.
4. If electrical contacts are made with the two semiconductor materials and the contacts are connected through an external electrical conductor, the free electrons will flow from the n-type material through the conductor to the p-type material.
5. The flow of electrons through the external conductor constitutes an electric current which will continue as long as more free electrons and holes are being formed by the solar radiation.

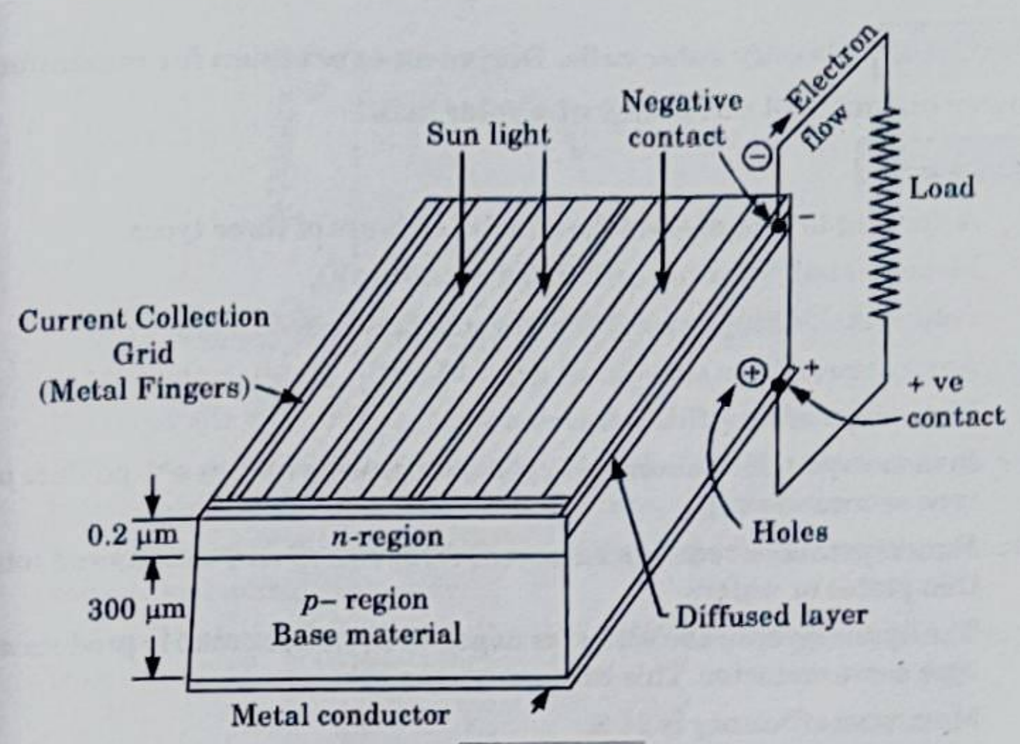

Fig. 1.10.1.

6. This is the basis of photovoltaic conversion, that is, the conversion of solar energy into electrical energy.
7. The combination of n-type and p-type semiconductors thus constitutes a photovoltaic (PV) cell or solar cell.
8. All such cells generate direct current which can be converted into alternating current if desired.

Que 1.11. Describe principle of solar photovoltaic conversion. Discuss the limitations of solar photovoltaic energy conversion.

Answer

A. Principle : Refer Q. 1.9, Page 1–14M, Unit-1.

B. Limitations :

1. When photons of light energy from the sun strike the cell, some of them (30 %) are reflected (since reflectance from semiconductors is high).
2. Photons of quantum energy $hv < Eg$ cannot contribute to photoelectric current production (h is the Planck's constant and v the frequency).
3. Photovoltaic cells are exposed directly to the sun. As the temperature rises, leakage across the cell increases. Consequently, there is reduction in power output relative to input of solar energy.
4. Incident active photons produce electron-hole pairs with high quantum efficiency. Better cell design is required to ensure 95 % absorption.
5. The semiconductor with optimum band gap should be used for maximum efficiency.

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Que 1.12. **Classify solar cells. Derive an expression for maximum power output and efficiency of a solar cell.**

Answer

1. According to type of crystal, the solar cells are of three types :

a. Monocrystalline silicon cells (band gap 1.12 eV),

b. Polycrystalline silicon cells (band gap 1.12 eV), and

c. Amorphous silicon cells (band gap 1.75 eV).

a. Monocrystalline Silicon Cell :

1. In monocrystalline silicon cells, silicon is doped with boron to produce *p*-type semiconductor.
2. Monocrystalline rods are extracted from silicon and then sawed into thin plates or wafers.
3. The upper layer of the wafers is doped with phosphorous to produce *n*-type semiconductor. This becomes *p-n* junction.
4. Maximum efficiency is 24 %.

b. Polycrystalline Silicon Cell :

1. In polycrystalline cells, liquid silicon is poured into blocks that are sawed into plates.
2. During solidification of the material, crystal structures of varying sizes are formed.
3. The size of crystallites mainly depends upon the cooling condition. If the molten silicon is cooled very slowly, the crystallites of larger size are obtained.
4. The silicon solar cells made from polycrystalline silicon are low cost but low efficiency (maximum efficiency is 17.8 %).

c. Amorphous Silicon Cell :

1. If a silicon film is deposited on glass or another substrate material, this is so called amorphous or thin layer cell.
2. The layer thickness is less than 1 μm, so production costs are lower due to the low material costs.
3. However, the efficiency of amorphous cells is much lower than that of the other cells. Because of this, they are primarily used in low power equipment such as watches, pocket calculators etc. Maximum efficiency is 13 %.

A. Efficiency of Solar Cell :

1. The electrical characteristics of a solar cell are expressed by the voltage current (*V-I*) curves plotted under a given illumination and temperature conditions as shown in Fig. 1.12.1.

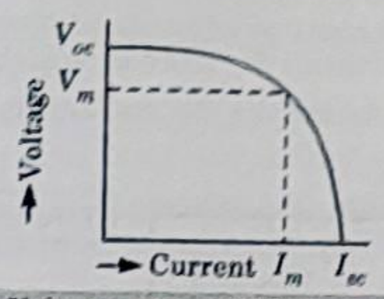

Fig. 1.12.1. Voltage-current (*V-I*) characteristic of a solar cell.

2. In Fig. 1.12.1, the maximum useful power of the cell is represented by the rectangle with the largest area.
3. When the cell yields maximum power, the current and voltage are represented by the symbols I_m and V_m respectively.
4. Leakage across the cell increases with temperature which reduces voltage and maximum power.
5. Cell quality is maximum when the value of 'fill factor' approaches unity, where the fill factor (*FF*) is expressed as

$$FF = \frac{I_m V_m}{I_{sc} V_{oc}}$$

Where, V_{oc} = Open circuit voltage, and
I_{sc} = Short circuit current.

6. Maximum efficiency of a solar cell is defined as the ratio of maximum electric power output to the incident solar radiation.

Mathematically :

$$\eta_{max} = \frac{I_m V_m}{I_s A_c}$$

Where, I_s = Incident solar flux, and
A_c = Cells area.

7. Maximum power output,

$$P_{max} = V_{max\,P} \times I_{max\,P}$$

Que 1.13. **Describe solar photovoltaic (SPV) module with neat sketch.** **AKTU 2018-19, Marks 10**

Answer

A. Solar Photovoltaic System :

1. It refers to a wide variety of solar electricity systems.
2. This system use solar array made of silicon to convert sunlight into electricity.
3. Components other than PV array are collectively known as balance of system (BOS) which includes storage batteries, an electronic charge controller and an inverter.

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4. Storage batteries with charge regulators are provided for back-up power supply during periods of cloudy day and during nights.
5. Batteries are charged during the day and supply power to loads as shown in Fig. 1.13.1.

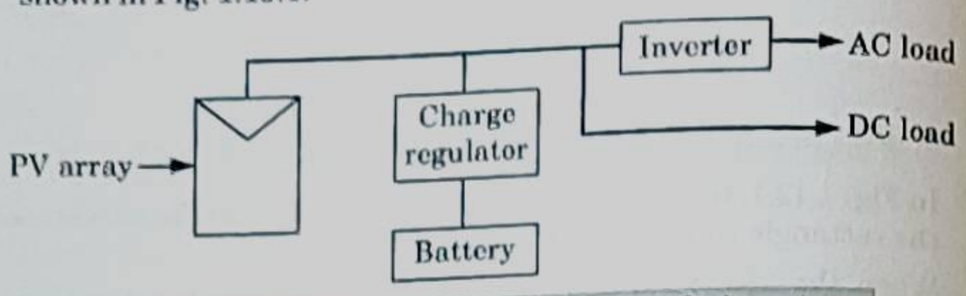

Fig. 1.13.1. Block diagram of solar photovoltaic system.

6. The capacity of a battery is expressed in ampere-hours and each cell of the lead-acid battery is of 2 volts.
7. Batteries are installed with a microprocessor based charge regulator to monitor the voltage and temperature.
8. It also regulates the input and the output current to eliminate overcharging and excessive discharge respectively.
9. An inverter is provided for converting DC power from battery or PV array to AC power.
10. It needs to have an automatic switch-off in case the output voltage from the array is too low or too high.
11. The inverter is also protected against over loading and short circuit.

PART-6

Solar Cell Array.

Questions-Answers

Long Answer Type and Medium Answer Type Questions

Que 1.14. **Write a short note on solar cell array.**

AKTU 2016-17, 2018-19; Marks 05

Answer

1. Solar cells are strung in series and thus form a solar module or array.
2. They may be tracking arrays or fixed arrays.

3. A tracking array is defined as one which is always kept mechanically perpendicular to the sun-array line so that all times it intercepts the maximum isolation.
4. Such arrays must be physically movable by a suitable prime-mover and are considerably more complex than fixed arrays.
5. A fixed array is usually oriented east-west and tilted up at an angle approximately equal to the latitude of the site.
6. Fixed arrays are mechanically simpler than tracking arrays. Thus the array designs fall into two broad classes :

a. Flat-Plate Arrays :

1. Wherein solar cells are attached with a suitable adhesive to some kind of substrate structure usually semi-rigid to prevent cells being cracked.
2. This technology springs from the space–related photovoltaic technology, and many such arrays have been built in various power sizes.

b. Concentrating Arrays :

1. Wherein suitable optics, e.g., Fresnel lenses, parabolic mirrors, compound parabolic concentrators (CPC), and others, are combined with photovoltaic cells in array fashion.
2. This technology is relatively new to photovoltaic in terms of hardware development, and comparatively fewer such arrays have actually been built.

PART-7

Solar Cell Power Plant.

Questions-Answers

Long Answer Type and Medium Answer Type Questions

Que 1.15. **With the aid of block diagrams explain :**

i. Autonomous solar power plant, and
ii. Combined solar power plant.

Answer

i. Autonomous Solar Power Plant :

1. This plant also known as grid independent or stand alone PV system for supplying the current having no connection with grid.
2. It is located at the load centre and dedicated to meet all the electrical loads of a village or community or a specific set of loads.

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3. Energy storage is generally essential.
4. It is most relevant and successful in remote and rural areas having no access to grid supply.
5. Indicative capacity of such a system is 10 W – 100 kW.
6. The main components of a general stand-alone solar PV system are shown in Fig. 1.15.1.

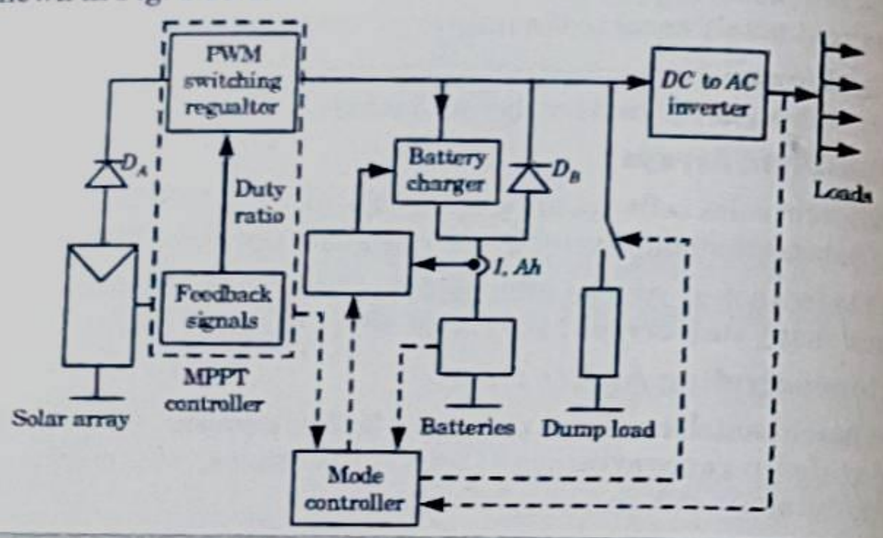

Fig. 1.15.1. A general stand-alone or autonomous solar PV system.

7. The MPPT senses the voltage and current outputs of the array and adjusts the operating point to extract maximum power under the given climatic conditions.
8. The output of the array after converting to AC is fed to loads.
9. The array output in excess of load requirement is used to charge the battery.
10. If excess power is still available after fully charging the battery, it may be shunted to dump heaters.
11. When the sun is not available, the battery supplies the load through an inverter.
12. The battery discharge diode D_B prevents the battery from being overcharged after the charger is opened.
13. The array diode D_A is to isolate the array from the battery to prevent battery discharge through array during nights.
14. A mode controller is a central controller for the entire system.
15. It collects the system signals and keeps track of charge or discharge state of the battery, matches the generated power and load and commands the charger and dump heater on-off operation.

ii. Combined Solar Power Plant or Hybrid PV System :

1. A hybrid PV system is essentially a system that employs at least one more source, other than the PV, to meet the electrical power demand of the loads.

2. The other sources that are generally used in conjunction with the PV source are diesel generators, wind generators, micro-turbines, fuel cells, etc.
3. The hybrid PV system can be classified depending on the type of source it uses, *e.g.*, PV-wind hybrid system, PV-diesel hybrid system and PV-fuel cell hybrid system.

a. PV-Wind Hybrid System :

i. In the case of the PV-wind hybrid system, the variation in the wind velocity results into large changes in the frequency and output power of the generator.

ii. As a result, it is advisable to convert the AC output to the DC and then convert it back to AC through the inverter.

iii. The PV and the rectified output of the wind generator are connected in parallel forming a DC link.

iv. However, the drawback of this system is that PV and wind both are the unreliable, sources and hence, in absence of the sun and wind, a large battery bank is required to meet the load demand.

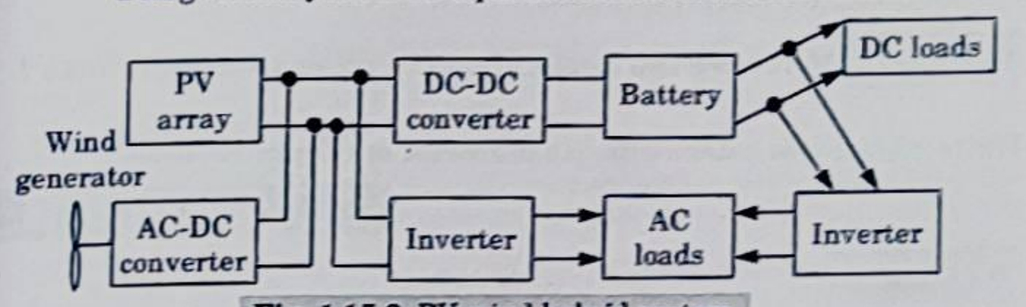

Fig. 1.15.2. PV-wind hybrid system.

PART-8

Limitations.

Questions-Answers

Long Answer Type and Medium Answer Type Questions

Que 1.16. What are the advantages and limitations of solar energy system ?

Answer

A. Advantages :

1. Noiseless and cheap energy conversion system.
2. Low cost of maintenance.

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3. Highly reliable.
4. Environment friendly.
5. Having long life.
6. Suitable for mobile loads such as cars, buses etc.
7. No fuel is required.
8. These systems are suitable for rural, remote and isolated areas.
9 Modularity in operation.
10. System modularity allows users to start with small system for single application and add on to their systems as their needs increase.

B. Limitations :

1. Higher initial cost.
2. Irregular supply of solar energy and do not generate power during cloudy season.
3. Require storage batteries for supply power during night.
4. Efficiency is low.
5. Large area required for plant.

Que 1.17. What are the applications of photovoltaic systems ?

OR

Write about the solar cells, its material and applications.

AKTU 2015-16, Marks 7.5

Answer

Solar Cell and Material : Refer Q. 1.8, Page 1–14M, Unit-1.

Application : The applications of photovoltaic systems are given below :

A. Solar Street Light :

1. It comprises of a compact fluorescent lamp, two 35 watt solar arrays and an 80 ampere-hour tubular cell battery as shown in Fig. 1.17.1.

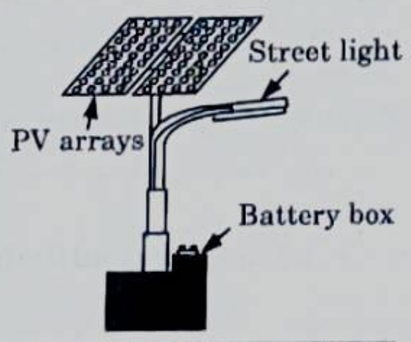

Fig. 1.17.1. Solar street light.

B. Home Lighting System :

1. These are the most popular solar PV units, typically designed to work with two light points and one TV point.

2. When necessary, a small DC fan can also be run from this system.

C. Water Pumping System :

1. It is another important application of photovoltaic systems.
2. These systems are mainly, employed in rural areas for agricultural applications, where power is not available easily and economically.

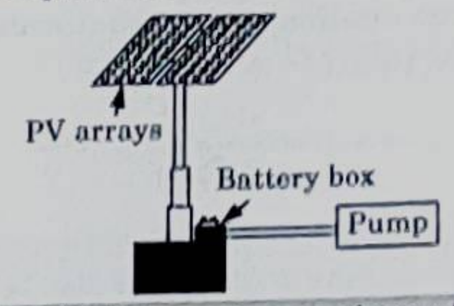

Fig. 1.17.2. Solar water pumping system.

3. Farmers use an 1800 watt PV array to operate a 2 hp DC motor pump set.
4. It can give water discharge of 140,000 litres per day from a depth up to 7 meters which is sufficient to irrigate 5-8 acres of land holding several crops.

D. Solar Vehicles :

1. Solar photovoltaic systems are also used in solar vehicles like solar cars, tractors etc. But solar vehicles are not used on commercial level until now.
2. Research and development is continuously trying to increase the efficiency of solar vehicles and to make it commercially workable.

VERY IMPORTANT QUESTIONS

Following questions are very important. These questions may be asked in your SESSIONALS as well as UNIVERSITY EXAMINATION.

Q. 1. Discuss the main features of various types of renewable and non-renewable energy sources. Also explain the importance of non-conventional energy sources in the context of global warming.

Ans. Refer Q. 1.3, Unit-1.

Q. 2. What are the conventional and non-conventional energy sources ? Write short notes on classification of energy resources.

Ans. Refer Q. 1.6, Unit-1.

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Q. 3. Explain the mechanism of photoconduction in a PV cell.

Ans. Refer Q. 1.10, Unit-1.

Q. 4. Write a short note on solar cell array.

Ans. Refer Q. 1.14, Unit-1.

Q. 5. What are the applications of photovoltaic systems ?

Ans. Refer Q. 1.17, Unit-1.

☺☺☺

2
UNIT

Solar Thermal Energy

CONTENTS

Part-1	:	Solar Radiation	**2–2M to 2–9M**
Part-2	:	Flat Plate Collectors	**2–9M to 2–11M**
Part-3	:	Material for Flat Plate Collectors	**2–11M to 2–11M**
Part-4	:	Application of Flat Plate Collectors	**2–11M to 2–12M**
Part-5	:	Performance of Flat Plate Collectors	**2–12M to 2–15M**
Part-6	:	Focusing of Collectors	**2–15M to 2–19M**
Part-7	:	Materials for Collectors	**2–19M to 2–20M**
Part-8	:	Application of Collectors	**2–20M to 2–21M**
Part-9	:	Performance of Collectors	**2–21M to 2–22M**
Part-10	:	Solar Thermal Power Plants	**2–22M to 2–25M**
Part-11	:	Thermal Energy Storage for Heating	**2–25M to 2–31M**
Part-12	:	Thermal Energy Storage for Cooling	**2–31M to 2–33M**
Part-13	:	Limitations of Solar Power Plants	**2–33M to 2–34M**

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PART-1

Solar Radiation.

Questions-Answers

Long Answer Type and Medium Answer Type Questions

Que 2.1. Write a short note on solar radiation.

Answer

1. The solar radiations received by the earth's surface vary with the location.
2. However radiation received outside the earth's atmosphere is different than what we receive on the earth surface because of absorption, reflection, scattering and attenuation by particulates and clouds present in the atmosphere.
3. The solar radiation is grouped in the following two categories :

a. Extraterrestrial Solar Radiation :

1. The intensity of the sun's radiation outside the earth's atmosphere is called extraterrestrial and has no diffuse components.

OR

Extraterrestrial radiation is the measure of solar radiation that would be received in the absence of atmosphere.

b. Terrestrial Solar Radiation :

1. The radiation we receive on the earth surface is called terrestrial radiation and is nearly 70 % of extraterrestrial radiation.
2. Solar radiations pass through the earth's atmosphere and are subjected to scattering and atmospheric absorption and a part of scattered radiations are reflected back into space.

A. Spectral Distribution of Solar Radiation :

1. Solar radiation spectral distribution shows the distribution pattern of extraterrestrial and terrestrial radiation of different wavelengths.
2. The maximum values of radiation are reaching in the range of 0.4 μm to 2.3 μm, with maximum value occurring at 0.48 μm.

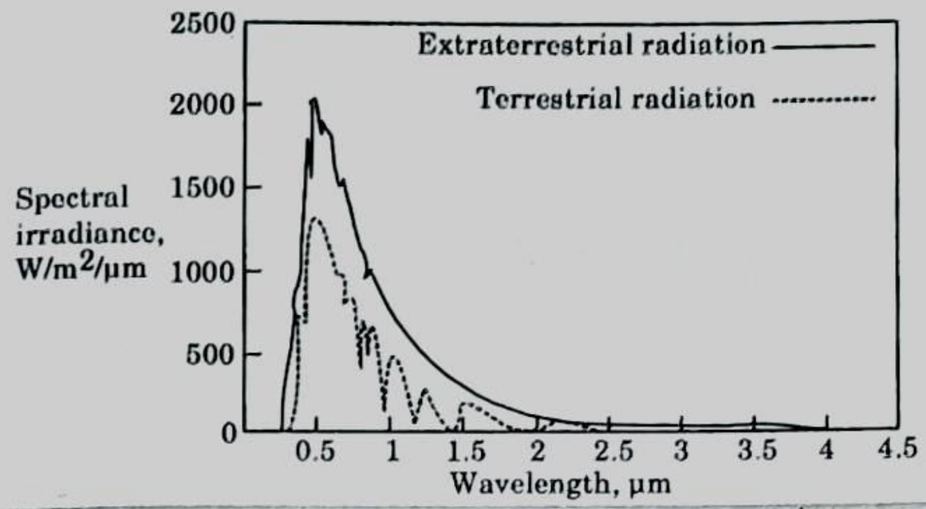

Fig. 2.1.1. Spectral distribution of extraterrestrial and terrestrial radiation.

B. Terms used in Solar Radiations :

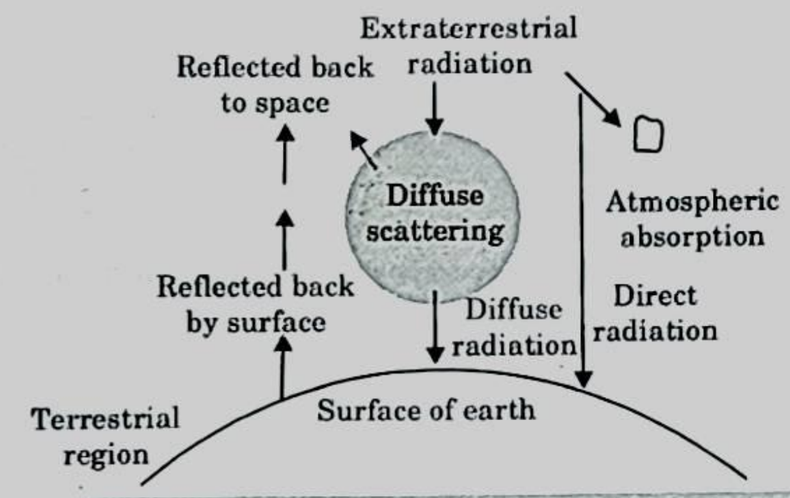

Fig. 2.1.2. Direct diffuse and total solar radiation.

i. **Beam Radiation (I_b) :** Solar radiation received on the earth's surface without change in direction is known as beam or direct radiation.

ii. **Diffuse Radiation (I_d) :** The radiation received on a terrestrial surface (scattered by aerosols and dust) from all parts of the sky dome is known as diffuse radiation.

iii. **Total Radiation (I_T) :** The sum of beam and diffuse radiation (*i.e.*, $I_b + I_d$) intercepted at the earth's surface per unit area of location is known as total radiation and it is also known as insolation. The radiations received by a collector surface are always global radiations.

iv. **Air Mass (m_a) :** It is the ratio of the path length of beam radiation through the atmosphere, to the length of path when sun is at over head or zenith.

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Que 2.2. **Describe the difference between the direct radiation and diffuse radiation.** **AKTU 2017-18, Marks 10**

Answer

S.No.	Direct Radiation	Diffuse Radiation
1.	Solar radiation received on the earth's surface without change in direction is known as direct radiation.	The radiation received on a terrestrial surface (scattered by aerosols and dust) from all parts of the sky dome is known as diffuse radiation.
2.	It has a unique path.	It does not have a unique path.
3.	Direct solar radiation is generally most intense at any one spot on the surface of the earth at solar noon.	It does not happen in diffuse radiation.
4.	It has the least amount of the atmosphere to travel through.	It has the more amount of the atmosphere to travel through.

Que 2.3. **Distinguish between global radiation and diffuse radiation.**

Answer

A. Differences between Global Radiation and Diffuse Radiation :

S.No.	Global Radiation	Diffuse Radiation
1.	Global solar radiation is the total amount of solar energy falling on a horizontal surface.	Diffuse solar radiation is the total amount of solar energy falling on a horizontal surface from all parts of the sky apart from the direct sun.
2.	Typical values for daily global solar radiation range from 1 to 35 MJ/m^2.	Typical values for daily diffuse solar radiation range from 1 to 20 MJ/m^2.
3.	The values are usually highest in clear sun conditions during the summer and lowest during winter or very cloudy days.	The values are usually highest during the cloudy conditions and lowest during clear sky days.

Que 2.4. **What do you mean by solar constant ?**

Answer

1. The solar constant is the energy received from the sun on a unit area perpendicular to sun's rays at the mean distance from the sun, outside the atmosphere.
2. The standard value of the solar constant based on experimental measurements is 1367 W/m^2 with an accuracy of ±1.5 %.
3. The value of solar constant remains constant throughout the year. However, this value changes with location because earth to sun distance changes seasonally with time.
4. The extraterrestrial radiation observed on different days is known as apparent extraterrestrial solar irradiance and can be calculated on any day of the year using the following equation :

$$I_o = I_{sc}\left[1 + 0.033\cos\frac{360n}{365}\right]$$

Where, I_o = Apparent extraterrestrial solar irradiance (W/m^2),

n = Number of days of the year counting January 1 as the first day of the year, and

I_{sc} = 1367 W/m^2.

Note : According to above equation, the apparent solar irradiance will be maximum during December last or first week of January, as the earth's centre is nearest to the sun during these days.

Que 2.5. **Explain solar radiation geometry.**

OR

Define the terms :

a. Altitude angle,
b. Incident angle,
c. Zenith angle,
d. Latitude angle, and
e. Hour angle.

Answer

1. The solar radiation comes from the sun on the earth's surface with arbitrary orientation and can be found out by knowing the beam radiation falling either on horizontal or perpendicular surface to sun's radiation.
2. The amount of incident beam flux on an inclined surface per unit time and per unit area is given as

$$I_N = I\cos\theta_i$$

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Where, I = Incident flux of beam radiation, and

θ_i = Angle of incident of beam radiation.

3. The various angles which are useful for conversion of beam radiation on the arbitrary surface are :

a. Incident Angle (θ) :

1. It is defined as the angle between the incident beam radiation and the normal to a plane surface.

b. Latitude Angle (ϕ) :

1. The latitude of a place is the angle subtended by the radial line joining the place to the centre of the earth, with the projection of the line on the equatorial plane.
2. **Note :** The latitude is taken as positive for any location towards the northern hemisphere and negative towards the southern hemisphere *i.e.*, the latitude at equator is 0° while at north and south poles are +90° and – 90° respectively.

c. Declination Angle (δ) :

1. The declination is the angle made by the line joining the centres of the sun and the earth with its projection on the equatorial plane.
2. The declination angle varies from a maximum value of +23.5° on June 21 to a minimum of –23.5° on December 21.
3. The declination (in degrees), for any given day may be calculated from the approximate equation of "Cooper".

$$\delta = 23.45 \sin\left[\frac{360}{365}(284 + n)\right]$$

where, n is the number of days in the year.

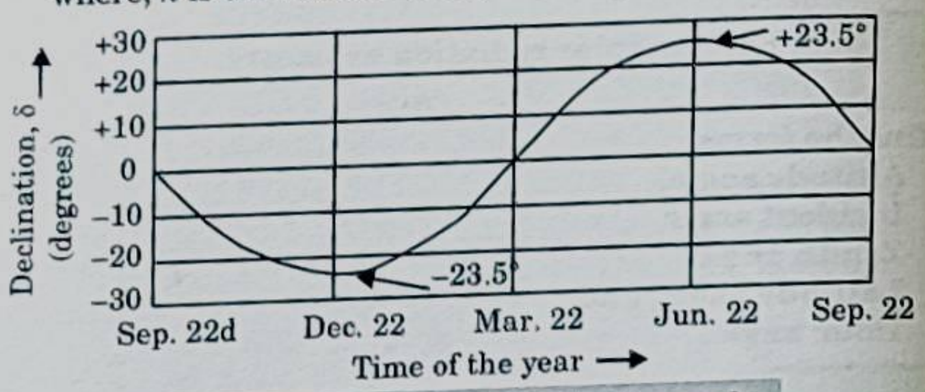

Fig. 2.5.1. Variation of declination angle.

d. Hour Angle (ω) :

1. It is the angle through which the earth must be rotated to bring the meridian of a point directly in line with the sun's ray.
2. In other words, it is the angular displacement of the sun, east or west of the local meridian, due to the rotation of the earth on its axis at an angle of 15° per hour.

3. The hour angle is zero at solar noon, negative in the morning and positive in the afternoon for the northern hemisphere (India) and vice-versa for southern hemisphere (Australia).
4. Mathematically hour angle can be expressed as

$$\omega = 15(\text{LST} - 12)$$

Where, LST = Local solar time.

e. Altitude Angle (α) :

1. It is a vertical angle between the projection of the sun's rays on the horizontal plane and the direction of the sun's rays.

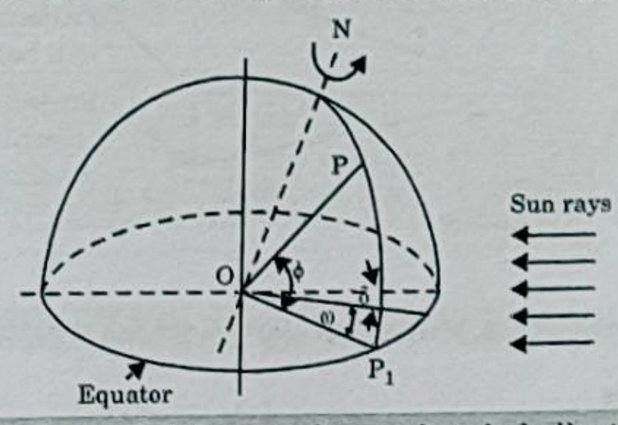

Fig. 2.5.2. Latitude ϕ, hour angle ω and sun's declination δ.

f. Zenith Angle (θ_z) :

1. It is the vertical angle between the sun's rays and line perpendicular to the horizontal plane through the point.

Mathematically :

$$\theta_z = \frac{\pi}{2} - \alpha$$

g. Surface Azimuth Angle (γ) :

1. It is the angle in the horizontal plane, between the line due south and the horizontal projection of the normal to the inclined plane surface.
2. By convention, the angle will be taken negative for northern hemisphere (India) and vice-versa for southern hemisphere (Australia).

h. Slope (β) :

1. It is the angle between the plane surface, under consideration, and with the horizontal.
2. It is taken to be positive for surface sloping towards south and negative for surfaces sloping towards north.

i. Solar Azimuth Angle (γ_s) :

1. It is the angle in a horizontal plane, between the line due south and the projection of beam radiation on the horizontal plane.

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2. Thus it gives the direction of the shadow cast in the horizontal plane by a vertical rod.

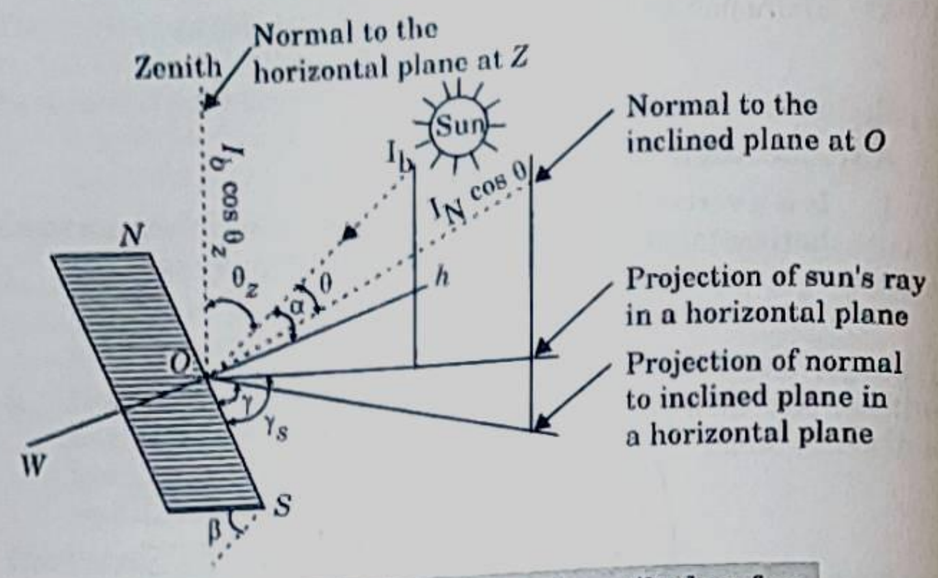

Fig. 2.5.3. Various sun-earth angle on tilted surface.

Que 2.6. **Write short note on :**

A. Day Length

B. Local Apparent Time (LAT).

OR

Write short note on local apparent time (LAT).

AKTU 2015-16, Marks 03

Answer

A. Day Length :

1. The time of sunrise, sunset and the duration of the day length depen upon the latitude of the location and the month in the year.
2. At sunrise and sunset, the sunlight is parallel to the ground surface wi a zenith angle of 90°.
3. The hour angle pertaining to sunrise or sunset (ω_s) is given below :

a. At solar noon, $\theta_z = 0$ and $\omega = \omega_s$

$$\cos \omega_s = -\tan\phi \tan\delta$$

b. On tilted surface,

$$\omega_s = \cos^{-1}[(-\tan(\phi - \beta)\tan\delta)]$$

4. The corresponding day length (in hours) is given by

$$N = \frac{2}{15}\cos^{-1}\left[-\tan(\phi-\beta)\tan\delta\right]$$

B. Local Apparent Time (LAT) :

1. The time used for calculating the hour angle ω is the local apparent time.
2. It is given by

LAT = Standard time + Equation of time correction + 4 (Standard time longitude – longitude of location)

PART-2

Flat Plate Collectors.

Questions-Answers

Long Answer Type and Medium Answer Type Questions

Que 2.7. **What are solar collectors ? How solar collectors are classified ?**

Answer

Solar Collectors :

1. Solar collectors are used to collect the solar energy and convert this energy into the thermal energy by absorbing them.
2. This thermal energy is further used for heating a collector fluid such as water, oil or air.
3. Solar collector surface is designed for high absorption and low emission.
4. Solar collectors are classified in two types :

a. Non-Concentrating Collector :

1. It is also known as flat plate solar collector.
2. In these collectors, the area of collector to grasp the solar radiation is equal to the absorber plate and has concentration ratio of 1.

b. Concentrating Collector :

1. It is also known as focusing type solar collector.
2. In these collectors, the area of the collector is kept less than the aperture through which the radiation passes, to concentrate the solar flux and has high concentration ratio.

Que 2.8. **Explain the principle of conversion of solar energy into heat. Explain a flat plate solar collector.**

AKTU 2017-18, Marks 10

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OR

Classify different types of solar thermal collector and show the constructional details of a flat plate collector. What are its main advantages ? **AKTU 2016-17, Marks 10**

Answer

A. Principle :

1. When solar radiation from the sun comes in the form of light (a short wave radiation) to the earth, visible sunlight is absorbed on the ground and transformed into heat energy ; the material becomes warm and stores the heat, conducts it to surrounding materials (air, water other solids or liquids) or reradiates it to other material of lower temperature.

B. Classification : Refer Q. 2.7, Page 2–9M, Unit-2.

C. Flat Plate Collector :

1. Flat plate collector is simplest in design and it is most important part of any solar thermal energy system.
2. In this collector both direct and diffuse radiations are absorbed and converted into useful heat.

D. Components of Flat Plate Collector :

i. Absorber plate,
ii. Transparent covers,
iii. Insulation, and
iv. Box.

i. Absorber Plate :

1. Absorber plate is used to grasp and absorb solar radiation.
2. The plate is usually metallic (copper, aluminum or steel), sometimes plastics have been used in some low temperature applications.

ii. Transparent Covers :

1. These are one or more sheets made of glass for trapping the heat received by the absorber plate.
2. It helps in reducing the convective and radiative heat losses.

iii. Insulation : It minimizes the heat losses by conduction.

iv. Box : It contains the above components and keep them into desired position.

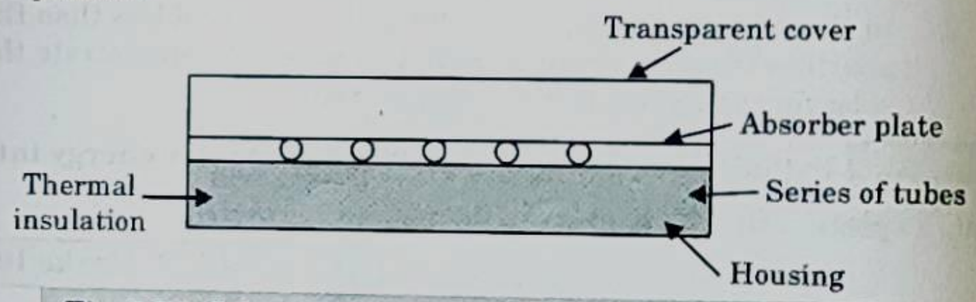

Fig. 2.8.1. Schematic diagram of a flat plate solar collector.

E. Advantages of Flat Plate Collector :

1. It absorbs both direct and diffuse radiations.
2. There is no need of tracking.
3. It has low cost and requires less maintenance.

PART-3

Material for Flat Plate Collectors.

Questions-Answers

Long Answer Type and Medium Answer Type Questions

Que 2.9. Write a short note on materials used for flat plate collectors.

Answer

1. The properties of the materials used for collectors can be classified as :
   a. Thermophysical properties such as thermal conductivity, heat capacity etc.
   b. Physical properties like density, tensile strength, melting point etc., and environmental properties like moisture penetration, corrosion resistance and degradation due to pollutants in atmosphere.
2. The material for absorber plate should have high thermal conductivity, adequate tensile strength and good corrosion resistance.
3. The most common material used for absorber plate is Copper because of high conductivity and resistance to corrosion.
4. Other materials which are used for absorber plate are Aluminium, Iron, Brass, Silver, Tin and Zinc.
5. The material for insulation should have low thermal conductivity, should be stable at high temperature.
6. Some commonly used materials are crown white wool, glass wool, calcium silicate, cellular foam etc.
7. For cover plate, tempered glass is most common material. Transparent plastic materials such as acrylic polycarbonate plastic, polyvinyl fluoride are used for cover plate.

PART-4

Application of Flat Plate Collectors.

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Questions-Answers

Long Answer Type and Medium Answer Type Questions

Que 2.10. Explain the disadvantages and application of flat plate collector.

Answer

A. Disadvantages of Flat Plate Collector :

1. Low temperature is achieved.
2. There is a large heat loss by conduction due to large area.
3. These are heavy in weight.

B. Applications of Flat Plate Collector :

1. It is used in solar water heating.
2. It is used in space heating and cooling.
3. It is used in low temperature power generation.

PART-5

Performance of Flat Plate Collectors.

Questions-Answers

Long Answer Type and Medium Answer Type Questions

Que 2.11. What do you understand by performance analysis of flat plate collector ?

Answer

A. Performance Analysis of Flat Plate Collector :

The performance of solar collector depends on the following factors :

a. Fin Efficiency Factor (F_e) : It is defined as the ratio of actual rate of heat transferred to the heat that would be transferred, if entire fins (plate area) are at base temperature.

Mathematically, $$F_e = \frac{Q_{actual}}{A_c[\alpha_o \tau_o I_t - U_L(T_p - T_a)]}$$

Where, Q_{actual} = Actual rate of heat transferred to the tube base,
A_c = Collector area,
I_t = Incident total radiations,
U_L = Overall heat loss coefficient,
T_p = Plate temperature,
T_a = Ambient temperature,
α_o = Absorptivity, and
τ_o = Transmitivity.

b. Collector Efficiency Factor (F_c) : It is defined as the ratio of useful heat removed by flowing fluid in the tubes to the rate of heat transferred to the fluid, if the fin is at local fluid temperature.
Mathematically,

$$F_c = \frac{Q_u}{A_c[\alpha_o \tau_o I_t - U_L(T_f - T_a)]}$$

Where, Q_u = Useful heat removed by flowing fluid in the tubes, and
T_f = Local fluid temperature.

c. Collector Heat Removal Factor (F_H) : It is defined as the ratio of actual useful energy gain by fluid to the rate of heat transferred to the fluid, if the fin is at inlet fluid temperature.
Mathematically,

$$F_H = \frac{Q_u}{A_c[\alpha_o \tau_o I_t - U_L(T_{fi} - T_a)]}$$

or, $$F_H = \frac{mC_f(T_{fo} - T_{fi})}{A_c[\alpha_o \tau_o I_t - U_L(T_{fi} - T_a)]}$$

Where, m = Mass flow rate of fluid per unit area of collector,
C_f = Specific heat of fluid,
T_{fo} = Outlet fluid temperature, and
T_{fi} = Inlet fluid temperature.

d. Collector Efficiency (η_c) : It is defined as the ratio of useful energy absorbed by collector to the incident solar energy over it.
Mathematically,

$$\eta_c = \frac{Q_u}{A_c I_t}$$

or $$\eta_c = \frac{F_R A_c[\alpha_o \tau_o I_t - U_L(T_{fi} - T_a)]}{A_c I_t}$$

$[\because Q_u = F_R A_c(\alpha_o \tau_o I_t - U_L(T_{fi} - T_a))]$

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$$\eta_c = F_R\alpha_o\tau_o - \frac{F_R U_L(T_{fi} - T_a)}{I_t}$$

or

$$\eta_c = mx + c \quad \text{(It is a linear equation)}$$

or

$$m = -F_R U_L \quad \text{(Effective heat loss coefficient)}$$

Where,

$$x = \frac{(T_{fi} - T_a)}{I_t}$$

and,

$$c = F_R\alpha_o\tau_o \quad \text{(Effective optical efficiency)}$$

Que 2.12. **What are the factors which affect the performance of flat plate collector ?**

Answer

1. The different factors which affect the performance of flat plate collector are :

a. Incident Solar Radiation :

1. The efficiency of collector is directly related with solar radiation falling on it and increases with rise in temperature.

b. Number of Cover Plate :

1. Increase in number of cover plate reduces the internal convective heat losses but also prevents the transmission of radiation inside the collector.
2. Therefore, the increase in cover plates will reduce the heat absorbed by the absorber.

c. Spacing between Absorber Plates and Glass Cover :

1. The more space between the absorber and the cover plate, the less is the internal heat losses.

d. Collector Tilt :

1. To achieve better performance, flat plate collector should be tilted at angle of latitude of the location.
2. The collector is placed with south facing at northern hemisphere to receive maximum radiation throughout the day.

e. Selective Surface :

1. It should be able to withstand high temperatures.
2. It should not oxidise.
3. It should be corrosion resistant.

f. Fluid Inlet Temperature :

1. On increasing the inlet temperature of the fluid there is an increase in operating temperature of the collector and this lead to decrease in efficiency.

g. Dust on Cover Plate :

1. The efficiency of collector decreases as dust particles increases on the cover plate.
2. Frequent cleaning is required to get the maximum efficiency of collector.

PART-6

Focusing of Collectors.

Questions-Answers

Long Answer Type and Medium Answer Type Questions

Que 2.13. **Explain the concentrating solar collector. Also, discuss its working.**

OR

Discuss the principle of a concentrating solar collector. How it differs with flat plate collector ? How collector coating can be used to improve the performance of collector with reference to the flat plate collector ? What is the concentrating ratio for focusing collector ?

AKTU 2017-18, Marks 10

Answer

A. Concentrating Solar Collector :

1. Concentrating solar collector is a device to collect solar energy with high intensity of solar radiation on the absorbing surface by the help of reflector or refractor.

OR

A concentrating solar collector is a modified form of flat-plate collector by introducing a reflecting or refracting surface between the solar radiation and absorber.

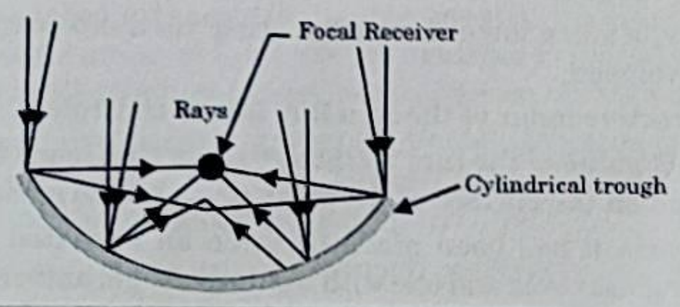

Fig. 2.13.1. Concentrating type of collector.

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B. Principle :

1. Concentrating solar collector uses reflective surface to concentrate sunlight to a small area, where it absorbed and converted to heat.

C. Working of Concentrating Solar Collector :

1. This collector system comprises of a concentrator and an absorber.
2. In these collectors radiation from sun falls on a relatively large area which is focused through concentrator on to an absorber of considerably smaller area.
3. As a result of energy concentration, fluid can be heated up to a temperature of 500 °C or more.
4. These systems have high collector efficiency, since the losses are much less as compared to non-concentrating type of collectors.

D. Difference between Flat and Focusing Collectors :

S. No.	Concentrating Collectors	Flat-Plate Collectors
1.	It requires less absorber area.	It requires more absorber area.
2.	It has high collection efficiency.	It has low collection efficiency.
3.	Suitable for large power generation.	Suitable for small power generation.
4.	Heat storage cost is lower.	Heat storage cost is higher.
5.	It has high initial and maintenance cost.	It has low initial and maintenance cost.
6.	No need of anti-freeze solutions.	It needs anti-freeze solutions.

E. Use of Collector Coating to Improve the Performance of Collector :

1. Most solar collectors employ a transparent cover plate often made of glass. These materials reflect around 8 % of the incident solar radiation, which leads to the reduction in the collector heat output.
2. The use of an antireflection coating could therefore improve the performance of such system by increasing the transmitted energy through the glass cover.
3. Recently, a silica low-reflection coating via a dip-coating process has been developed.
4. The refractive index of the thin film is well controlled.
5. The exact value of the film refractive index that leads to a minimum of reflection on the surface of the glass cover can be achieved.
6. A comparison has been made between an uncoated flat-plate solar collector glass cover and one with a porous sol-gel anti-reflection coating. Using the porous sol-gel coating with the index of refraction of $n = 1.23$ on the glass cover of the solar collector increases the useful energy by a factor of approximately 1.05.

F. Concentration Ratio (C) : It is defined as the ratio of the effective aperture ratio to the absorber tube area.

Mathematically,

$$C = \frac{\text{Effective aperture ratio}}{\text{Absorber tube area}}$$

$$= \frac{W - D_o}{\pi D_o}$$

Where, W = Aperture, and

D_o = Outer diameter of absorber tube.

Que 2.14. Enumerate the different types of concentrating collector.

Answer

1. The different types of concentrating collector are as follows :

a. Parabolic Trough Collector :

i. Principle : The principle of parabolic trough collector is, when the solar radiations falls on the area of parabolic reflector are concentrated at the focus of parabola.

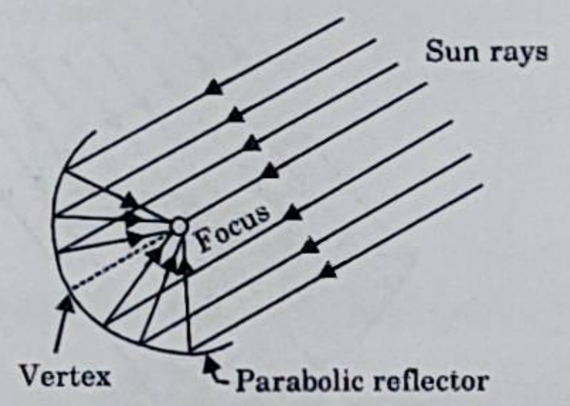

Fig. 2.14.1. Cross section of parabolic trough collector.

ii. Working :

1. In the parabolic trough collector mostly cylindrical parabolic concentrators are used in which the absorber is placed along the focus axis where collector pipe is used as an absorber with a selective coating.
2. The solar radiation coming from particular direction is collected over the area of collector and then focused to the collector pipe which is placed along the focus line to heat up the fluid.
3. The orientation of the parabolic trough collector is kept in the east-west or north-south directions.

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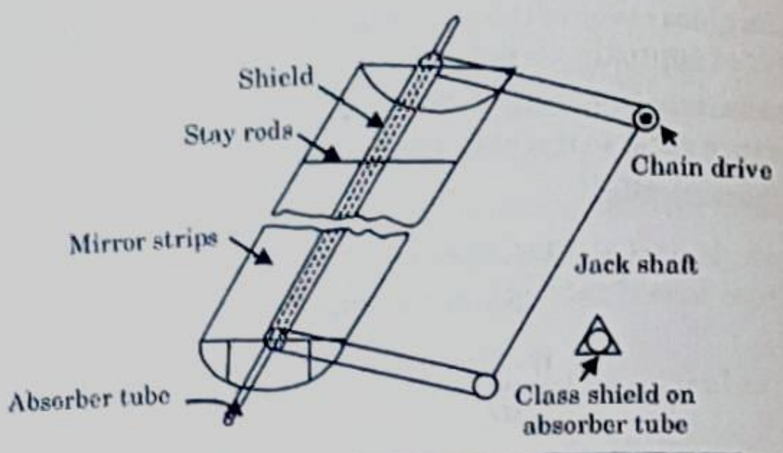

Fig. 2.14.2. A typical cylindrical parabolic system.

b. Mirror-Strip Reflector :

1. It has a number of planes or slightly curved or concave mirror strips which are mounted on a base.
2. These individual mirrors are placed at such angles that the reflected solar radiations fall on the same focal line where the absorber pipe is placed.
3. In this system, collector pipe is rotated so that the reflected rays on the absorber remain focused with respect to changes in sun's elevation.

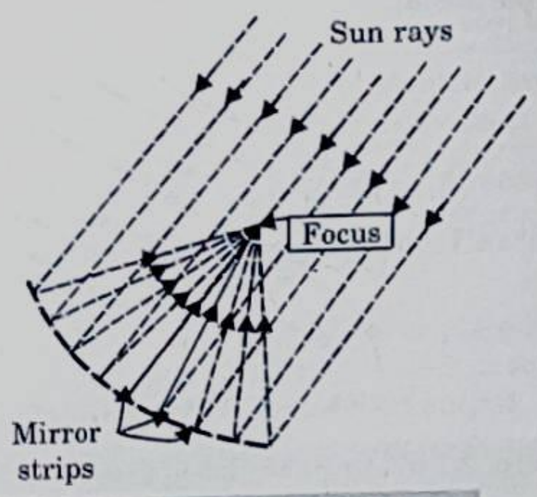

Fig. 2.14.3. Mirror-strip solar collector.

c. Fresnel Lens Collector :

1. In this collector, a Fresnel lens is used in which linear grooves are present on one side and flat surface on the other side.
2. The solar radiations which fall normal to the lens are refracted by the lens and are focused on the absorber (tube) as shown in Fig. 2.14.4.

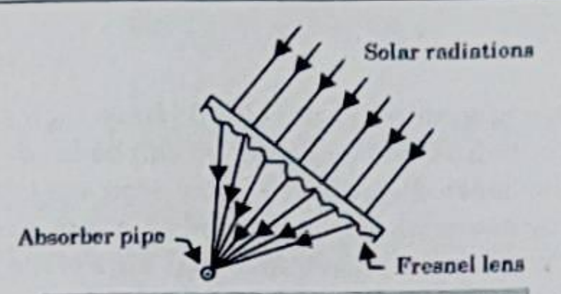

Fig. 2.14.4. Fresnel lens collector.

d. Compound Parabolic Concentrator (CPC) :

1. It is also known as Winston collector.
2. It is a trough like arrangement of two facing parabolic mirrors which are attached to an absorber.
3. In this system, solar radiations from all directions are reflected towards the absorber kept at the bottom, which collects both direct and diffused solar radiations to heat up the fluid.

i. Advantages :

1. It has high concentration ratio.
2. There is no need of tracking.
3. The efficiency for accepting diffuse radiation is high.

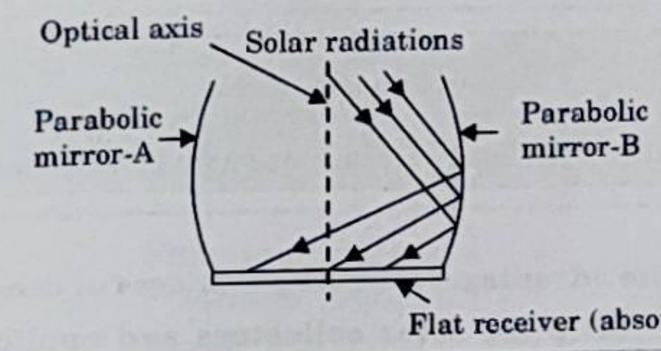

Fig. 2.14.5. Compound parabolic concentrator (CPC).

PART-7

Materials for Collectors.

Questions-Answers

Long Answer Type and Medium Answer Type Questions

Que 2.15. **Write a short note on materials used for concentrators.**

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Answer

1. The reflector of a concentrator should have high reflectivity therefore mirror glass or front surface mirrors can be used.
2. Glass is the most durable with low iron content and is used as a transmitting material. Now-a-days plastics are also in use.
3. Acrylic is found to be a good material for Fresnel lenses. Polymethyl methacrylate is generally used.
4. Glass and transparent plastic films are generally used as cover material for receivers. Glass should have low iron content to reduce absorption.
5. Coatings are required to have strong solar absorptivity, weather resistance, stability at high temperature. Black paints are good. Black chrome is also suitable. It can be electroplated on steel, copper, aluminium etc.
6. Other metal oxide coatings are black copper oxides, black nickel etc. Insulation is required to reduce heat losses. Fiber glass with and without binder, urethane foams and mineral fibre blankets are commonly used for insulation.

PART-8

Application of Collectors.

Questions-Answers

Long Answer Type and Medium Answer Type Questions

Que 2.16. Write advantages and disadvantages of concentrating collector over flat plate solar collectors and applications of concentrating collectors.

Answer

A. Advantages of Concentrating Collector :

1. It gives high concentration ratio.
2. High fluid temperature can be achieved (up to 500 °C).
3. Thermal heat loss is less.
4. Efficiency of this system increases at high temperature.
5. In expensive process.

B. Disadvantages of Concentrating Collector :

1. These collectors are best suited for places having more number of clear days in a year.
2. It has non-uniform flux on absorber.
3. It needs costly tracking device.
4. It has high initial cost.

5. It needs maintenance in order to retain the quality of reflecting surface against dirt and oxidation.

C. Applications :

1. They are used for which need high amount of heat such as power generation.
2. They used in solar power plant.
3. They used to collected large solar energy which is used to convert water into steam.

PART-9

Performance of Collectors.

Questions-Answers

Long Answer Type and Medium Answer Type Questions

Que 2.17. What do you understand by performance analysis of concentrator collector ?

Answer

A. Performance Analysis : The performance of concentrator collector depends on the following factors :

a. Concentration Ratio (C) : It is defined as the ratio of the effective aperture ratio to the absorber tube area.

Mathematically,

$$C = \frac{\text{Effective aperture ratio}}{\text{Absorber tube area}}$$

$$= \frac{W - D_o}{\pi D_o}$$

Where, W = Aperture, and

D_o = Outer diameter of absorber tube.

b. Intercept Factor (γ) : It is defined as the ratio of radiation intercepted by absorber tube to the total reflected radiation.

Mathematically, $\gamma = \frac{\text{Radiation intercepted by absorber tube}}{\text{Total reflected radiation}}$

c. Collector Efficiency Factor (F_c) : Collector efficiency factor is given by the relation :

$$F_c = \frac{1}{U_l\left[\frac{1}{U_l} + \frac{D_o}{D_i h_f}\right]}$$

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Where, U_l = Overall heat loss coefficient,
D_o and D_i = Outer and inner diameter, and
h_f = Heat transfer coefficient of the tube.

d. Instantaneous Collector Efficiency (η_i) : Instantaneous collec efficiency is given by the relation :

$$\eta_i = \frac{q_u}{(I_b r_b + I_d r_d)WL}$$

Where, q_u = Useful heat gain,
$I_b r_b$ = Beam radiation normally incident on apertu
$I_d r_d$ = Diffuse radiation, and
L = Length of concentrator.

PART-10

Solar Thermal Power Plants.

Questions-Answers

Long Answer Type and Medium Answer Type Questions

Que 2.18. Explain the different types of solar thermal power plan

Answer

A. Solar Thermal Power Plant :

1. Solar thermal power generation involves the collection of solar h which is utilized to increase the temperature of a fluid in a turb operating on a cycle such as Rankine or Brayton.
2. Solar thermal power plants can be classified as low, medium and hi temperature cycles.
3. Low temperature cycles operate at about 100 °C, medium temperatu cycles up to 400 °C, while high temperature cycles work above 500 °

a. Low Temperature Solar Power Plant :

1. A low temperature solar power plant uses flat-plate collector arr shown in Fig. 2.18.1.
2. Hot (above 90 °C) water is collected in an air insulated tank. It flo through a heat exchanger, through which the working fluid of t energy conversion cycle is also circulated.
3. The working fluid is either methyl chloride or butane having a l boiling temperature up to 90 °C.
4. Vapours so formed operate a regular Rankine cycle by flowing throu a turbine, a condenser and a liquid pump.

5. As the temperature difference between the turbine outlet and the condensed liquid flowing out is small, *i.e.*, about 50 °C, the overall efficiency of the generating system is about 2 % (8 % pumped cycle efficiency × 25 % collector system efficiency).
6. Finally, the organic fluid is pumped back to the evaporator for repeating the whole cycle.

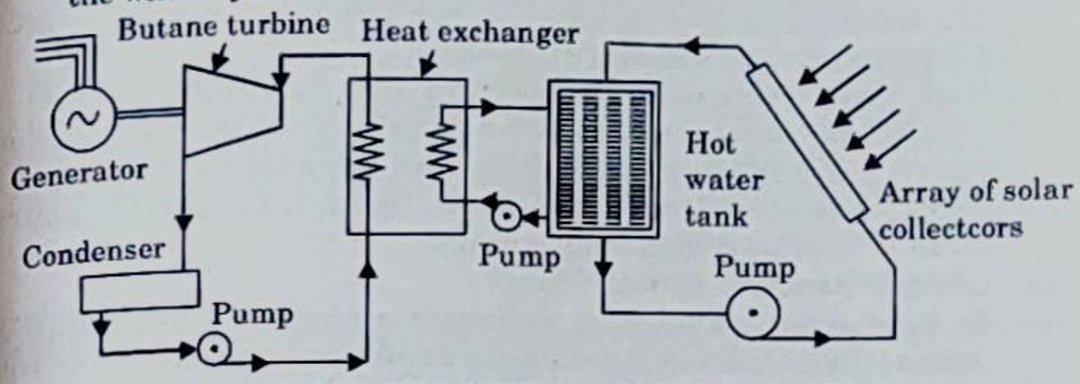

Fig. 2.18.1. Low temperature solar power plant.

7. Such plants up to 150 kW capacities are operative in Israel for the last 25 years.

b. Medium Temperature Solar Power Plant :

1. Solar thermal power plants operating on medium temperatures up to 400 °C use the line focusing parabolic collector for heating synthetic oil flowing in the absorber tube as shown in Fig. 2.18.2. A schematic diagram of a typical plant is shown in Fig. 2.18.2.

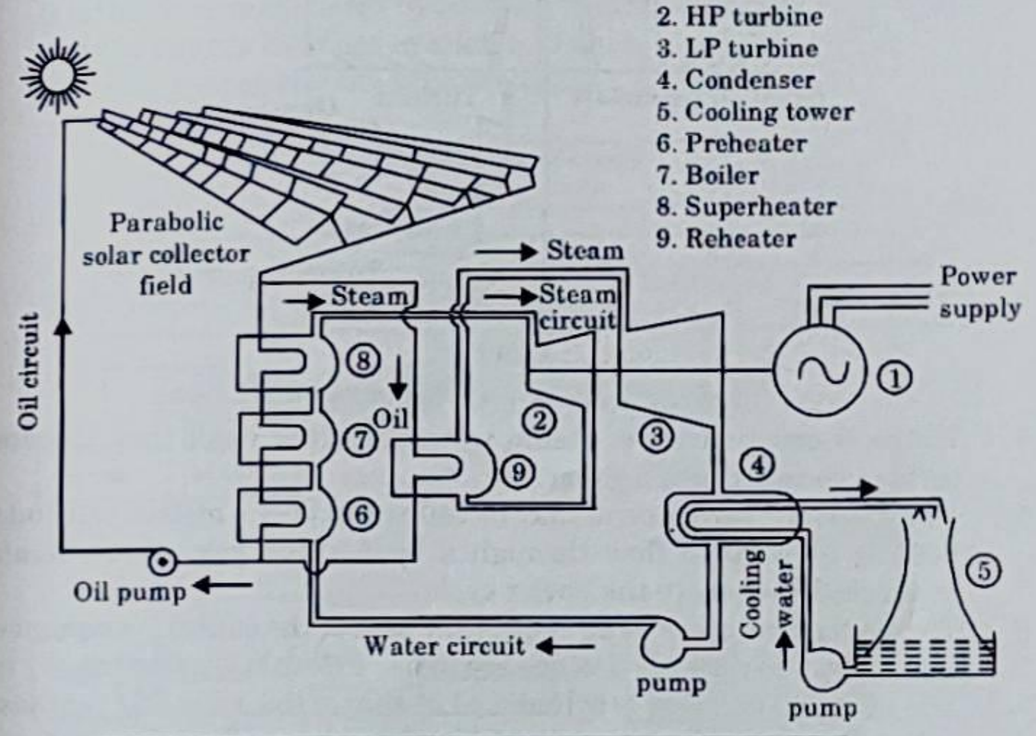

Fig. 2.18.2. Medium temperature solar power plant.

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2. A suitable sun-tracking arrangement is made to ensure that maximum quantity of solar radiation is focused on the absorber pipeline.
3. Preheater and superheater are used to increase the inlet steam temperature for the high pressure (HP) turbine.
4. Reheaters are used to raise the steam temperature for low pressure (LP) turbine.
5. The system generates superheated high pressure steam to operate Rankine cycle with maximum efficiency.

c. High Temperature Solar Thermal Power Generator :

1. For efficient conversion of solar heat into electrical energy, the working fluid needs to be delivered into turbine at a high temperature.
2. There are two possible systems-the 'paraboloidal dish' and the 'central receiver' to achieve high temperatures.

d. Central Receiver Power Plants :

1. In these power plants, solar radiations are reflected from arrays of mirrors (called heliostats) installed in circular arcs around provided with a tracking tower.
2. Reflected radiations concentrate on to the receiver.
3. The array is provided with a tracking control system that focuses beam radiation towards the receiver as shown in Fig. 2.18.3.

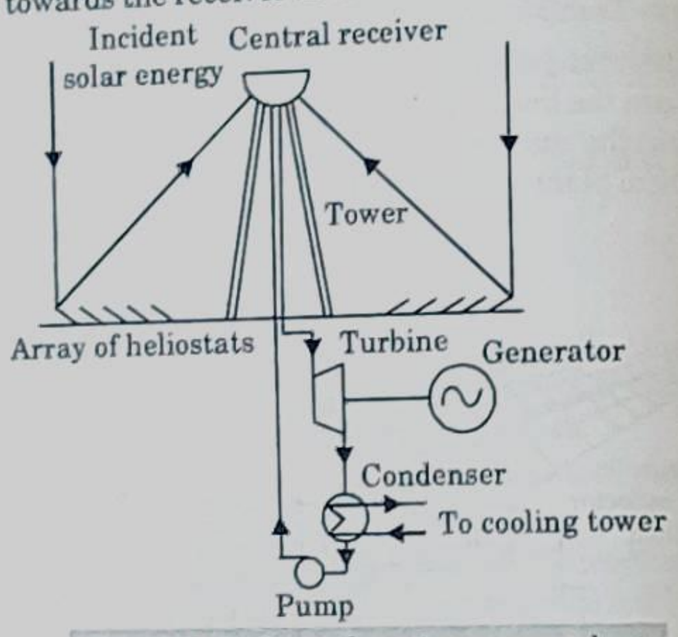

Fig. 2.18.3. Central receiver power plant.

4. Water is converted into steam in the receiver itself that operates turbine coupled with a generator.
5. Alternatively, the receiver may be utilized to heat a molten salt and the fluid is allowed to flow through a heat exchanger where steam is generated to operate the power cycle.
6. The 'central receiver' is an important part of the collection equipment.
7. Typically, two receiver designs are in use external type and cavity type. The external receiver is cylindrical in shape; the solar flux reaches the outer surface and heat is absorbed by the receiver fluid flowing through the tubes on the inner surface.

8. In a 'cavity receiver', the solar flux enters through several apertures, where the radiant energy is transferred to the receiver fluid.

PART-11

Thermal Energy Storage for Heating.

Questions-Answers

Long Answer Type and Medium Answer Type Questions

Que 2.19. Classify various energy storage systems. Describe steam storage system for solar thermal energy.

OR

Explain sensible heat storage, latent heat storage and thermochemical storage of solar energy.

Answer

A. Energy Storage System : It stores the excess amount of energy than requirement of the demand and supplies stored energy when the demand exceeds the supply of energy of the system.

a. Thermal Energy Storage :

1. It is the storage of energy by heating, melting or vaporization of material and the energy becomes available as heat.
2. Thermal energy storage is of two types :

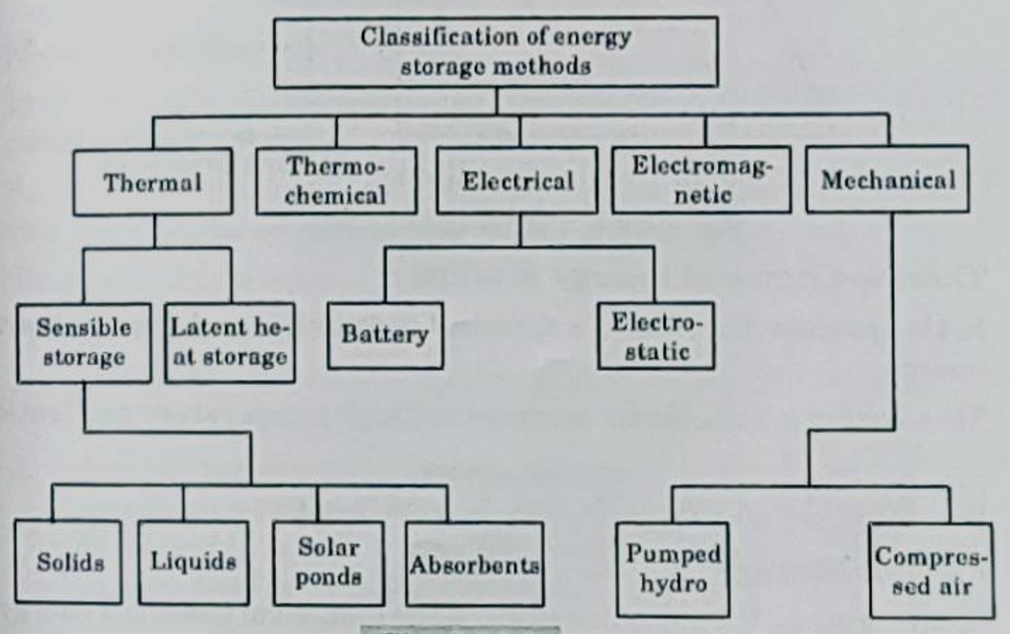

Fig. 2.19.1.

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i. Sensible Heat Storage :

1. The energy stored in this system is due to rise in temperature of the storage medium (solid or liquid).
2. When the phase does not change on heating solid or liquid, then this type of storage is called sensible heat storage.
3. The sensible energy (E) is given by the relation :

$$E = m\int_{T_1}^{T_2} c_p.dT$$

Where, m = Mass,
c_p = Specific heat at constant pressure,
T_1 = Initial temperature, and
T_2 = Final temperature.

ii. Latent Heat Storage :

1. The energy stored in this system is in the form of latent heat caused by phase change during heating either from solid to liquid or liquid to vapour.

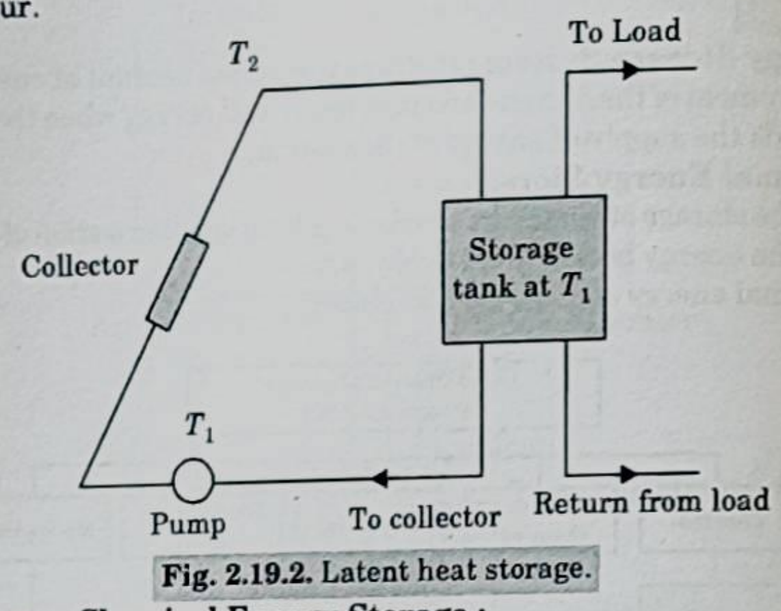

Fig. 2.19.2. Latent heat storage.

b. Thermo-Chemical Energy Storage :

1. In this process, the heat of a chemical reaction is used to store thermal energy.
2. This system is suitable for medium and high temperature applications.

For example : $A + B \xrightarrow[\downarrow \quad \downarrow]{\text{Solar heat absorbed}} C + D$

c. Electric Storage :

1. In this process, the capacitor stores large amount of electrical energy for long periods.

2. The total energy stored is given by

$$\text{T.E.} = \frac{1}{2} V\varepsilon E^2$$

Where, V = Volume of the dielectric,
ε = Dielectric constant, and
E = Electric field strength.

3. These are of two types :

i. **Battery Storage :** A rechargeable storage battery (called secondary battery) receives electrical energy as direct current which is stored in the form of chemical energy by a reversible electro-chemical reaction.

ii. **Electrostatic Energy Storage :** It is the energy stored in the large capacity capacitors. The total energy stored is given by :

$$H = \frac{1}{2} \varepsilon.V.E^2$$

Where, ε = Dielectric constant,
V = Volume of the dielectric, and
E = Electric field strength.

d. Electromagnetic Energy Storage : In this process, the energy is stored in the magnetic field of a superconducting coil, carrying direct current. The energy stored is given by :

$$E = \frac{1}{2} LI^2$$

Where, L = Inductance, and
I = Current.

e. Mechanical Energy Storage :

1. In this process, the energy can be stored in the form of kinetic energy or potential energy.
2. So, in this process the storage of solar energy firstly converts into these two forms of energy before utilization.
3. These are of two types :

i. Pumped Hydro Storage :

1. Pumped hydro storage system consists of two reservoirs *i.e.*, upper and lower reservoir.
2. A pump-turbine combination is installed between these reservoirs which can work as a pump or turbine and the energy to be stored is developed by a solar engine which is further used to drive the pump to raise the water from lower reservoir to upper reservoir.

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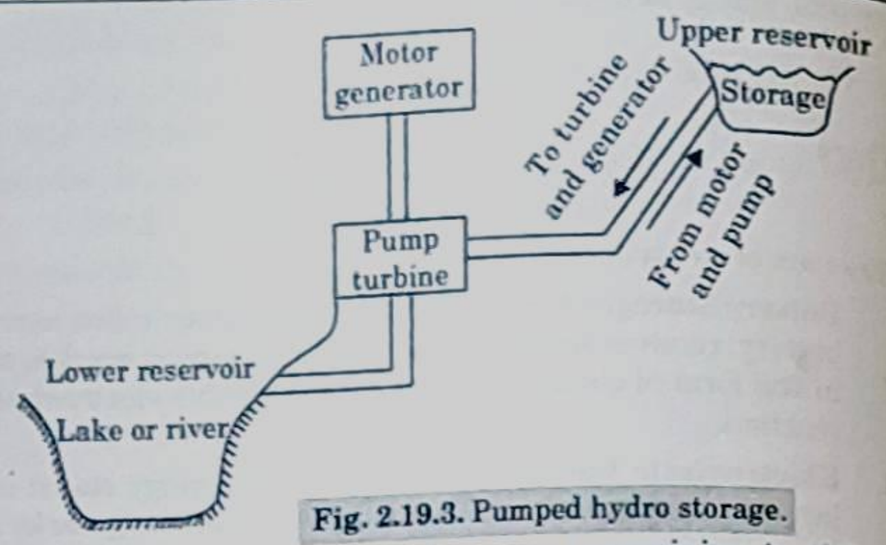

Fig. 2.19.3. Pumped hydro storage.

3. Therefore, the energy stored in upper reservoir is potential energ
4. When water flows down from upper reservoir to the lower reservoi the pump and turbine combination acts as a turbine to conver potential energy into the mechanical energy.
5. This mechanical energy can be further converted into electrica energy by a generator which is coupled to the turbine.

b. Compressed Air Storage :

i. Compressed air energy storage has similar principles as pumpe storage system.

ii. The excess electrical energy generated by turbine during low load can be stored by compressing the air in large vessels at high pressures.

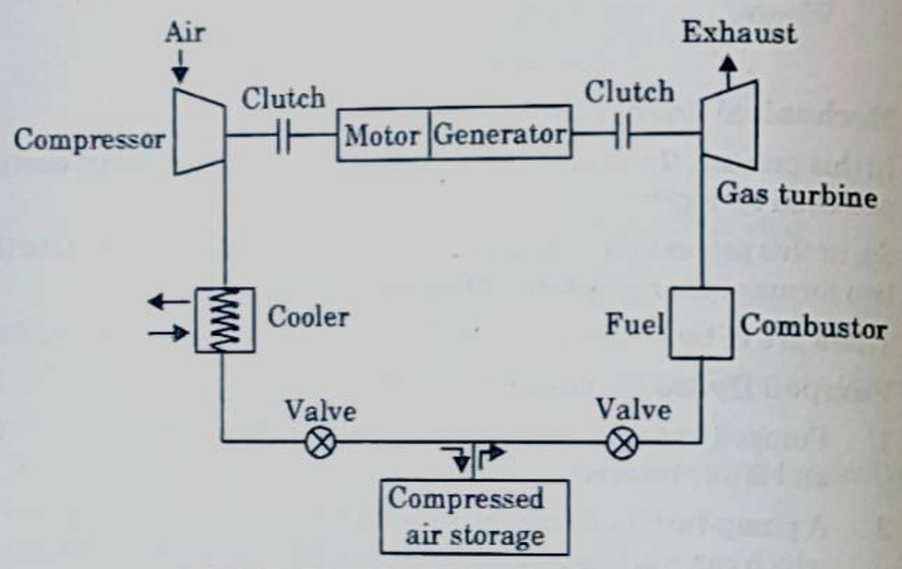

Fig. 2.19.4. Compressed air energy storage.

Que 2.20. **How many ways we can store solar energy ? Explain in brief about thermal energy storage and mechanical storage.**

Answer

1. These are different ways through which we can store solar energy :

a. **Solar Pond :** It is a natural or artificial body of water for collecting and absorbing solar radiation energy and storing it as heat.

b. **Solar Pumping :** Solar water pump work with the help of solar cell, which convert solar energy into electrical energy.

c. **Solar Cooker :** A solar cooker is a device which uses sunlight as its energy source and these sun rays are absorbed by a blackened metal tray which kept inside the solar box. Solar cooking is done by the help of solar cooker.

d. **Solar Still :** The process used to convert saline water into pure water by using solar energy is called solar distillation and the device used is called solar still.

A. **Thermal Energy Storage and Mechanical Energy Storage :** Refer Q. 2.19, Page 2–25M, Unit-2.

Que 2.21. **Define solar air heater with neat sketch and also write its application.** **AKTU 2018-19, Marks 10**

Answer

A. Solar Air Heater :

1. Solar air heater is a technology in which the energy from the solar insolation is captured by an absorbing medium and used to heat air.
2. Solar air heaters use roof, wall or window mounted solar collectors to heat the air that passes through them.

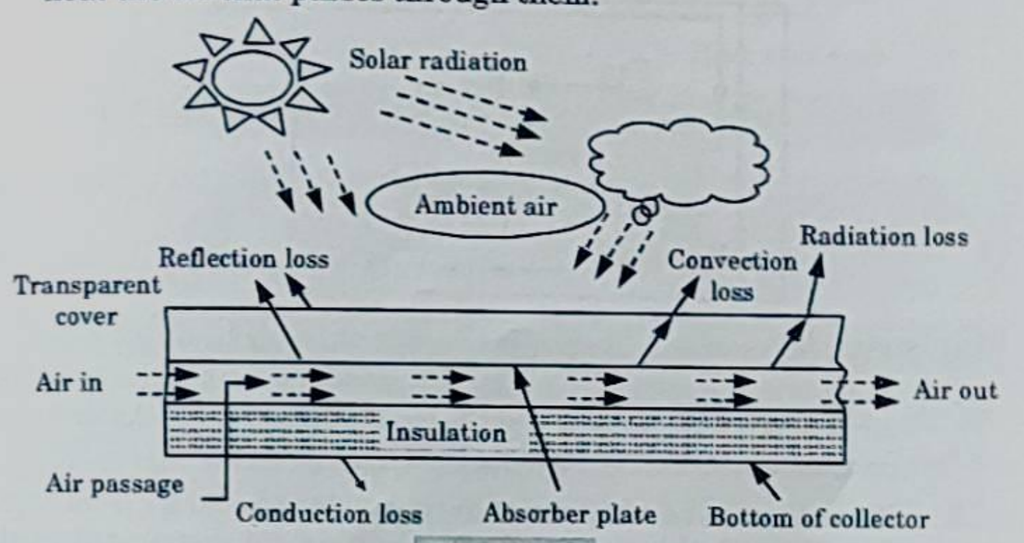

Fig. 2.21.1.

3. A conventional solar air heater is essentially a flat plate collector with an absorber plate.

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4. It is a transparent cover system at the top and insulation at the bottom and on the sides.
5. The whole assembly is enclosed in a sheet metal container.

B. Applications of Solar Air Heater :

1. Solar air heater provides ventilation and process air heating.
2. Solar air heater serves as weather cladding.

Que 2.22. Explain thermal energy storage for solar heating.

Answer

A. Thermal Energy Storage for Space Heating :

1. The solar energy is utilized for space heating in winter or in colder countries.
2. The different methods adapted for space heating by utilizing the solar energy are passive and active method.
3. Solar space heating reduces the considerable heating load on air conditioning apparatus during winter.

a. Active Space Heating :

1. The active method of space heating utilizes perforated collectors (air heater) through which the cold air is circulated, gets heated and supplied to the living space directly.

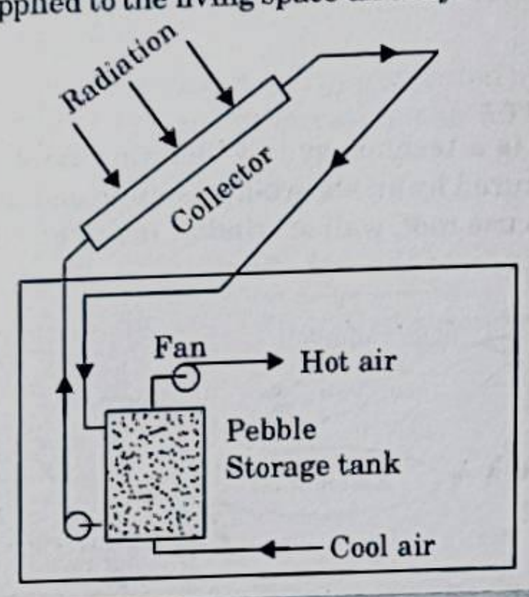

Fig. 2.22.1. Schematic diagram of active space heating by air collector.

2. In other arrangement to utilize this available heat in the night, hot air is first circulated through the tank packed with rocks, gravel or pebbles that serve as a thermal storage.
3. When there is no sunshine or during night the cold air from rooms flow through this pack and after heating get distributed in the living rooms as shown in Fig. 2.22.1.

4. In other arrangement the water heating arrangement is provided through the collectors placed at top of the roof as shown in Fig. 2.22.2.
5. The hot water stored in a storage tank is circulated through the tubes attached to black absorber surface of collector.
6. The heat is delivered to living space by a fan blowing the room air through a heating coil in the heat exchanger, through which hot water from storage tank is circulated in the coils.
7. The heat is extracted by re-circulated air through the exchanger and transferred to the room.
8. The water storage system is more compact than pebble packed system due to higher specific heat capacity of water, but costly because of pump and secondary heat exchanger.

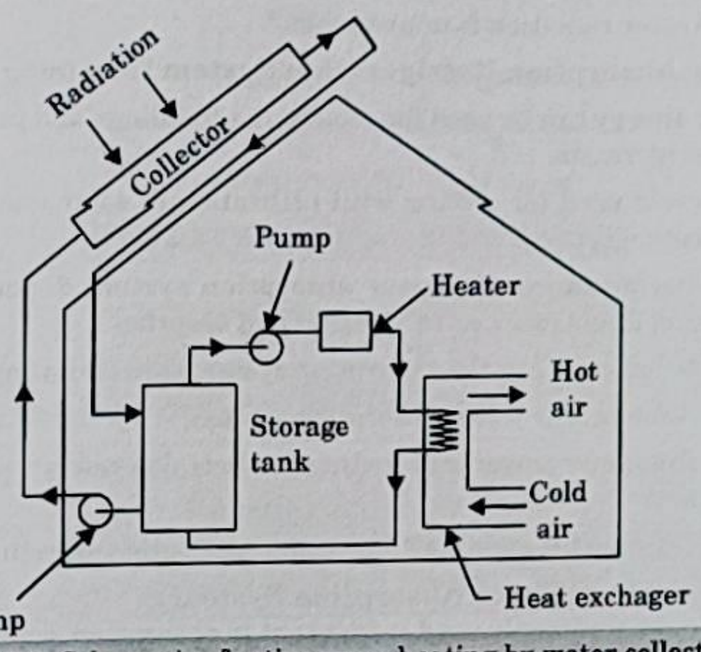

Fig. 2.22.2. Schematic of active space heating by water collector.

PART-12

Thermal Energy Storage for Cooling.

Questions-Answers

Long Answer Type and Medium Answer Type Questions

Que 2.23. Describe solar absorption refrigeration system for space cooling.

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OR

Define solar air conditioning and refrigeration system.

AKTU 2018-19, Marks 10

Answer

A. Solar Air Conditioning System :

1. A solar air conditioning system is simply a system of cooling and heating that utilises solar power, rather than electricity.
2. This can be done through passive solar, solar thermal energy conversion and photovoltaic conversion.
3. It is eco-friendly as well as energy efficient.
4. It can be operated with green resources of heat as backup resources when solar radiation is unavailable.

B. Solar Absorption Refrigeration System for Space Cooling :

1. Solar energy can be used for cooling the buildings and preserving food by refrigeration.
2. The cycle used for cooling with utilization of solar energy is vapour absorption cycle.
3. The performance of vapour absorption system depends upon the working fluids pair *i.e.*, refrigerant and absorbent.
4. On the basis of this, the absorption systems are classified as :
   a. Ammonia or water absorption system,
   b. Ammonia or water or hydrogen electrolux refrigeration system, and
   c. Lithium bromide (LiBr) or water absorption system.

a. Ammonia or Water Absorption System :

1. The most commonly used refrigerant in absorption system is ammonia.
2. It is cheap, readily available and has great affinity with water.
3. It is used as absorbent and absorbs NH_3 very fast.
4. The ammonia-water absorption system is used for cooling – 50 °C.
5. The main components of this system are shown in Fig. 2.23.1.
6. The heat required in generator is supplied by water heated in flat plate collector.
7. The hot water transfers the heat to the mixture of absorbent and refrigerant pair, rich of refrigerant.
8. The refrigerant in vapour form enters in the condenser and weak solution returns back in the absorber.
9. In some designs the absorbents are used in the bed of collector.

10. On heating they liberate the water vapour stored in the insulated tank and used for heating the generator during sunshine and off sunshine hours.

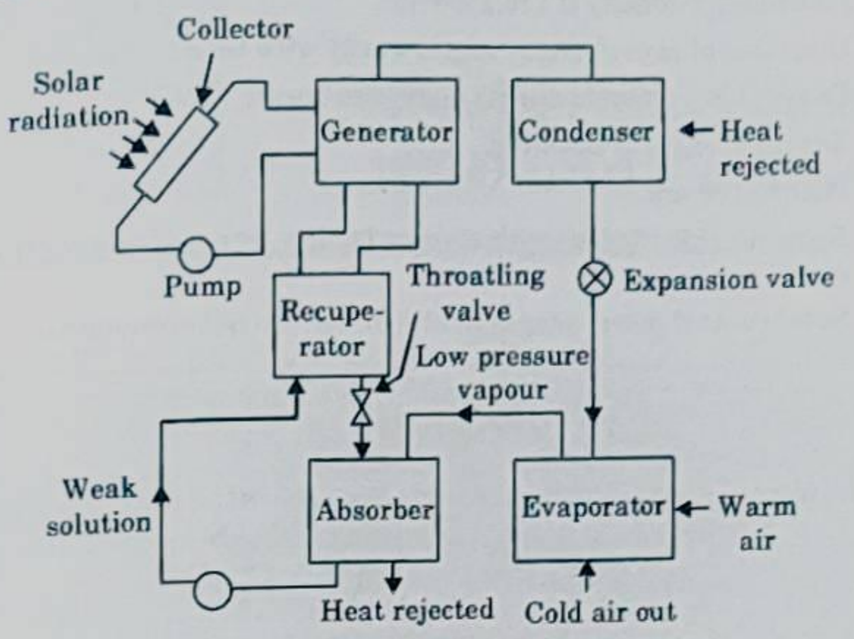

Fig. 2.23.1. Absorption cooling system.

PART-13

Limitations of Solar Power Plants.

Questions-Answers

Long Answer Type and Medium Answer Type Questions

Que 2.24. Explain thermal energy storage for solar heating and cooling. What are limitations of solar plants ?

AKTU 2017-18, Marks 10

Answer

A. Thermal Energy Storage for Solar Heating : Refer Q. 2.22, Page 2–30M, Unit-2.

B. Thermal Energy Storage for Solar Cooling : Refer Q. 2.23, Page 2–31M, Unit-2.

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C. Limitations :

1. Large area required for collecting solar thermal energy.
2. Low energy density 0.1 to 1 kW/m^2.
3. Direction of rays changes continuously with time.
4. Energy not available during night and during clouds.
5. Energy storage is essential.
6. High initial cost.
7. Requires hybrid plant with storage facility for supplying energy during night.
8. Solar central power plants in MW range are not economical.

VERY IMPORTANT QUESTIONS

Following questions are very important. These questions may be asked in your SESSIONALS as well as UNIVERSITY EXAMINATION.

Q. 1. Describe the difference between the direct radiation and diffuse radiation.

Ans. Refer Q. 2.2, Unit-2.

Q. 2. Classify different types of solar thermal collector and show the constructional details of a flat plate collector. What are its main advantages ?

Ans. Refer Q. 2.8, Unit-2.

Q. 3. Define solar air heater with neat sketch and also write its application.

Ans. Refer Q. 2.21, Unit-2.

Q. 4. Explain thermal energy storage for solar heating and cooling. What are limitations of solar plants ?

Ans. Refer Q. 2.24, Unit-2.

☺☺☺

3
UNIT

Geothermal Energy, MHD and Fuel Cells

CONTENTS

Part-1	:	Geothermal Energy and its Resources	**3–2M to 3–5M**
Part-2	:	Thermodynamics of Geothermal Energy Conversion, Electrical Conversion, Non Electrical Conversion	**3–5M to 3–9M**
Part-3	:	Environmental Consideration	**3–9M to 3–10M**
Part-4	:	Principle of Working of MHD Power Plant	**3–10M to 3–14M**
Part-5	:	Performance and Limitations of Geothermal Energy	**3–14M to 3–15M**
Part-6	:	Principle of Working of Various Types of Fuel Cells	**3–15M to 3–22M**
Part-7	:	Performance and Limitations of Fuel Cells	**3–23M to 3–25M**

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PART-1

Geothermal Energy and its Resources.

Questions-Answers

Long Answer Type and Medium Answer Type Questions

Que 3.1. **What is geothermal energy ? Give the classification of geothermal energy resources.**

OR

Describe the various types of identified geothermal energy resources and mention its application at different temperatures.

Answer

Geothermal Energy :

1. The enormous amount of energy available inside the earth in the form of heat is known as geothermal energy. Geothermal energy is a form of renewable energy and independent of sun, having the source of natural heat inside the earth.
2. Surface manifestation of this heat energy is indicated by hot water springs and geysers discovered at several places.
3. Heat can be experienced from the temperature rise of the earth's crust with increasing depth below the surface.
4. Radial temperature gradient increases proportionally to depth at a rate of about 30°C per km. At a depth of 3–4 km, water bubbles up, while at a depth of 10–15 km the earth's interior is as hot as 1000° to 1200°C.
5. The core of the earth consists of a liquid rock known as 'Magma' having a temperature of about 4000°C.
6. This geothermal heat is transferred to the underground reservoir of water which also circulates under the earth's crust. Its heat dissipates into the atmosphere as warm water and the steam vents up through the fissures in the ground as hot springs and geysers.
7. Limitless heat content in magma plus the heat generated by radioactive decay of unstable elements such as K_{40}, Th_{232} and U_{235} which are abundant in earth's crust are forms of geothermal energy and considered as a renewable energy resource.

Classification : Geothermal resources are of following types :

A. **Hydrothermal Resources :**

1. These are the deposits of hot water and steam at lesser depths and these can be extracted by means of production well.

2. High temperature water and steam (300° F to 700° F) is used for the generation of electricity, otherwise it is used for space heating.
3. It may be seen that only a part of the rock is permeable constituting the geo-fluid reservoir, so the field is able to produce commercially a viable resource.
4. Examples of Hydrothermal Resource sites :

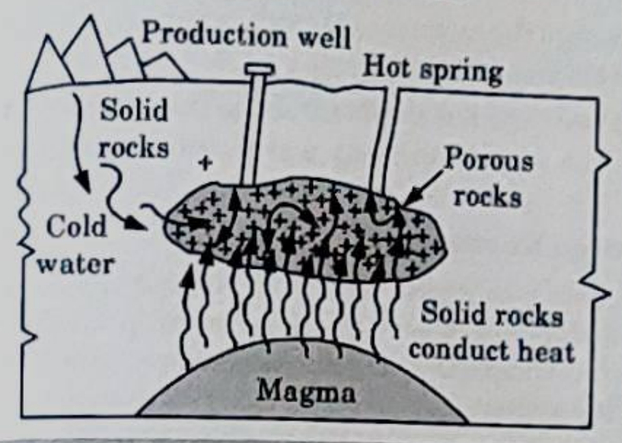

Fig. 3.1.1. Schematic view of hydro-geothermal energy resources.

a. **Hot Water Fields :**

1. At these locations hot water below 100 °C emit out as hot spring and the geothermal aquifers being covered by confining layers to keep the hot water under pressure.
2. Examples of hot water fields are Sahestra dhara near Dehradun, Sacred kund at Badrinath in Uttarakhand, Manikaran in Kullu valley (Himachal Pradesh) and Internationally known fields are Pannonian basin (Hungary), Po river valley (Italy) and Klamath Falls Oregon (USA).

b. **Wet Steam Fields :**

1. The pressurized water is at more than 100 °C and contains small quantities of steam and vapour in the geothermal reservoir (at 370 °C).
2. Sites where the steam escapes through cracks in the surface are called fumaroles.
3. An impermeable cap-rock prevents the fluid from escaping into atmosphere and drilling is carried out to bring the fluid to the surface.
4. The fluid is used to produce steam and boiling water in predominant phase.
5. Examples of wet steam fields : Los Azufre (Mexico), Puna (Hawaii, USA), Deing (Indonesia), Azores (Portugal) and Latera (Italy).

B. **Vapour-Dominated Resources :**

1. Vapour dominated reservoirs produce dry saturated steam of pressure above atmospheric pressure and at high temperature about 350 °C.
2. Water and steam coexist, but steam is in dominant phase and regulates pressure in the reservoir.

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3. Steam obtained from such a geothermal field directly drives a turbine.
4. Examples : Malsukawa (Japan), The Geysers California (USA), Kamojang (Indonesia) etc.

C. Hot Dry Rock Resources :

1. This is a geological formation with high temperature rocks at 650 °C heated by conductive heat flow from magma but contains no water.
2. To trap its energy the impermeable rock is fractured and water is injected to create an artificial reservoir.
c. Water circulates and hot fluids return to the surface through the other drilled well as steam and hot water, which are used to generate electricity.

D. Geopressured Resources :

1. These resources contains moderate temperature brines (160 °C) containing dissolved methane and these are trapped under high pressure in a deep sedimentary formation sealed between impermeable layers of shale and clay at depths.
2. When trapped by boring wells, three sources of energy are available:
   a. Thermal,
   b. Mechanical as pressure, and
   c. Chemical as methane.

E. Magma :

1. Magma is a molten rock at temperature ranging from 900 °C to 1600 °C. This hot viscous liquid comes out from active volcanic vents and solidifies.
2. It may form reservoirs at some depth from the earth's surface and magma chambers represent a huge energy source.
3. The existing technology does not allow recovery of heat from these resources.

Que 3.2. What is meant by dry steam, wet steam and hot water geothermal system ? **AKTU 2015-16, 2017-18; Marks 10**

Answer

A. Hot Water Fields :

1. Hot water field, containing a water reservoir at temperature ranging 50-100 °C.
2. Such fields without much steam content can be useful for house heating and agricultural purposes the temperature gradient in this field is less.
3. The reservoir contains water in the liquid phase below the boiling point of water under pressure.

4. On the surface, there are often thermal springs whose temperature is near the boiling point of water. These fields occur at depth less than 2 km.
5. The geyser plant of USA is the largest plant in the world today.

B. Wet Steam Field :

1. The wet steam fields contain pressurized water in reservoir at temperature higher than 100 °C.
2. When hot water at high pressure is brought to the surface, its pressure is sufficiently reduced and some water will get flashed into steam and remaining in the form of boiling water.
3. The resulting mixture is a mixture of water and steam. Such fields are suitable for power generation.
4. When the well is drilled at such locations, the pressurized water rises into well because of less pressure above the well.
5. The vapour is used directly for producing power while the hot water gets separated in the separator and is used for thermal applications.
6. The percentage of steam generated depends upon the available geothermal fields and more than 90 % of hydrothermal reservoirs exploited on industrial scales are this type.

C. Dry Steam Field :

1. These fields are similar to wet, steam fields but heat transfer from the depth is much higher.
2. These reservoirs produce superheated steam at pressure above atmosphere.
3. The permeability of these fields is lower than wet fields.
4. When the well is drilled up to the reservoir and extraction of fluid starts, a depressed zone is formed at the bottom of the well that enhances the boiling of water surrounding the rocks.
5. The steam flows through the dry bottom area and starts expanding and gets cool. But the heat added by surrounding rocks at high temperature keeps the steam at superheated state. The degree of superheating may reach up to 100 °C.

PART-2

Thermodynamics of Geothermal Energy Conversion, Electrical Conversion, Non Electrical Conversion.

Questions-Answers

Long Answer Type and Medium Answer Type Questions

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Que 3.3. **What is geothermal energy ? Discuss different systems used for generating the power using geothermal energy.**

OR

Explain geothermal power plants. Write the application of geothermal energy.

Answer

A. Geothermal Energy : Refer Q. 3.1, Page 3–2M, Unit-3.

B. Different Systems used for Generating the Power :

a. Vapour-Dominated Power Plant :

1. In a vapour-dominated power plant, steam is extracted from geothermal wells, passed through a separator to remove particulate contents and flows directly to a steam turbine.

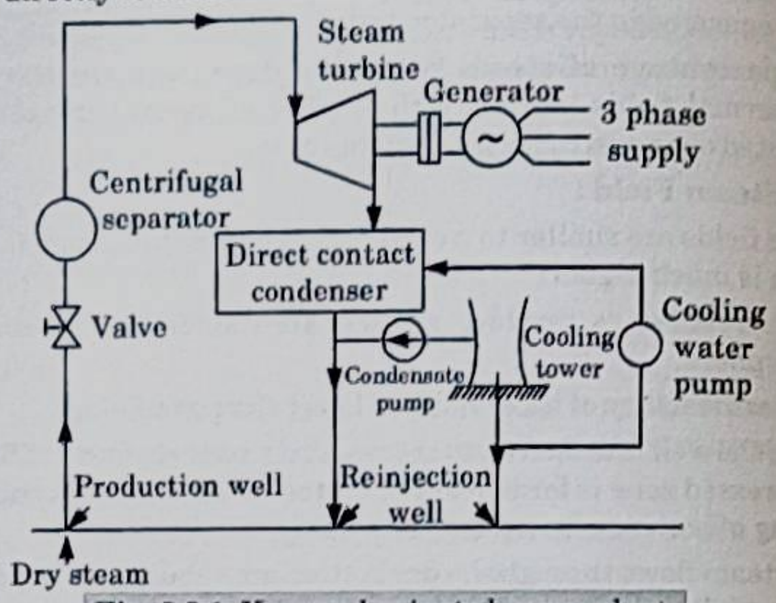

Fig. 3.3.1. Vapour-dominated power plant.

2. Steam then operates the turbine coupled with the generator is at a temperature of about 245 °C and pressure 7 bar which are less than those in conventional steam cycle plants.
3. Thus, the efficiency of geothermal plants is low, *i.e.*, about 20 %.
4. Exhaust steam from the turbine passes through a condenser and the water so formed circulates through the cooling tower.
5. It improves the efficiency of the turbine and controls environmental pollution associated with the direct release of steam into the atmosphere.
6. Waste water from the cooling tower sump is re-injected into the geothermal well to ensure continuous supply.

b. Liquid-Dominated Power Plants :

1. These plants are also called wet steam plants because they give wet steam *i.e.*, a mixture of hot water and steam under high pressure.

2. There are two types of liquid-dominated power plants :
   i. Flashed steam system, and
   ii. Binary cycle system.

i. Flashed Steam System :

1. Flashed system is preferred for high temperature mixture of geothermal brine and steam, with low dissolved impurities.
2. Geothermal fluid (mixture of brine and steam) passes through a flash chamber where a large part of the fluid is converted to steam.
3. Dry saturated steam passes through the turbine coupled with the generator to produce electric power.
4. Hot brine from the flash chamber and the turbine discharge from the condenser are re-injected into the ground and reinjection of the spent brine ensures a continuous supply of geothermal fluid from the well.

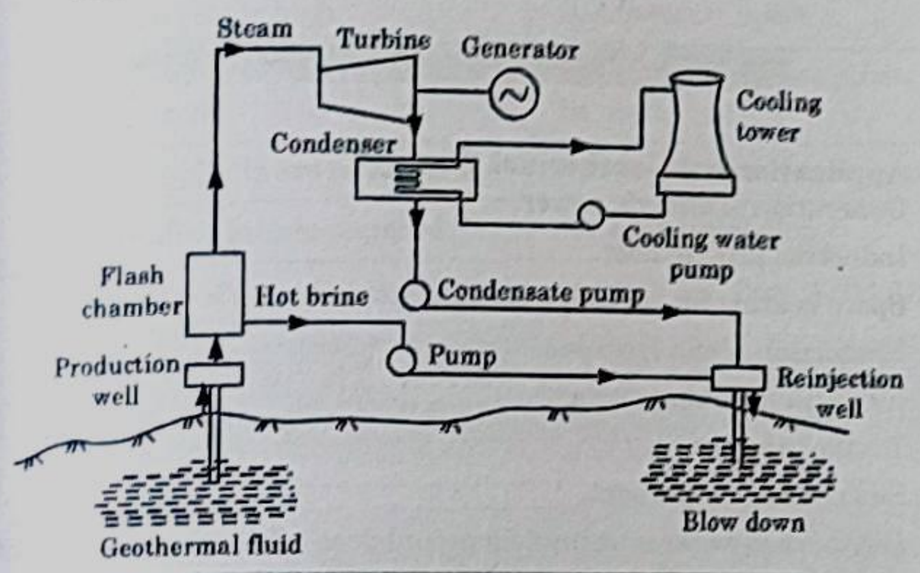

Fig. 3.3.2. Flashed steam geothermal power plant.

ii. Binary Cycle System :

1. A binary cycle is used where geothermal fluid is hot water with temperature less than 100 °C.
2. This plant operates with a low boiling point working fluid (Isobutene or Freon) in a thermodynamic closed Rankine cycle.
3. Hot brine from underground reservoir circulates through a heat exchanger and is pumped back to the ground.
4. In heat exchanger, hot brine transfers its heat to the organic fluid thus converting it to a superheated vapour that is used in a standard closed Rankine cycle.

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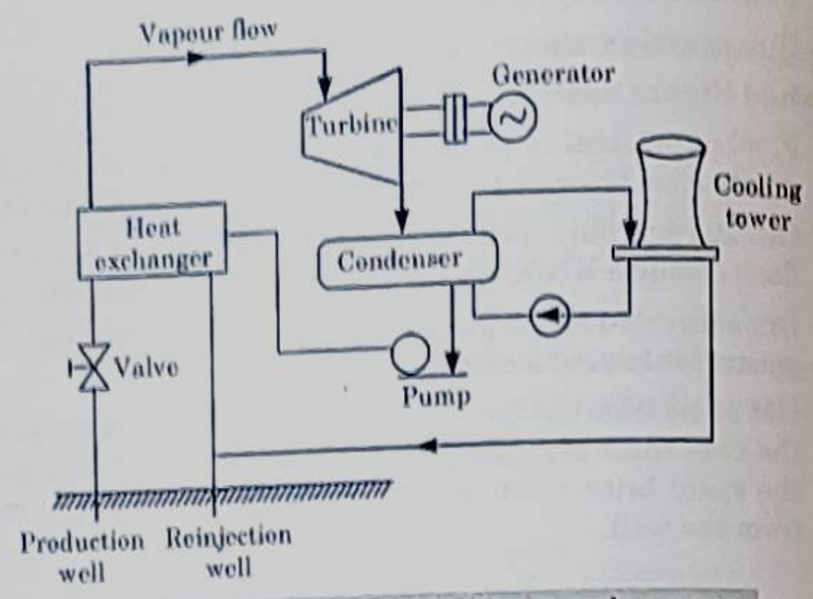

Fig. 3.3.3. Schematic view of binary cycle system.

C. Applications of Geothermal Energy :

1. Generation of electric power.
2. Industrial process heat.
3. Space heating for buildings.
4. Production of salt from sea.
5. Extraction manufacturing.
6. Textile industry.
7. Sewage heat treatment.
8. Geothermal water is utilized for greenhouse cultivation using discharg water from a geothermal drill hole.

Que 3.4. **Discuss the difference between a geothermal power plan and thermal power plant. Categorize resources of geotherma energy.** **AKTU 2017-18, Marks 10**

OR

Write the difference between a geothermal power plant and therma power plant. **AKTU 2015-16, Marks 10**

Answer

A. Differences :

S. No.	Geothermal Power Plant	Thermal Power Plant
1.	It uses inexhaustible source of energy.	It uses exhaustible source of energy.
2.	It is more environment friendly	It is less environment friendly.
3.	These power plants in some dangerous cases can cause earthquakes.	There is no such problem.
4.	It is mainly used for power generations process.	It can be used for various industrial processes.
5.	Setup cost is high.	Setup cost is low.
6.	Byproducts of these plants are not used.	Byproducts of these plants can be used.
7.	These plants are less flexible.	These plants are more flexible.
8.	Specified area is required.	No such restriction.

B. Resources of Geothermal Energy : Refer Q. 3.1, Page 3–2M, Unit-3.

PART-3

Environmental Consideration.

Questions-Answers

Long Answer Type and Medium Answer Type Questions

Que 3.5. **Describe the various operational and environmental problems encountered in obtaining the geothermal energy.** **AKTU 2015-16, Marks 7.5**

Answer

Operational and Environmental Problems of Geothermal Energy Sources :

1. Geothermal energy is not completely pollution free energy.

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2. The main adverse environmental effects are air pollution (waste steam is sometimes vented directly to the atmosphere), thermal pollution (pumping more thermal energy to the atmosphere), surface disturbance, physical effects (land subsidence) caused by fluid withdrawal.
3. At geothermal site, the air pollution is the major problem because of emission of poisonous gases such as hydrogen sulphide (H_2S), ammonia, methane, Carbon dioxide (CO_2) etc.
4. The extraction of energy from hot dry rocks or molten magma, it is necessary to force water down boreholes as a working fluid and return it to surface to use in turbine.
5. If the underground reservoir is highly permeable, there is no way to know how much water will need to be injected before a useful amount of steam or hot water is returned to the surface.
6. A large volume of flash steam escaping into the atmosphere could cause dense fog to occur.
7. At geothermal site, some harmful substances may escape into the air.
8. These may contain radioactive materials also thus systematic monitoring is advisable.
9. Geothermal water contains dissolved solids.
10. The amount of dissolved solids is in the range of 300–1500 ppm of which silica amounts to 25–50 %.
11. The possible solution is reinjection or disposal into sea through ducts and channels and also the use of evaporator ponds.

PART-4

Principle of Working of MHD Power Plant.

Questions-Answers

Long Answer Type and Medium Answer Type Questions

Que 3.6. **Write short note on magneto hydrodynamics (MHD).**

AKTU 2015-16, Marks 03

Answer

A. Magneto Hydrodynamics (MHD) :

1. The magneto hydrodynamics deals with generation of electric field, when an ionized fluid at high temperature passes through the applied magnetic field. The direct current is generated from the system with the expense of thermal energy.

2. MHD power generation is based on the Faraday's law of electromagnetic induction.
3. The MHD generator should meet the following requirements :
   a. The magnet material should have high melting point.
   b. The electrodes are made of SiC or ZrC material to withstand high temperature for preventing the chemical erosion.
   c. To prevent the chemical erosion from hot gases the ceramics are chosen to construct the duct.
   d. Duct material should have high electrical and thermal insulation.
   e. The insulation and conducting materials should be able to withstand high temperature around 2500 °C.

Que 3.7. **Explain the working principle of magneto hydrodynamic power generation.**

Answer

1. In MHD power generation conversion process depends upon Faraday's law of electromagnetic induction, which states that when a conductor and a magnetic field move relative to each other, a voltage is induced in the conductor. This induced voltage produces an electric current.
2. The conductor may be solid, liquid or gas.
3. In MHD generator solid conductors are replaced by hot ionized gas.
4. The hot ionized gas (3000 °C) is passed through the MHD duct across which a strong magnetic field is applied.
5. Since the gases are hot and ionized they form an electrically conducting medium moving in a magnetic field, thus a voltage is generated.
6. The power generated by MHD generator is in the direct current form.
7. Now, if the electrodes are placed in a suitable position then generated current can be extracted.

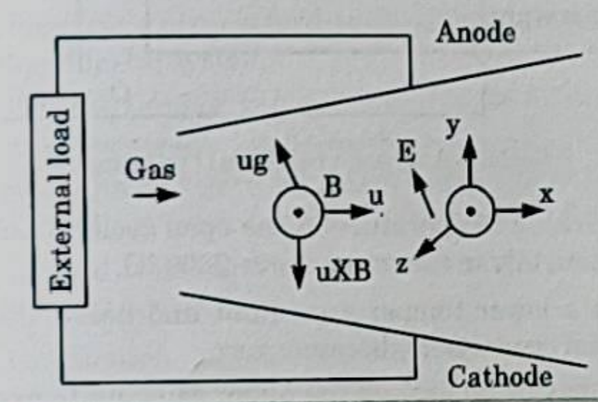

Fig. 3.7.1. Working principle of MHD generator.

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Que 3.8. **Write short notes on the following :**

a. Open cycle MHD system, and
b. Closed cycle MHD system.

OR

With the help of a schematic diagram, explain the operation of closed cycle MHD generating system. **AKTU 2016-17, Marks 10**

OR

Draw schematic diagram of an MHD power generating system having heat recovery steam generator. Explain the functioning of the steam. **AKTU 2018-19, Marks 10**

Answer

A. Working Principle of MHD : Refer Q. 3.7, Page 3–11M, Unit-3.

B. Classification of Hydrodynamic System :

a. Open Cycle MHD System :

1. The open cycle MHD generator uses coal as a fuel as it produces more conductive plasma. This is because of more carbon atom as compared to hydrogen atom (as the presence of hydrogen is undesirable in MHD).
2. Fig. 3.8.1 shows the schematic diagram of an open cycle MHD generator.

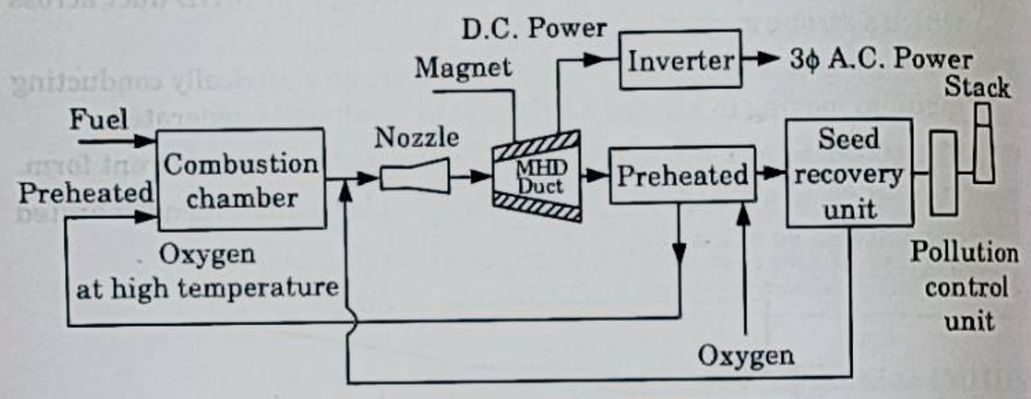

Fig. 3.8.1. Open cycle MHD generator.

4. The working temperature in the open cycle MHD generator lies approximately in the range above 2300 °C.
5. This is a lower temperature limit and below this the effective electrical conductivity becomes zero.
6. There may be no limit in the upper working temperatures; so far the materials can stand with the high heat fluxes under high electric field.

b. Closed Cycle MHD System :

1. Fig. 3.8.2 shows the schematic diagram of closed cycle MHD generator.

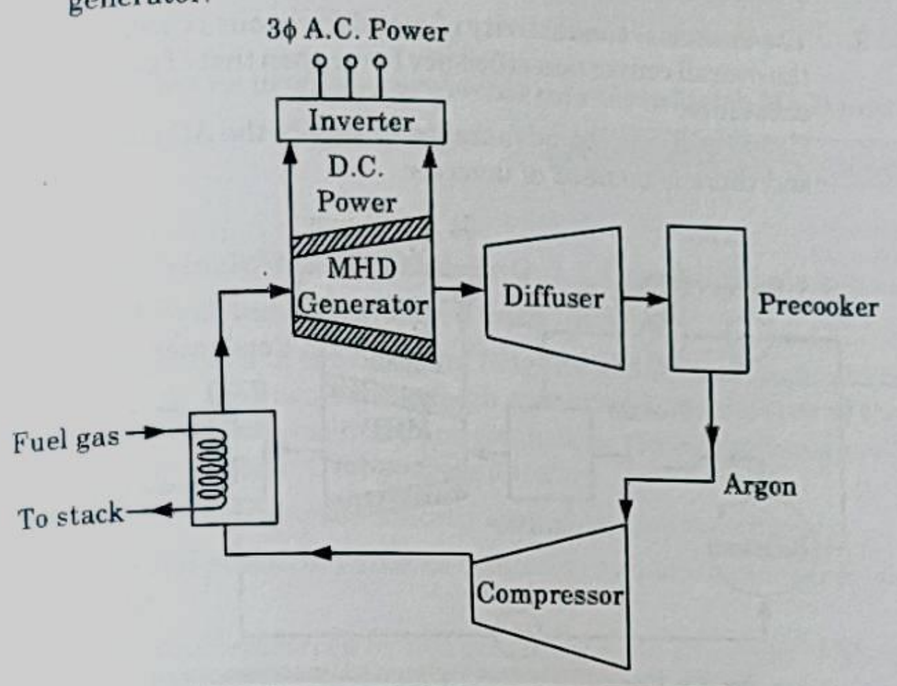

Fig. 3.8.2. Closed cycle MHD generator.

2. The very high thermal efficiency is achieved with low cycle cost in closed cycle plant and provides more useful power at low temperature at 1600 °C. The duct of these units is small because of high pressure.
3. Helium or argon is used as working fluid, heated in heat exchanger and gets ionised.
4. Less ionised substances such as alkali metal is mixed with inert gas to provide the necessary conductivity in closed cycle plant, where recovery is possible.
5. The closed cycle plant is further classified in seeded inert gas system and liquid metal system.
6. The working fluid (argon or helium) in closed cycle is seeded with cesium and circulates in a close loop.
8. The gas burned in the combustor is supplied in the heat exchanger, where the heat is transferred to the working fluid.
9. The ionised working fluid passes through the magnetic field to produce DC power.
10. The combustion products are discharged to the atmosphere after removal of heat in heat exchanger.

i. Closed Cycle Liquid Metal MHD Generator :

1. Fig. 3.8.3 shows the schematic diagram of closed cycle liquid metal MHD generator.

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2. The superheated metallic vapour is expanded through the supersonic nozzle and enters in the generator in liquid form with velocity of 150 m/s.
3. The electrical conductivity of metallic vapour is poor. That brings the overall conversion efficiency lower than that of gas as a working substance.
4. However it has the advantages to supply the AC current directly and there is no need of inverter.

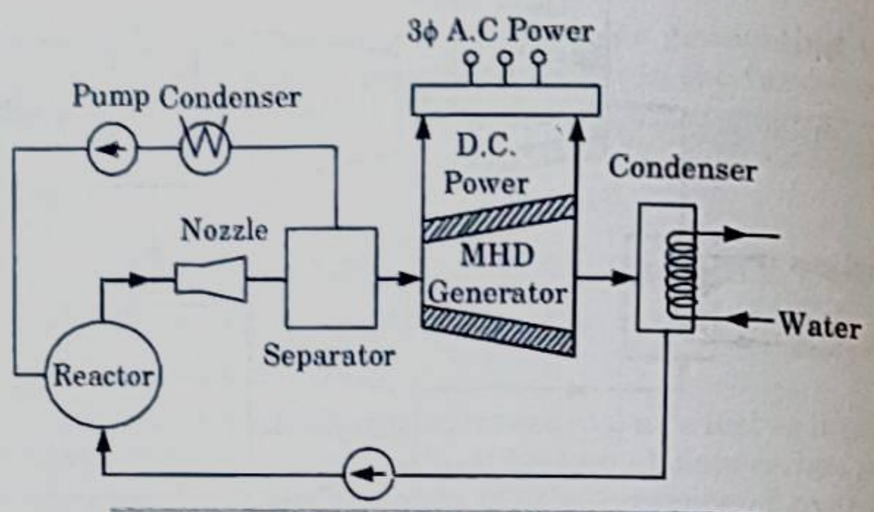

Fig. 3.8.3. Closed cycle liquid MHD generator.

6. In nuclear fired MHD generator the high temperature nuclear reactor is used to utilize solid fuel elements to meet the requirements.
7. The ceramic coated electrodes are film cooled by hydrogen to protect them from unusual build up of uranium droplets.
8. The cyclonic separators are used to remove the uranium droplet from the hydrogen gas and the hydrogen flows in the compressor expands through the turbine, then is cooled in heat exchangers in multistage compression.

C. Function of Steam : The steam is used partly for driving a steam turbine operating the compressor and partly expanded in a steam turbine driving a three phase alternator.

PART-5

Performance and Limitations of Geothermal Energy.

Questions-Answers

Long Answer Type and Medium Answer Type Questions

Que 3.9. **Explain the working principle of MHD generator. Also discuss the practical problems associated with MHD power generator.** **AKTU 2017-18, Marks 10**

OR

Write short notes on practical problems associated with MHD power generation. **AKTU 2016-17, Marks 05**

Answer

A. Working Principle of MHD : Refer Q. 3.7, Page 3–11M, Unit-3.

B. Practical Problems Associated with MHD :

1. The main problem in the design of long-life MHD generators is to find the materials that can survive high operating temperatures of these generators. Both the insulator and conducting materials should sustain temperature of 2500 °C for prolong duration.
2. Electrode materials are chemically eroded by combustion gases.
3. Seed material potassium attacks insulating materials and makes them conducting.
4. The major problem forced by this generator is the economics. Although the overall thermal efficiency is 60 %, against 40 % for conventional thermal plant, additional investment in the magnet, generator, duct, compressors, scrubbers, seed recovery plant and DC to AC converters may increase the plant cost and it may be much higher than conventional plant.

PART-6

Principle of Working of Various Types of Fuel Cells.

Questions-Answers

Long Answer Type and Medium Answer Type Questions

Que 3.10. **What do you understand by fuel cell ? How fuel cells are classified ? Describe an H_2O_2 fuel cell with a sketch showing reactions.**

OR

Describe the principle of working of a H_2O_2 cell. Also give its limitations. **AKTU 2018-19, Marks 10**

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OR

Describe the principle of working of a fuel cell with reference to hydrogen oxygen cell. Also discuss advantages and limitations of fuel cells. **AKTU 2015-16, Marks 10**

OR

What is a fuel cell ? Describe the principle and working of a H_2O_2 fuel cell. Give also limitations. **AKTU 2017-18, Marks 10**

OR

What is fuel cell ? Define its working, operation and advantages. **AKTU 2018-19, Marks 10**

Answer

A. Fuel cell :

1. A fuel cell is an electro-chemical device that converts chemical energy into electricity and heat without combustion.
2. The conversion of chemical energy into electrical power in case of fuel cell is an isothermal process.
3. Main components of a fuel cell are :
   1. Anode (Fuel electrode), 2. Cathode (Oxidant electrode),
   3. Electrolyte, 4. Container,
   5. Separators, 6. Sealings,
   7. Fuel supply, and 8. Oxidant supply.

B. Type of Fuel Cell :

1. Alkaline fuel cells (AFC),
2. Direct methanol fuel cells (DMFC),
3. Phosphoric acid fuel cells (PAFC),
4. Proton or polymer exchange membrane fuel cells (PEMFC),
5. Molten carbonate fuel cells (MCFC),
6. Solid oxide fuel cells (SOFC),
7. Zinc air fuel cells (ZAFC), and
8. Regenerative fuel cells (RFC).

C. Working Principle of Hydrogen Oxygen Cell :

1. A basic hydrogen-oxygen fuel cell with phosphoric acid as electrolyte is shown in Fig. 3.10.1.
2. In fuel cells, platinum coated special graphite plates are used as the electrodes, separated by an electrolyte.
3. The fuel is hydrogen gas which is supplied at the anode side where the hydrogen molecules are effectively reduced to hydrogen ions which move to the electrolyte.

$$H_2(gas) \longrightarrow 2H^+(ion) + 2e^-$$

4. Electrons liberated at the anode build up a negative potential and travel towards the cathode through an externally connected circuit.
5. Oxygen gas is supplied at the cathode where it is reduced by hydrogen ions to produce water.

$$4H^+ + O_2 + 4e^- \longrightarrow 2H_2O$$

6. Electrochemical reactions coupled with movement of hydrogen ions through the electrolyte generate an electric potential, which causes electric current to flow through the load.

$$2H_2 + O_2 \xrightarrow{\text{Fuel cell}} 2H_2O + \text{Electric energy generated} + \text{Heat energy released}$$

This reaction is exothermic, which results in heating up the cell.

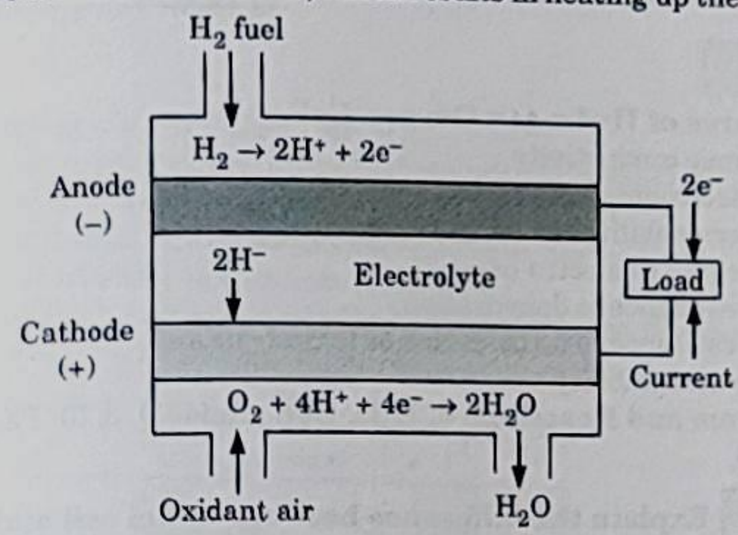

Fig. 3.10.1. Fuel cell operation.

7. A stream of air is circulated on the cathode side of the cell which absorbs enough heat to maintain outlet air and steam at 180 °C which is optimum for best performance of the cell.

C. Advantages of Fuel Cell :

1. Fuel cell has high efficiency.
2. Fuel cell is simple and safe.
3. Fuel cell does not have moving parts.
4. Fuel cells are compact and noiseless.
5. They are pollution free.
6. No cooling water needed.
7. The capacity can be increased as the demand grows.
8. Space requirement is much less.
9. It has long life.
10. It is odourless and quiet in application.

D. Limitations of Fuel Cell :

1. The reactivity and invariance are the two general requirements for all fuel cells. To satisfy the reactivity requirement, it is necessary that we have proper stoichiometry and also require high electrode activity, which results in large current densities.

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2. The second requirement, invariance, means that a fuel cell should only be a converter of energy and should remain unlike a conventional battery, invariant throughout its life. This requirement implies no corrosion or side reactions, no change in the electrolyte and no change in the electrodes.
3. If the fuel cell is operated near room temperature, then it would deliver little electricity.

Que 3.11. Explain the essential features of a hydrogen oxygen cell. Draw a suitable diagram of this cell and give the reactions took place at the electrodes. **AKTU 2016-17, Marks 10**

Answer

A. Features of Hydrogen Oxygen Cell :

i. High ionic conductivity,
ii. Zero electronic conductivity,
iii. Low permeability of fuel and oxidant,
iv. Low degree of electro-osmosis,
v. High resistance to dehydration,
vi. High resistance to its oxidation or hydrolysis and,
vii. Mechanical stability.

B. Diagram and Reactions of H_2O_2 Cell : Refer Q. 3.10, Page 3–15M, Unit-3.

Que 3.12. Explain the difference between a fuel cell and battery. What are the uses and advantages of fuel cells ? **AKTU 2017-18, Marks 10**

Answer

A. Difference between Fuel Cell and Battery

S. No.	Fuel Cell	Battery
1.	The fuel cell is a primary cell and cannot be recharged but can be refueled.	The battery is rechargeable.
2.	The fuel and oxidizer do not mix together.	Fuel and oxidizer are not used separately.
3.	Fuel and oxidizer need continuous replacement as per requirement.	Battery stores fixed charges of chemical, used up during reaction.
4.	It produces electricity continuously as long as fuel oxidizer is supplied.	Battery stores energy.

B. Advantages of Fuel Cells : Refer Q. 3.10, Page 3–15M, Unit-3.

C. Applications of Fuel Cell :

1. It is used in automotive vehicles.
2. It is used in domestic power unit.
3. It is used in central power station.
4. It is used in military and aerospace.
5. For remote and inaccessible locations, fuel cell can be used unattended for a long period.

Que 3.13. Explain the working of molten carbonate fuel cells using appropriate diagram and write the various chemical reactions involved in this type of fuel cell.

Answer

A. Molten Carbonate Fuel Cell (MCFC) :

1. It uses an electrolyte, which is a molten mixture of carbonate salts.
2. Two mixtures commonly used are :
   a. Lithium carbonate and potassium carbonate, and
   b. Lithium carbonate and sodium carbonate.
3. Since, these salts can act as electrolytes only in liquid phase; the operating temperature should be as high as 650 °C.

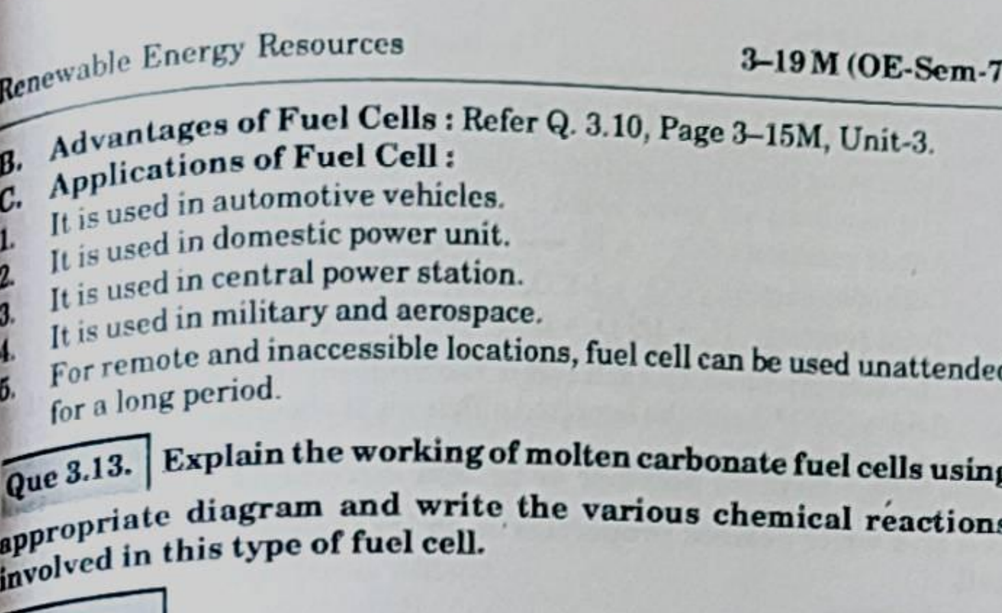

Fig. 3.13.1. Molten carbonate fuel cell.

4. Due to high temperature, these salts melt and become conductive to carbonate ions (CO_3^{--}).
5. These ions flow from the cathode to the anode where they combine with hydrogen to give water, carbon dioxide and electrons.

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6. The electrons flow through external circuit and reaches to cathode, generating electricity and byproduct heat.
7. The reactions are given below :

Anode reaction : $CO_3^{--} + H_2 \longrightarrow H_2O + CO_2 + 2e^-$

Cathode reaction : $CO_2 + 1/2\ O_2 + 2e^- \longrightarrow CO_3^{--}$

Total reaction : $H_2 + 1/2\ O_2 + CO_2 \longrightarrow H_2O + CO_2$

8. The emf produced by each cell is theoretically 1 V and actual emf of 0.8 V at 700 °C and the expected efficiency is about 60 %.

Que 3.14. Explain polymer or proton electrolyte membrane fuel cell and write desired properties of an ideal electrolyte of this fuel cell.

Answer

A. Proton Electrolyte Membrane Fuel Cell (PEMFC) :

1. In this type of fuel cell, electrolyte is a solid polymer membrane of an organic material such as polystyrene sulphonic acid and this polymer is permeable to protons when it is saturated with water but it does not conduct electrons.
2. In PEMFC, the fuel is hydrogen and charge carriers are hydrogen ions (protons).

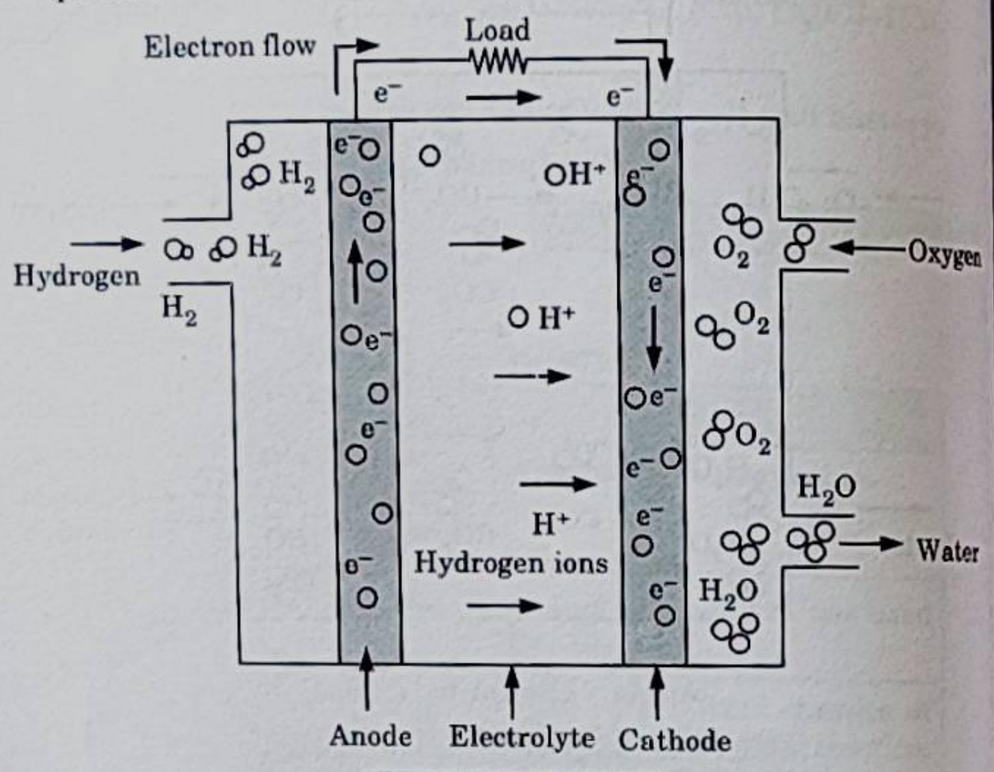

Fig. 3.14.1. PEMFC.

3. At the anode, the hydrogen molecule is split into hydrogen ions and electrons.

4. The hydrogen ions penetrate across the electrolyte to cathode while the electrons flow through an external circuit and produce electric power.
5. Oxygen is supplied to the cathode and combines with electrons and the hydrogen ions to produce water.
6. The reactions are given below :

Anode reaction : $2H_2 \longrightarrow 4H^+ + 4e^-$

Cathode reaction : $O_2 + 4H^+ + 4e^- \longrightarrow 2H_2O$

Total reaction : $2H_2 + O_2 \longrightarrow 2H_2O$

7. The membrane is coated on its both sides with finely powdered platinum which acts as a catalyst. These cells are also called ion exchange membrane cell.

B. Advantages of PEMFC :

1. It has high power density.
2. It can be start rapidly.
3. Less expensive.
4. It has less problems with corrosion.
5. It has longer life.
6. It operates at low temperature *i.e.*, usually below 100 °C.

C. Disadvantage :

1. The main disadvantage of this type of cell is that due to low operating temperature these are not enough to perform useful cogeneration.

D. Desirable Properties of an Ideal PEMFC Electrolyte :

1. High ionic conductivity.
2. Zero electronic conductivity.
3. Low permeability of fuel and oxidant.

Que 3.15. Describe solid oxide fuel cells.

OR

Sketch and explain the functioning of solid oxide fuel cells.

Answer

A. Solid Oxide Fuel Cells (SOFC) :

1. The electrolyte in this cell is a solid, non porous metal oxide that is conductive to oxygen ions and these cells are operate at high temperature between 650 °C to 1000 °C.
2. At the cathode, the oxygen molecules from the air are split into oxygen ions with the addition of four electrons.
3. The oxygen ions are conducted through the electrolyte and combine with hydrogen at the anode releasing four electrons.
4. The electrons move through the external circuit producing electric power and byproduct heat.
5. The reactions are as followed :

Anode reaction : $2H_2 + 2O^{--} \longrightarrow 2H_2O + 4e^-$

Cathode reaction : $O_2 + 4e^- \longrightarrow 2O^{--}$

Total reaction : $2H_2 + O_2 \longrightarrow 2H_2O$

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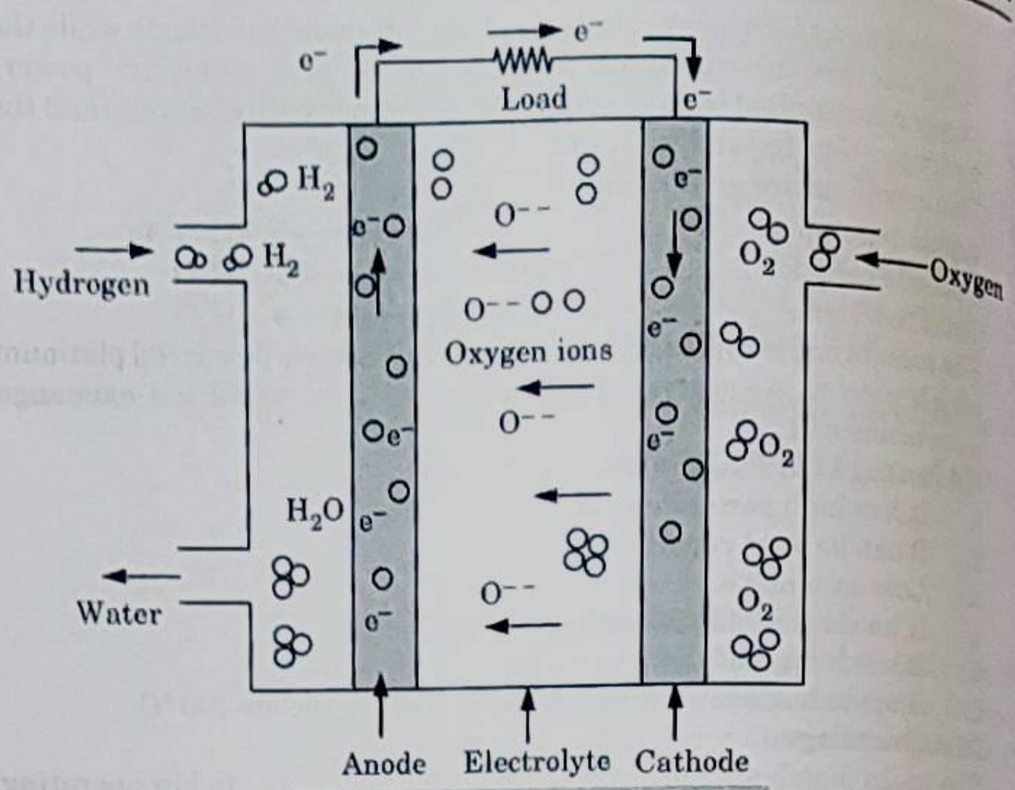

Fig. 3.15.1. Solid oxide fuel cells.

6. The output voltage is about 0.65 V at about 800 °C and a tubular type of SOFC system has been developed which operates at high temperature of about 900 °C – 1000 °C.

Que 3.16. **Differentiate between AFC and PEMFC.**

Answer

Difference between AFC and PEMFC :

S. No.	AFC	PEMFC
1.	It operates between the temperature ranges of 40 °C - 200 °C.	It operates between the temperature ranges of 40 °C - 90 °C.
2.	It uses H_2 as a fuel.	It uses H_2, CH_4, CH_3OH as a fuel.
3.	KOH used as electrolyte.	Polymer used as electrolyte.
4.	The power density of this cell is in the range of 0.1 – 0.3 W cm^2.	The power density of this cell is in the range of 0.7 W cm^2.
5.	It is mainly used for power generation on spacecrafts.	It is mainly used for transport applications.

PART-7

Performance and Limitations of Fuel Cells.

Questions-Answers

Long Answer Type and Medium Answer Type Questions

Que 3.17. **Describe the performance analysis for fuel cell.**

OR

Show that a hydrogen-oxygen fuel cell has a maximum efficiency of 83 %.

Answer

A. Performance Analysis for Fuel Cell :

1. In a fuel cell, a chemical reaction takes place where the reactants are converted into products in a steady flow process.
2. The work is obtained by combining the first and second laws of thermodynamics.
3. According to first law of thermodynamics for a steady flow process,

$$\Delta Q - \Delta W = \Delta H + (\Delta KE) + (\Delta PE)$$

where,
ΔQ = Heat transferred to the steady flow stream from the surroundings,
ΔW = Workdone by the flow stream from entrance to exit,
ΔH = Change in enthalpy of the flow stream from entrance to exit,
KE = Kinetic energy of the stream, and
PE = Potential energy of the stream.

4. In this case, KE and PE are usually negligible. Hence equation can be written as,

$$\Delta W = \Delta Q - \Delta H \quad ...(3.17.1)$$

5. According to second law of thermodynamics,

$$\Delta Q = T\Delta S \quad ...(3.17.2)$$

6. On combining equation (3.17.1) and (3.17.2), we get

$$\Delta W_{max} = T\Delta S - \Delta H \quad ...(3.17.3)$$

7. The Gibbs free energy is given by

$$\Delta G = \Delta H - T\Delta S \text{ (where } T \text{ is constant)} \quad ...(3.17.4)$$

8. From equation (3.17.3) and equation (3.17.4), we get

$$\Delta W_{max} = -\Delta G$$

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B. Efficiency :

1. The efficiency (η_F) of energy conversion of a fuel cell is defined as the ratio of the useful work to the heat of combustion of the fuel.

   Mathematically,

$$\eta_F = \frac{\Delta W_{max}}{(-\Delta H)} = \frac{-(\Delta G)}{-(\Delta H)} \quad ...(3.17.5)$$

2. For hydrogen-oxygen fuel cell, ΔG is (–237191) J/kg-mol at 25 °C while its heat of reaction ΔH is about (–285838) J/kg-mol.

$$\eta_{max} = \frac{237191}{285838} = 0.829 \approx 83\ \%$$

C. EMF of a Fuel Cell :

1. The electromotive force that will drive electrons through external load is proportional to Gibbs free energy change.

   Mathematically,

$$E = \frac{-\Delta G}{nF} \quad ...(3.17.6)$$

   Where, E = Electromotive force,
   ΔG = Change in Gibbs free energy (J/mol),
   n = Number of electrons per mole of fuel, and
   F = Faraday's constant = 96500 coulombs/mole.

2. From equation (3.17.5) and equation (3.17.6), we get

$$\eta_{max} = \frac{nFE}{\Delta H} = \frac{-I.t.E}{\Delta H}$$

   Where, I = Current, and
   t = Time for which current flows.

3. The overall efficiency of fuel cell is given as

$$\eta_{overall} = \eta_F \times \text{Loss factor}$$

4. The power output of a reversible fuel cell is given by

$$P_{rev.} = \frac{\Delta G_{max}}{\text{Molar mass of hydrogen}}$$

   Where, molar mass of hydrogen = 2.018 kg/mole

5. Actual electrical power output, $P = P_{rev.} \times \eta_{overall}$

6. The rate of heat released, $Q = P_{rev.} - P$

Que 3.18. Discuss the *V-I* and *P-I* characteristics of fuel cells.

Answer

A. Current-Voltage Characteristics :

1. The actual cell potential is decreased from its equilibrium potential because of irreversible losses, activation polarization, internal polarization and concentration polarization. These losses are called polarization voltage losses.

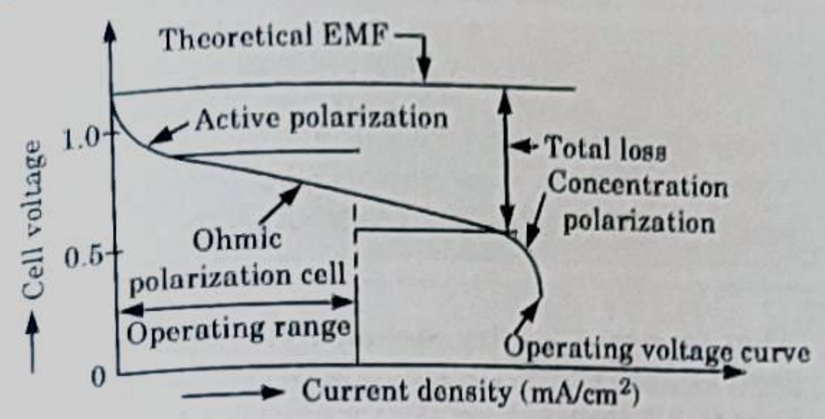

Fig. 3.18.1. Current-voltage characteristics of fuel cell.

2. These losses cause cell voltage (V_C) to get reduced.

3. The cell voltage drops with increase in current density because of polarization losses and it is given by

$$V_C = E - \Delta V_p$$

   Where, E is an ideal or no load voltage potential of cell.

5. With increase in load on the cell, the internal losses increase and drop in cell terminal voltage occurs.

B. Power-Current Characteristics :

1. The power of fuel cell increases in cell current density and decreases after saturation point is reached, because of polarization effect.

2. The power of cell starts decreasing with increase in cell current after saturation point, because of high thermal losses. The power of cell is given by :

$$P = V_c \times I_c$$

4. Therefore, power of cell increases with increase in cell current, but this will also increase the polarization losses and saturation point will be reached where power of the cell will be maximum.

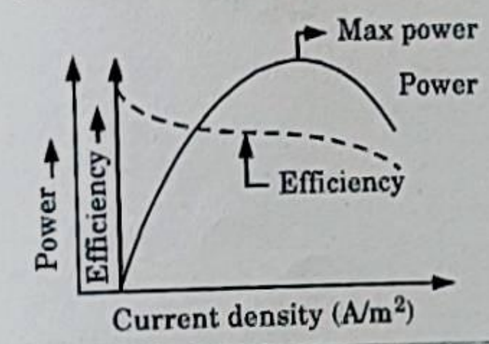

Fig. 3.18.2. Current density (A/m²).

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VERY IMPORTANT QUESTIONS

Following questions are very important. These questions may be asked in your SESSIONALS as well as UNIVERSITY EXAMINATION.

Q. 1. What is meant by dry steam, wet steam and hot water geothermal system ?

Ans. Refer Q. 3.2, Unit-3.

Q. 2. Discuss the difference between a geothermal power plant and thermal power plant. Categorize resources of geothermal energy.

Ans. Refer Q. 3.4, Unit-3.

Q. 3. Draw schematic diagram of an MHD power generating system having heat recovery steam generator. Explain the functioning of the steam.

Ans. Refer Q. 3.8, Unit-3.

Q. 4. Describe the principle of working of a H_2O_2 cell. Also give its limitations.

Ans. Refer Q. 3.10, Unit-3.

Q. 5. Explain the difference between a fuel cell and battery. What are the uses and advantages of fuel cells ?

Ans. Refer Q. 3.12, Unit-3.

☺☺☺

4 UNIT

Thermoelectrical and Thermionic Conversion and Wind Energy

CONTENTS

Part-1	:	Thermoelectrical Conversion Principle of Working	4–2M to 4–5M
Part-2	:	Performance and Limitation	4–5M to 4–9M
Part-3	:	Thermoionic Conversions, Principle of Working	4–9M to 4–12M
Part-4	:	Performance and Limitations	4–12M to 4–14M
Part-5	:	Wind Power and its Source	4–15M to 4–15M
Part-6	:	Site Selection Criterion	4–15M to 4–17M
Part-7	:	Momentum Theory, Classification of Rotors	4–17M to 4–20M
Part-8	:	Classification of Rotors	4–20M to 4–25M
Part-9	:	Concentrations and Augments	4–25M to 4–26M
Part-10	:	Wind Characteristics Performance	4–27M to 4–29M
Part-11	:	Limitations of Energy Conversion System	4–29M to 4–29M

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PART-1

Thermoelectricl Conversion Principle of Working.

Questions-Answers

Long Answer Type and Medium Answer Type Questions

Que 4.1. Explain the following : Joule effect, Thomson effect, Peltier effect and Seebeck effect

Answer

a. Seebeck Effect :

1. It states that if a closed circuit is made of two dissimilar metals and the two junctions are maintained at different temperatures than an emf is setup in circuit.
2. The magnitude of current depends on both the metals (1 and 2) and the temperature difference between the junctions.
3. The emf produced is function of the difference in temperature between hot and cold junction, and is given by

$$E = \alpha_s \Delta T$$

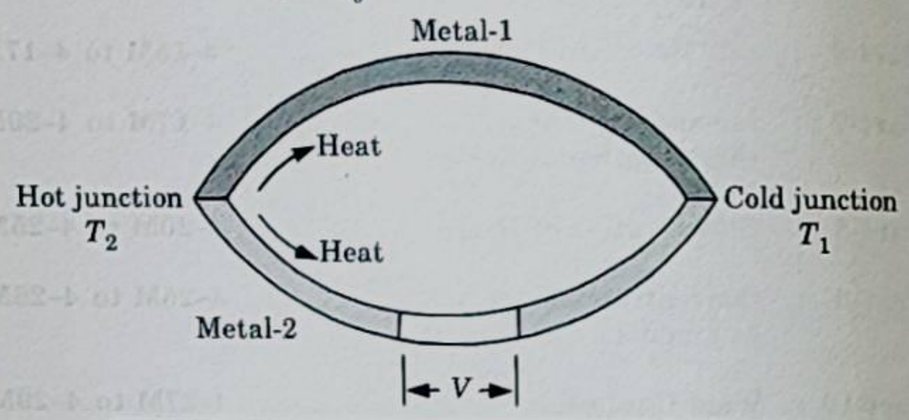

Fig. 4.1.1. Seebeck effect principle of thermocouple.

Where, ΔT = Difference between hot and cold junction, and
α_s = Seebeck coefficient.

b. Peltier Effect :

1. When an electric current flows across an isothermal junction of two dissimilar materials, there is either an evolution or absorption of heat at the junction. This effect is called the Peltier effect.

2. The Peltier coefficient α_p is defined as the heat evolved or absorbed at the junction per unit current flow per unit time.
3. Mathematically,

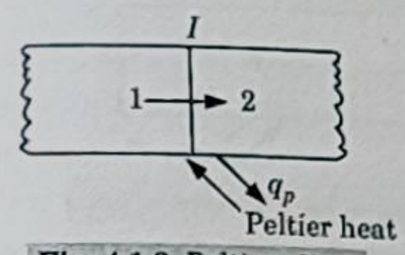

Fig. 4.1.2. Peltier effect.

$$\alpha_{p1\text{-}2} = \alpha_{p1} - \alpha_{p2} = \frac{q_p}{I}$$

and $$q_p = \alpha_{p1\text{-}2} I$$

Where, I = Peltier heat per unit time.

c. Joule Effect :

1. In a closed electrical circuit, if the current I flows through a resistance R, the heat generated by the resistance is equal to I^2R. It is known as Joule effect.
2. The expression is given by

$$Q = I^2R$$

d. Thomson Effect :

1. It states that "any current carrying conductor with a temperature difference between two points will either absorb or emit heat, depending upon the material."
2. The Thomson coefficient (σ) is defined as the heat absorbed (or evolved) per unit time per unit electric current and per unit temperature gradient.
3. Mathematically,

$$\sigma = I \frac{dq_T/dx}{dT/dx}$$

Where, $\frac{dq_T}{dx}$ = Heat interchange per unit time per unit length of conductor,

$\frac{dT}{dx}$ = Temperature gradient, and

I = Current

4. Hence, the Thomson heat per unit time is given by

$$\frac{dq_T}{dx} = \frac{\sigma I dT}{dx}$$

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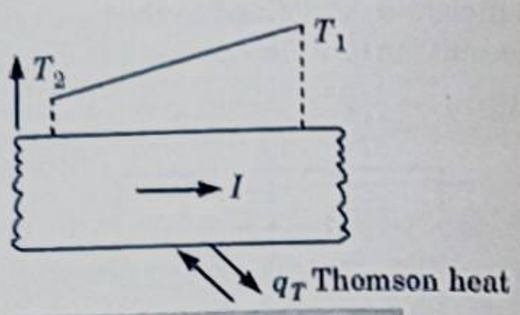

Fig. 4.1.3. Thomson effect.

A. Relationships between the Seebeck, Peltier and Thomson Coefficients as Derived by Kelvin are as follows :

1. The first Kelvin relation exists between the Seebeck coefficient and Peltier coefficient by the relation :

$$\alpha_p = \alpha_s . T$$

or $$\alpha_{p1\text{-}2} = \alpha_{s1\text{-}2}\, T$$

2. The second Kelvin relation exists between the Seebeck coefficient and Thomson coefficient by relation :

$$\sigma = T\frac{d\alpha_s}{dT}$$

or $$\sigma_{1\text{-}2} = T\frac{d\alpha_{s_{1\text{-}2}}}{dT}$$

Que 4.2. Explain the principle working and construction of thermoelectrical generator.

Answer

A. Principle : The principle is based on Seebeck effect.
Refer 4.1, Page 4–2M, Unit-4.

B. Construction :

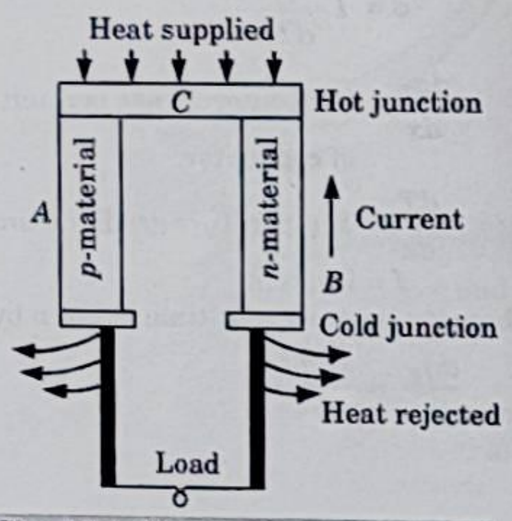

Fig. 4.2.1. Thermoelectric generator.

1. It consists of two dissimilar metals *A* and *B* respectively with their end joined together at point *C* which is kept at higher temperature.
2. The other end of metals *A* and *B* is kept at low temperature and induced voltage which is measured by connecting the potentiometer at these free ends.
3. The simplest thermoelectric generator consists of a thermocouple, comprising a *p*-type and *n*-type material.

C. Working :

1. Consider a metal bar whose one side is kept at a higher temperature than the other.
2. If the free electrons in the metal are considered to behave as a gas, the kinetic theory of gases predicts that the free electrons in the hot side of the bar will be on higher kinetic energy and will be moving at greater speed than those in the cold side of the bar.
3. As the faster moving electrons flow from the hot side to the cold side of the bar, it results in an accumulation of negative charge at the cold side and prevents further charge build up until circuit is closed.
4. In a closed circuit, the current will flow to reduce the charge build up and will continue to flow as long as the temperature gradient is maintained.

Que 4.3. Write short notes on thermoelectric material.

AKTU 2018-19, Marks 05

Answer

1. A material that can be used to convert thermal energy into electric energy or provide refrigeration directly from electric energy is known as thermoelectric material.
2. The good thermoelectric materials should possess :

i. Large Seebeck coefficients.
ii. High electrical conductivity.
iii. Low thermal conductivity.

3. The examples of thermoelectric materials are :

i. Lead telluride (PbTe).
ii. Silicon germanium (SiGe).
iii. Bismuth antimony (BiSb).
iv. Bismuth telluride (Bi_2Te_3).

PART-2

Performance and Limitations.

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Questions-Answers

Long Answer Type and Medium Answer Type Questions

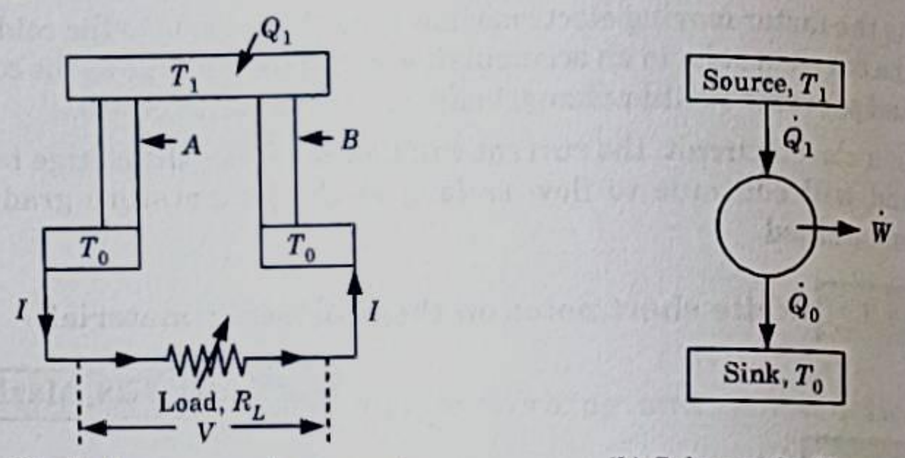
Que 4.4. **Describe the working of a thermoelectric generator. Derive an expression for its power output.**

AKTU 2015-16, Marks 7.5

Answer

A. Working : Refer Q. 4.2, Page 4–4M, Unit-4.

B. Expression for Power Output :



(*a*) Single thermocouple generator. (*b*) Schematic diagram.

Fig. 4.4.1. Efficiency of thermoelectric generator.

1. Let, T_1 = Temperature of source (K),
   T_0 = Temperature of sink (K),
   L = Length of thermoelectric materials (m),
   A = Area of thermoelectric materials (m^2),
   k = Thermal conductivity of materials (W/m-K),
   ρ = Electrical resistivity of materials (ohm-m),
   K = Thermal conductance of materials, $\frac{kA}{L}$ (W/K),
   R = Electrical resistance of elements, $\frac{\rho.L}{A}$ (ohm),
   α = Absolute value of Seebeck coefficients (V/K), and
   π = Absolute value of Peltier coefficient (V).
   Suffix A and B are used for respective materials and suffix L is used for load resistance.

2. Consider hot plate as control volume and the Seebeck voltage, $V = \alpha_{AB} \cdot (T_1 - T_0)$ is generated as a result of current flow I in the circuit when the load R_L is placed in the circuit.

3. The heat rate into hot plate $\dot{Q}_1$ is balanced by the heat conducted into two legs $\dot{Q}_k$ and the Peltier heat $\dot{Q}_p$ at the junction due to current flow in the circuit and the Joulian heat $\frac{\dot{Q}_J}{2}$ flows into each junction assuming that half the heat appears at each junction.

4. According to 1st law of thermodynamics, the energy balance equation can be written as :

$$\dot{Q}_1 + \frac{\dot{Q}_j}{2} = \dot{Q}_p + \dot{Q}_k \quad \ldots(4.4.1)$$

Where,
$$\dot{Q}_p = \pi_{AB} \times I = \alpha_{AB} \times I \times T_1 \quad \ldots(4.4.2)$$

$$\dot{Q}_j = (R_A + R_B) I^2 = \left[\frac{\rho_A \cdot L_A}{A_A} + \frac{\rho_B \cdot L_B}{A_B}\right] \cdot I^2 \quad \ldots(4.4.3)$$

$$\dot{Q}_k = (K_A + K_B)(T_1 - T_0) = \left[\frac{k_A \cdot A_A}{L_A} + \frac{k_B \cdot A_B}{L_B}\right](T_1 - T_0) \quad \ldots(4.4.4)$$

5. On substitution,

$$\dot{Q}_1 = \alpha_{AB} \cdot I \cdot T_1 + \left[\frac{k_A \cdot A_A}{L_A} + \frac{k_B \cdot A_B}{L_B}\right](T_1 - T_0) - \frac{1}{2}\left[\frac{\rho_A \cdot L_A}{A_A} + \frac{\rho_B \cdot L_B}{A_B}\right] I^2 \quad \ldots(4.4.5)$$

6. Useful power generated is,

$$\dot{W}_L = I^2 \cdot R_L = \frac{V_L^2}{R_L} \quad \ldots(4.4.6)$$

Where, $V_L = \alpha_{AB}(T_1 - T_0) - I(R_A + R_B)$ ...(4.4.7)

7. By Kirchhoff's law, $I = \frac{V}{R} = \frac{\alpha_{AB}(T_1 - T_0)}{R_A + R_B + R_L}$ ...(4.4.8)

8. Let the resistance ratio, m be defined as

$$m = \frac{R_L}{R_A + R_B} \quad \ldots(4.4.9)$$

Hence, $(1 + m) = \frac{R_A + R_B + R_L}{R_A + R_B}$

9. On putting the value from equation (4.4.9) in equation (4.4.8), we get

$$I = \frac{\alpha_{AB}(T_1 - T_0)}{(1+m)(R_A + R_B)} \quad \ldots(4.4.10)$$

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10. Power output from equations (4.4.6), (4.4.9) and (4.4.10) can be written as,
Power generated,

$$\dot{W}_L = \frac{[\alpha_{AB}(T_1 - T_0)]^2 \times (R_A + R_B)m}{(1+m)^2(R_A + R_B)^2}$$

i.e., $$\dot{W}_L = \frac{m}{(1+m)^2} \times \frac{[\alpha_{AB}(T_1 - T_0)]^2}{(R_A + R_B)} \quad ...(4.4.11)$$

11. From equation (4.4.5), the rate of heat input becomes,

$$\dot{Q}_1 = \alpha_{AB}\frac{\alpha_{AB}(T_1 - T_0)}{(1+m)(R_A + R_B)} \cdot T_1 + (K_A + K_B)(T_1 - T_0)$$

$$-\frac{1}{2}(R_A + R_B) \cdot \frac{\alpha_{AB}^2(T_1 - T_0)^2}{(1+m)^2(R_A + R_B)^2}$$

$$\dot{Q}_1 = \frac{\alpha_{AB}^2 T_1(T_1 - T_0)}{(1+m)(R_A + R_B)} + (K_A + K_B)(T_1 - T_0)$$

$$-\frac{1}{2}\frac{\alpha_{AB}^2}{(1+m)^2} \times \frac{(T_1 - T_0)^2}{(R_A + R_B)} \quad ...(4.4.12)$$

12. Efficiency of the generator can be written as :

$$\eta = \frac{\dot{W}_L}{\dot{Q}_1}$$

On substituting the values of $\dot{W}_L$ and $\dot{Q}_1$ from the equation (4.4.11) and equation (4.4.12) and on solving we get,

$$\eta = \left(\frac{T_1 - T_0}{T_1}\right) \frac{m}{(1+m) - \frac{1}{2}\left(\frac{T_1 - T_0}{T_1}\right) + \frac{(K_A + K_B)(R_A + R_B)(1+m)^2}{\alpha_{AB}^2 \cdot T_1}}$$

13. We define figure of merit as :

$$Z = \frac{\alpha_{AB}^2}{(R_A + R_B)(K_A + K_B)}$$

Z consists of material properties of semiconductors A and B only.

14. Thus efficiency will increase with increase in Z.
15. The Seebeck effect can be +ve or –ve. High magnitude of Z is required with high value of Seebeck coefficient (opposite polarity), low resistivity and low thermal conductivity.
16. For high efficiency, $(Z.T)$ should be as high as possible.
17. The maximum power output and maximum efficiency are given by the relations :

$$Z_{(opt)} = \left[\frac{\alpha_{AB}}{\sqrt{\rho_A \cdot k_A} + \sqrt{\rho_B \cdot k_B}}\right]^2$$

$$(\dot{W}_L)_{max} = \frac{1}{4}\frac{\alpha_{AB}^2 \cdot (T_1 - T_0)^2}{(R_A + R_B)}$$

$$\eta_{max} = \left(\frac{T_1 - T_0}{T_1}\right) \cdot \frac{1}{2 - \frac{1}{2}\left(\frac{T_1 - T_0}{T_1}\right) + \frac{4}{Z_{opt} \cdot T_1}}$$

Que 4.5. Give advantages and disadvantages of thermoelectrical generator.

Answer

A. Advantages :

1. Absence of rotating parts.
2. It is a compact device.
3. It operates at high temperatures.
4. It is quiet in operation, long life, low cost and has low maintenance.
5. It can operate in remote areas and harsh environment.
6. It has higher conversion efficiency.
7. It can be developed for very low power to very high power generation.

B. Disadvantages :

1. It needs high operating temperature at anode.
2. It needs special seal to protect the cathode from corrosive gases.
3. It needs the cesium vapour in the tube to reduce the space charge.
4. Metal is costly as it has to withstand high temperatures.

C. Applications :

1. It can be installed anywhere at the place of use.
2. Thermionic power can be developed at centralized location and distributed for various commercial and residential applications.
3. It is suitable for military and space applications.
4. These generators are suitable for use with nuclear reactor or radio-isotope heat sources.

PART-3

Thermionic Conversions, Principle of Working.

Questions-Answers

Long Answer Type and Medium Answer Type Questions

Que 4.6. What is a thermionic converter ? How does it works ? What are its major limitations ?

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OR

Describe the principle of operation and constructional details of basic thermionic generator.

Answer

A. Principle :

1. A thermionic converter (or generator) converts heat energy directly into electrical energy by utilizing thermionic emission effect.
2. In this device, electrons act as the working fluid in place of a vapour (or gas) and electrons are emitted from the surface of heated metal.

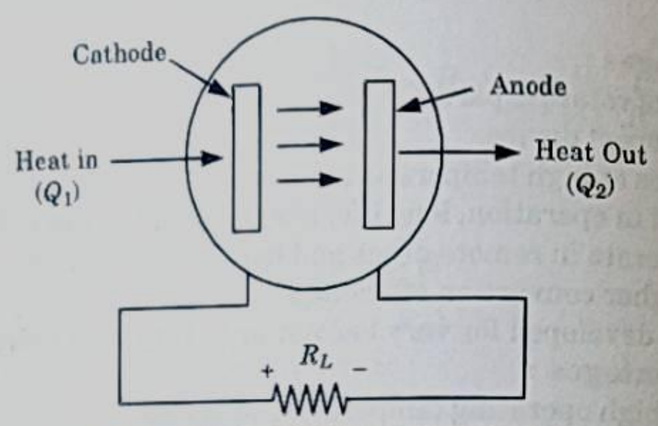

Fig. 4.6.1. Thermionic converter.

B. Work Function :

1. It is defined as the energy required to extract an electron from the metal.
2. The value of work function varies with the nature of metal and surface conditions.

C. Construction : It consists of the following :

a. Anode (Negatively Charged Electrode) :

1. It is a colder collector electrode into which vapourized electrons are condensed after conduction through the inter-electrode plasma and it must be cooled to avoid back emission of electrons.

b. Cathode (Positively Charged Electrode) :

1. It is a hot emitter electrode from which electrons are vapourized by thermionic emission at high temperatures due to heat input and it has higher work function than anode.

D. Working :

1. A metal electrode called emitter is heated until it releases electrons from its surface and flow towards the anode.
2. The electrons cross the small gap between the electrodes and accumulate on a cooled metal electrode called collector.
3. The work function of emitter should be higher than that of collector.
4. In order to minimize the energy losses, as the electrons move across the gap, the space between the electrodes is either maintained at high vacuum or it is filled by a high conducting plasma like ionized cesium vapour.

5. The electrons enter the collector and return through the external circuit back to the emitter, thus it produces electrical power.
6. The energy of electrons emitted by a cathode is partially rejected to the heat sink from anode and the remainder is converted into electrical power.

Que 4.7. Describe the principle of working and constructional details of basic thermionic generator. What is the basic difference between thermoelectric and thermionic conversion systems ? Also, explain the working of thermoelectric generators.

AKTU 2016-17, Marks 15

OR

What is the basic difference between thermoelectric and thermionic conversion systems ? Also, explain the working of thermoelectric generators.

AKTU 2018-19, Marks 10

Answer

A. Principle of Working and Constructional Details of Basic Thermionic Generator : Refer Q. 4.6, Page 4–9M, Unit-4.

B. Difference between Thermoelectric and Thermionic :

S. No.	Thermoelectric Conversion System	Thermionic Conversion System
1.	A thermoelectric converter is a form of heat engine which takes up heat at an upper temperature (hot junction) converts it partly into electrical energy and discharges the remaining part at a lower temperature (cold junction).	A thermionic converter is a form of heat engine that uses an electron gas as a working substance.
2.	The efficiency of thermocouple as in case with other heat engines increases by increasing the upper temperature and decreasing the lower temperature.	A thermionic converter works because of the phenomenon of "thermionic emission" of electrons from the metal when it is heated.
3.	Since the lower temperature is usually that of environment, the efficiency of a thermocouple practically depends upon the hot junction temperature.	A thermionic converter consists of two metals or electrode with different work function is maintained at a higher temperature than one with the smaller work function.

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C. **Working of Thermoelecric Generators :** Refer Q. 4.2, Page 4–4M, Unit-4.

PART-4

Performance and Limitations.

Questions-Answers

Long Answer Type and Medium Answer Type Questions

Que 4.8. **Derive the expression for power and efficiency of a thermionic generator.** **AKTU 2017-18, Marks 10**

Answer

Power and Efficiency of Thermionic Converter :

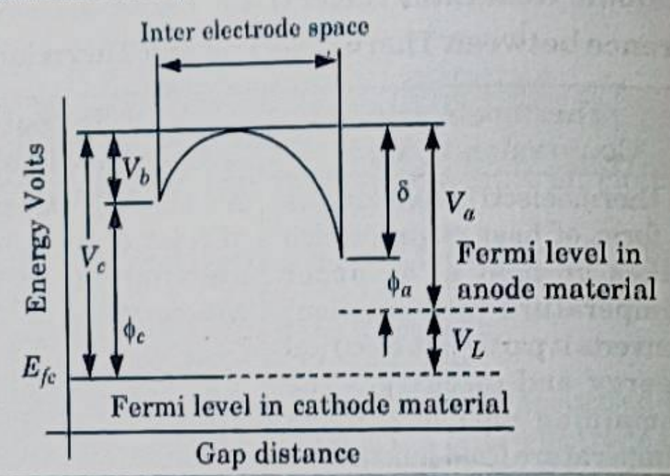

Fig. 4.8.1. Characteristic curve of thermionic converter.

1. Let ϕ_c = Work function of cathode,
   ϕ_a = Work function of anode,
   V_b = Kinetic energy barrier at cathode,
   V_L = Load voltage,
   δ = Space charge, and
   T_c and T_a = Temperature of cathode and anode respectively.
2. The total energy required at cathode, $V_c = \phi_c + V_b$
3. Current density of cathode is given by,

$$J_c = AT_c^2 \exp\left(\frac{-V_c}{KT_c}\right)$$

4. Similarly, current density of anode is given by,

$$J_a = AT_a^2 \exp\left(\frac{-V_a}{KT_a}\right)$$

4. Load voltage (output), $V_L = V_c - V_a = \phi_c - \phi_a$
5. Energy required to reaching the electrons at anode,

$$Q_1 = (\phi_c + V_b) \times J_c = V_c \times J_c$$

6. The electrons also carries the energy in the form of kinetic energy $(2KT_c)$ is,

$$Q_2 = J_c \frac{2KT_c}{e}$$

7. Therefore, total energy needed for electron to reach to the anode,

$$Q_c = Q_1 + Q_2 = J_c V_c + J_c \frac{2KT_c}{e}$$

$$= J_c\left(V_c + \frac{2KT_c}{e}\right)$$

8. Similarly, back emission from anode carries the energy is given by,

$$Q_a = J_a V_a + J_a \frac{2KT_a}{e}$$

$$= J_a\left(V_a + \frac{2KT_a}{e}\right)$$

9. The net energy supplied to cathode,

$$Q_{net} = J_c\left(V_c + \frac{2KT_c}{e}\right) - J_a\left(V_a + \frac{2KT_a}{e}\right)$$

10. Power output $(P) = V_L(J_c - J_a)$

11. The efficiency of converter $(\eta) = \dfrac{P}{Q_{net}}$

$$= \frac{V_L(J_c - J_a)}{J_c\left(V_c + \frac{2KT_c}{e}\right) - J_a\left(V_a + \frac{2KT_a}{e}\right)}$$

$$= \frac{(V_c - V_a)(J_c - J_a)}{J_c\left(V_c + \frac{2KT_c}{e}\right) - J_a\left(V_a + \frac{2KT_a}{e}\right)}$$

Let $\dfrac{V_c}{KT_c} = \gamma_c$

$\dfrac{V_a}{KT_a} = \gamma_a$

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And $$\frac{T_a}{T_c} = T'$$

$$\therefore \quad \eta = \frac{(\gamma_c - T'\gamma_a) \times [1 - T'^2 \exp(\gamma - \gamma_a)]}{(\gamma_c + 2) - T'^2(\gamma_a + 2T')\exp(\gamma_a - \gamma_a)}$$

12. If $\gamma_a = \gamma_c = \gamma$ then

$$\eta_{max} = (1 - T')\frac{\gamma}{\gamma + 2}\left[\frac{1 - T'^2}{1 - \frac{T'^2(\gamma + 2T')}{\gamma + 2}}\right]$$

$$= (1 - T')\frac{\gamma}{\gamma + 2} \qquad \left[\because \frac{1 - T'^2}{1 - \frac{T'^2(\gamma + 2T')}{\gamma + 2}} \approx 1\right]$$

Que 4.9. Explain the advantages, disadvantage and application of thermionic conversion.

Answer

A. Advantages :

1. Absence of rotating parts.
2. It is a compact device.
3. It operates at high temperatures.
4. It is quiet in operation, long life, low cost and has low maintenance.
5. It can operate in remote areas and harsh environment.
6. It has higher conversion efficiency.
7. It can be developed for very low power to very high power generation.

B. Disadvantages :

1. It needs high operating temperature at anode.
2. It needs special seal to protect the cathode from corrosive gases.
3. It needs the cesium vapour in the tube to reduce the space charge.
4. Metal is costly as it has to withstand high temperatures.

C. Applications :

1. It can be installed anywhere at the place of use.
2. Thermionic power can be developed at centralized location and distributed for various commercial and residential applications.
3. It is suitable for military and space applications.
4. These generators are suitable for use with nuclear reactor or radio-isotope heat sources.

PART-5

Wind Power and its Source.

Questions-Answers

Long Answer Type and Medium Answer Type Questions

Que 4.10. What do you understand by "Wind" ? What are the causes of wind ?

Answer

1. The movement of air is called wind which is created by the pressure difference between two uneven heated places.
2. Wind is air in motion and it derives energy from solar radiation
3. Wind energy is a clean, eco-friendly, safe and renewable source of energy.

A. Causes of Wind : Basically there are two causes of wind :

1. Uneven heating of the earth's surface (local winds), and
2. The rotation of earth around its axis (planetary winds).

a. Local Winds : Local winds are caused by unequal heating and cooling of ground surfaces and water bodies in day and night *i.e.*, lake, sea, desert etc.

b. Planetary Winds :

i. Planetary winds are caused by the rotation of the earth about its axis and the combined effect of difference in temperature at equator and polar region.

ii. Because of this, warm air from tropical regions flows upward and moves towards the poles and cool air from poles move towards the tropical region.

PART-6

Site Selection Criterion.

Questions-Answers

Long Answer Type and Medium Answer Type Questions

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Que 4.11. **Describe main considerations in selecting a site for wind farm. Discuss merits and demerits of wind energy.**

AKTU 2017-18, Marks 10

Answer

A. Main Criteria for Site Selection :

1. The area should be open and away from cities.
2. Flat open area should be selected, as the wind velocities are high in flat open area.
3. The proposed altitude is to be selected by taking average wind speed data.
4. Minimum wind speed is available throughout the year.
5. Ground surface should be stable and high soil strength.
6. To minimize the transmission losses, the wind power should be near the consumers.
7. It should be at least 5 km away from the cities to reduce the effect of sound pollution.
8. Low land cost.
9. Approach roads up to site.
10. Height of tower will increase where trees are present.

B. Design Consideration for Wind Turbine : The wind turbine must be able to meet the following criteria :

1. It should be small in size and suitable for roof mounting in urban areas.
2. No risks for its neighborhood.
3. Have good efficiency.
4. Insensitive to turbulence.
5. Suitable for mass production at low price.

C. Advantages :

1. The electricity generation by wind turbines is pollution free and does not release carbon dioxide.
2. It doesn't involve the consumption of water unlike many other conventional energy sources.
3. It is renewable and available at free of cost.
4. Helpful for supplying energy in rural, offshore, onshore areas.
5. It is reliable and cost effective for large units.
6. Wind does not require any transportation.
7. Low operating cost.
8. Economically competitive.

D. Disadvantages :

1. Wind energy has low power density and variable with time and locations.
2. Wind energy can be useful only at remote areas away from cities.
3. The efficiency of turbine rotor is less (10 – 45 % only).
4. Large transmission losses as the wind farms are located in remote areas.
5. The capital cost per kWh is more in small unit with respect to larger unit (but smaller units are more reliable).
6. Wind cannot be stored as a conventional source.
7. Weight of power system is high and requires large area.
8. Wind energy generates noise pollution.
9. Direction of wind changes and is never constant and reliable.

PART-7

Momentum Theory, Classification of Rotors.

Questions-Answers

Long Answer Type and Medium Answer Type Questions

Que 4.12. **What are the most favorable sites for installing wind turbines ? Using Betz model of a wind turbine, derive the expression for power extracted from wind. Under what condition does the maximum theoretical power can be extracted from the wind turbine ?**

AKTU 2016-17, Marks 15

AKTU 2018-19, Marks 10

OR

Describe the basic principle of wind energy conversion and derive the expression for power developed due to wind.

AKTU 2017-18, Marks 10

Answer

A. Site Selection : Refer Q. 4.11, Page 4–16M, Unit-4.

B. Principle of Power Generation in Wind Mills :

1. The basic principle of wind energy is to convert the kinetic energy of wind into rotational motion to operate an electric generator.

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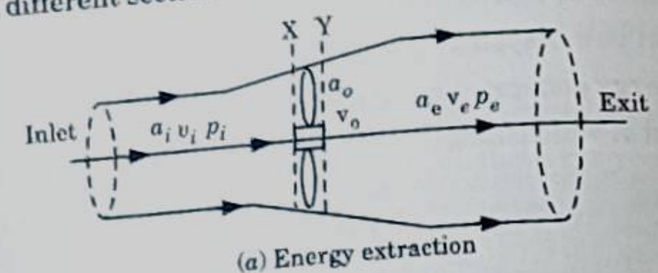
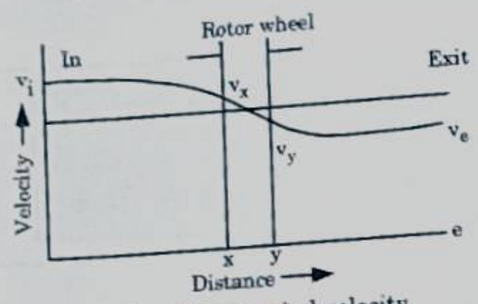
C. Expression for maximum efficiency :

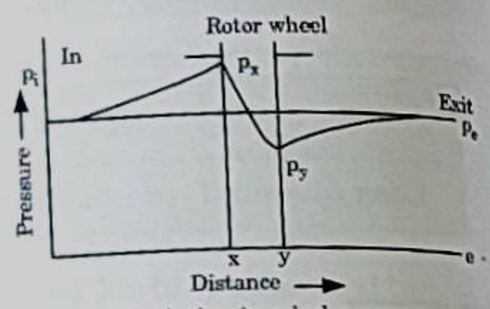
1. The power in the wind can be extracted by allowing it to pass through moving wings that exert torque on a rotor.
2. The amount of power transferred is directly proportional to the density of the air, the area swept out by the rotor, and the cube of the wind speed.
3. Fig. 4.12.1 shows the air flow diagram on rotor, with variation of wind speed at different sections.



(*a*) Energy extraction



(*b*) Variation in wind velocity.



(*c*) Variation in wind pressure.

Fig. 4.12.1. Air flow through the rotor.

4. As the air passes through the rotor disk, there is a drop in static pressure such that pressure is below the atmospheric pressure while it leaves the blade.
5. The speed of the wind also gets reduced in the section (wake) and after this section the atmospheric pressure and speed of air again increases.
6. Let a_i and a_e = Inlet and outlet area of air enclosure,
   a_o = Rotor swept volume,
   v_i and v_e = Velocity of wind at inlet and outlet of enclosure,
   v_o = Velocity of rotor,
   ρ = 1.25, air density, and
   $\dot{m}$ = Mass flow rate of air over rotor.
7. The thrust on the turbine by moving air as it passes over the rotor,

$$F = \dot{m}(v_i - v_e) \quad \text{...(4.12.1)}$$

8. The power extracted by turbine,

$$P_T = \dot{m}(v_i - v_e) \times v_o \quad \text{...(4.12.2)}$$

9. Instantaneous loss in kinetic energy of wind as it passes through rotor,

$$P'_W = \frac{1}{2}\dot{m}(v_i^2 - v_e^2) \quad \text{...(4.12.3)}$$

10. From equation (4.12.2) and equation (4.12.3),

$$\dot{m}(v_i - v_e) \times v_o = \frac{1}{2}\dot{m}(v_i^2 - v_e^2)$$

or
$$v_o = \frac{v_i + v_e}{2} \quad \text{...(4.12.4)}$$

11. From equation (4.12.2) and equation (4.12.4),

$$P_T = \dot{m}(v_i - v_e)\frac{(v_i + v_e)}{2} \quad \text{...(4.12.5)}$$

12. The mass flow rate through turbine rotor,

$$\dot{m} = \rho a_o v_o = \rho a_o \left(\frac{v_i + v_e}{2}\right) \quad \text{...(4.12.6)}$$

13. From equation (4.12.5) and equation (4.12.6),

$$P_T = \rho a_o \left(\frac{v_i + v_e}{2}\right)(v_i - v_e)\left(\frac{v_i + v_e}{2}\right)$$

$$P_T = \frac{1}{4}\rho a_o (v_i + v_e)(v_i^2 - v_e^2) \quad \text{...(4.12.7)}$$

14. For maximum power,

$$\frac{\partial P}{\partial v_e} = 0$$

$$3v_e^2 + 2v_i v_e - v_i^2 = 0$$

$$v_e = \frac{1}{3}v_i \quad ; \quad v_e = v_i \text{ (not consider)}$$

$$P_{max} = \frac{8}{27 g_c}\rho A v_i^3$$

$$= \frac{16}{27 g_o} \times \frac{1}{2}\rho A v_i^2 = 0.593 \times \left(\frac{1}{2} \times \frac{\rho A v_i^3}{g_c}\right)$$

$$= 0.593\, P_{total}$$

15. Power coefficient = C_P = $\frac{\text{Power output from wind machine}}{\text{Power available in wind}}$

16. The torque on rotor, $T = \frac{P_T}{\omega}$, where $\omega = 2\pi N$ ...(4.12.8)

17. The axial force on turbine,

$$F = \dot{m}(v_i - v_e)$$

$$= \rho a_o \left(\frac{v_i + v_e}{2}\right)(v_i - v_e)$$

$$= \frac{\pi}{8}\rho D^2 (v_i^2 - v_e^2) \quad \text{...(4.12.9)}$$

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18. For a given turbine power, lower the angular velocity of rotor, higher the torque and conversely higher the angular velocity lower the torque.

PART-8

Classification of Rotors.

Questions-Answers

Long Answer Type and Medium Answer Type Questions

Que 4.13. **What do you mean by the nature of wind ? Describe the construction and working of a wind energy conversion system (WECS) with the help of a neat sketch.**

AKTU 2017-18, Marks 10

OR

Classify rotors employed for wind generation.

OR

Write short note on vertical axis wind mills.

AKTU 2016-17, Marks 05

Answer

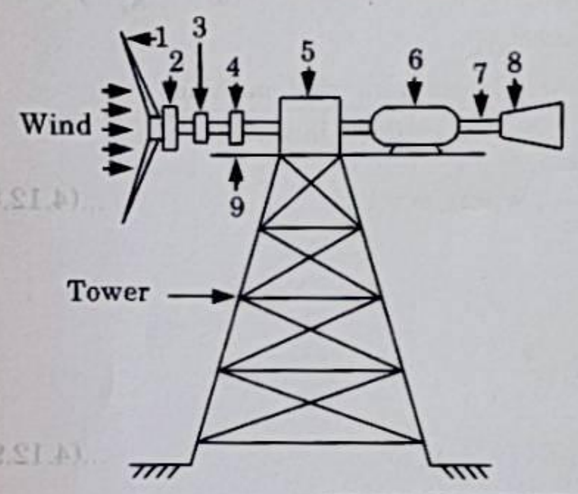
A. Nature of Wind : Refer Q. 4.10, Page 4–15M, Unit-4.

B. Wind Energy Conversion System and Classification of Rotors :

There are two types of wind energy conversion system :

a. Horizontal Axis Wind Turbine (HAWT) Generator :

1. The components of a horizontal axis wind turbine are shown in Fig. 4.13.1.



1. Rotor with blades
2. Electromagnetic brakes
3. Controller
4. Mechanical brakes
5. Gear box
6. Generator
7. Shaft
8. Flap or tail vane
9. Tower top

Fig. 4.13.1. Wind generator.

2. In horizontal axis wind turbine, the propeller are of two types *i.e.*,
   a. Up wind, and b. Down wind.
3. In up wind type, the blades are slanted and wind approaching the blades from front side and nacelle is placed on rear side of the blade (nacelle having propeller, gears and generator).
4. In down wind design, wind approaching blades from nacelle side.
5. HAWT are divided according to the blade mounted on hub :
   i. Mono-blade, ii. Twin-blade,
   iii. Three-blade, and iv. Multi-blade.

i. Mono-Blade :

1. The mono-blade HAWT are simple in construction and lighter in weight, low price, easy to install and have easy maintenance.
2. Mono-blade turbines are equipped with synchronous generator and experience minimum stress on bearing.
3. This HAWT produce low power of 15 kW to 50 kW with length of the blade varying from 15 – 25 m.

Applications : Pumping, battery charging and power supply to farmhouses.

ii. Twin-Blade :

1. Twin-blade HAWT is less costly than three-blade type but have large vibrations while running.
2. The teething control is provided with these machines to reduce the fatigue on main shaft.
3. These machines are rated from 1 to 3 MW.

iii. Three-Blade :

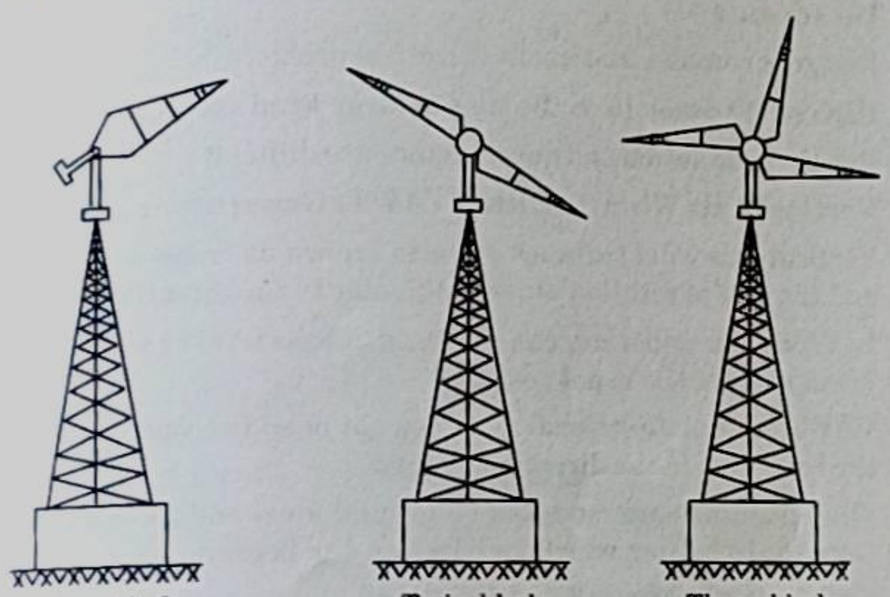


Fig. 4.13.2. Horizontal axis propeller type wind turbine generator unit.

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1. Three-blade HAWT has a 3-phase synchronous generator or a 3-phase induction generator to operate at constant speed and axis of nacelle is oriented so that plane of turbine blades is perpendicular to wind direction.
2. To change the generator shaft speed, a gear chain is provided from turbine shaft.
3. This HAWT produce smaller to larger unit *i.e.*, from 15 kW to 3 MW units.

iv. Multi-Blade :

1. The multi-blade turbines are high solidity turbine used for pumping the water because of high starting torque characteristics.
2. The multi-blade rotors are less efficient because of interference of blades in each other but they are less noisy.

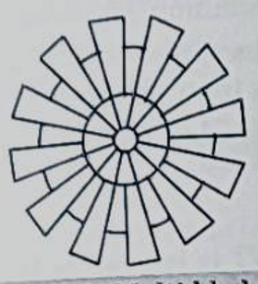

Fig. 4.13.3. Multi-blade.

Advantages :

i. Energy cost/kWh is lower.
ii. Higher power generation.

Disadvantages :

i. Design is complex and involves control problems.
ii. High stress develops at the time of large wind speed.
iii. Repair, installation and maintenance are difficult.

b. Vertical Axis Wind Turbine (VAWT) Generator :

1. Vertical axis wind turbines are also known as cross-wind axis turbine and the axis of rotation is perpendicular to the direction of the wind.
2. In VAWT, the generator can be placed at base level or within the support tower and nacelle is not required.
3. VAWT is omni directional and does not need the yaw control to orient the rotor axis in the direction of wind.
4. These turbines are mounted on ground level and have blades that go from top to bottom which look like an egg beaters.
5. VAWT stands 30 m tall and 15 m wide.

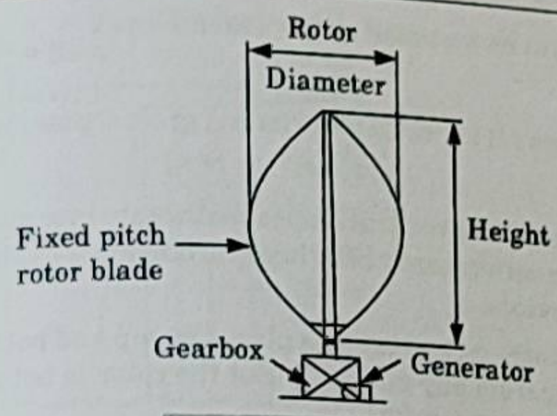

Fig. 4.13.4. Vertical axis.

6. VAWT are classified into three categories :

i. **Savonius :**

1. The rotor of these turbines are 'S' shaped and supported at top and bottom by two circular plates.

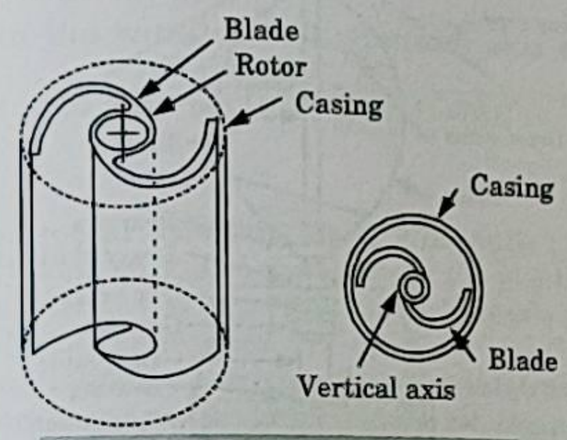

Fig. 4.13.5. Savonius wind turbine.

Fig. 4.13.6. Savonius rotor (S-shaped).

2. Savonius rotor rotates for air flow in any direction and they are self starting, and having low speed and low efficiency.
3. This is a drag type VAWT turns relatively slow, but yields a high torque.

Advantages :

1. This type of rotor operates at low velocity of wind.
2. There is no need of yaw and pitch control.

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3. Generator can be mounted at the ground level.
4. Low cost.

Applications : It is useful for grinding grains, pumping water etc.

ii. Darrieus :

1. These turbines have vertical blades that rotate into and out of the wind and by using aerodynamic lift, these turbines can capture more energy than drag devices.
2. The rotor rotates on two bearing placed on top and bottom of pipe, with wind blowing from any direction. But the rotor is not self starting and needs auxiliary starter.
3. The wind turbine is anchored on ground by wire ropes and generator is mounted on the ground.

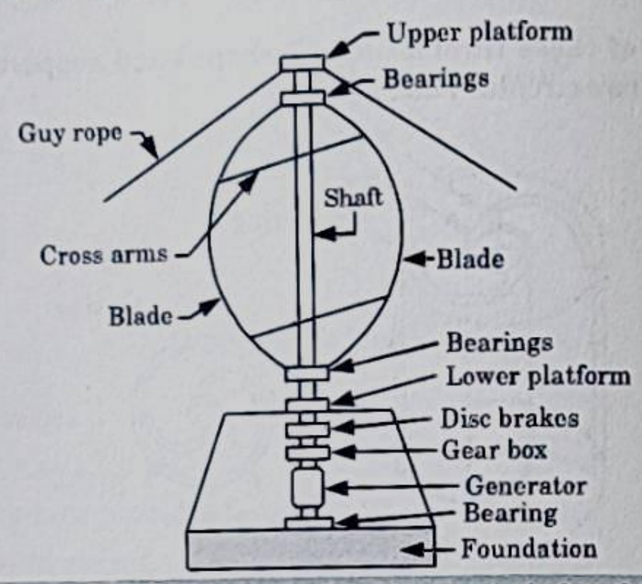

Fig. 4.13.7. Vertical axis darrieus type wind turbine.

Advantages :

1. The generator, gear box etc. are placed on the ground.
2. No need of yaw mechanism to turn the rotor against the wind.

Disadvantage :

1. These turbines suffer a problem of stalling when wind speed is very high.

iii. H-shape Rotor :

1. In this type of turbine, two vertical blades are joined by a horizontal beam to form H-shape and the horizontal beam is mounted on the tower with hub at the centre of the horizontal beam.
2. H-rotor and horizontal beam are designed as aerodynamic foils and H-shape rotates around the vertical axis.
3. In this type, the generator is placed at the upper part and coupled with the rotor through gear system.

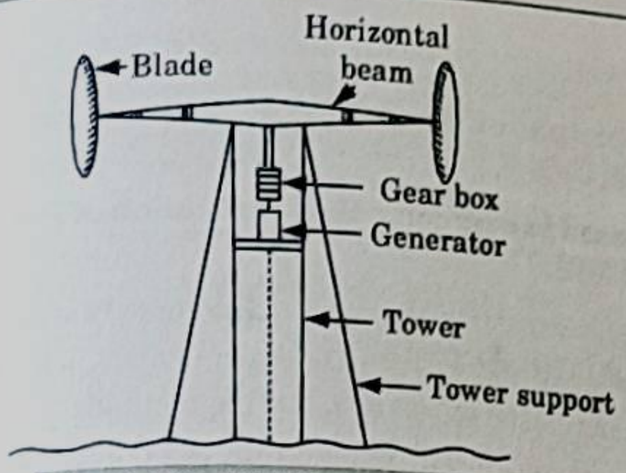

Fig. 4.13.8. VAWT with H-shape rotor.

4. The blades of the rotor are either fixed type or variable shape in terms of angle.
5. For smaller ratings fixed type is used and for higher ratings variable shape blades are used.

Que 4.14. Explain the working of horizontal axis two blade windmills with suitable diagram. **AKTU 2018-19, Marks 10**

Answer

Working of Horizontal Axis Two Blade Windmills :

1. The high velocity wind strikes on the blades at the plant and makes them rotate; this rotation will cause the gears to be rotate and mashed with each other inside the gear box.
2. The motion at gears inside the gear box will cause to rotate the generator attached and hence electricity produced with the rotation of generator.
3. This electricity will be passed to the switchyard through transformers.

Figure : Refer Q. 4.13 (Fig. 4.13.2), Page 4–20M, Unit-4.

PART-9

Concentrations and Augments.

Questions-Answers

Long Answer Type and Medium Answer Type Questions

Que 4.15. What is the basic principle of wind energy conversion ? What methods are used to overcome the fluctuating power generation of a windmill ? **AKTU 2015-16, Marks 10**

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Answer

A. Basic Principle of Wind Energy Conversion : Refer Q. 4.12, Page 4.17M, Unit-4.

B. Methods used to overcome the Fluctuation of Power Generation of a Wind Mill.

1. As the wind speed varies, the speed of the generator varies and produces fluctuations in the electricity.
2. The problems can be solved by following methods.

a. To have constant speed turbines where the blades adjust the pitch, by turning slightly to the side, to adjust with wind speeds.

b. To use variable speed turbines, where the blades and generator change speed with the wind and power control fix the fluctuations of the electrical output.

c. To use low-speed generators.

The various mechanical controls provided with wind machine are as follow :

a. Tethering Control :

1. It is provided with mono and twin blade type horizontal axis turbine to prevent failure because of vibration (fatigue) during orientation of nacelle.
2. The axis of the turbine gets positioned in such a way that the propeller blades revolve in slanting plane at higher speed. The slants get reduced at low speed and get increased at higher speed.
3. This type of control is not provided with three blade rotor.

b. Yaw Control :

1. The yaw control is provided to position the nacelle automatically in the direction of wind with the help of hydraulic mechanism and continuously orient the rotor in the direction of wind.
2. The axis is oriented in such a way that rotor swept area is perpendicular to the wind flowing either in upward or downward direction.

c. Pitch Control :

1. The blade tips are adjusted automatically to provide feathering action. This reduces the speed and power of turbine to match with generator speed.
2. The pitch angle has wide control between $0° - 30°$.

PART-10

Wind Characteristics Performance.

Questions-Answers

Long Answer Type and Medium Answer Type Questions

Que 4.16. Write short notes on :

A. Wind characteristics.

B. Environmental impacts of wind energy.

Answer

A. Wind Characteristics :

1. Performance of the wind turbine is determined by non dimensional characteristic curve from which actual performance can be determined.
2. These characteristic curves are :

a. Wind Speed-Power Characteristics :

1. The power output from wind turbines vary with wind speed.

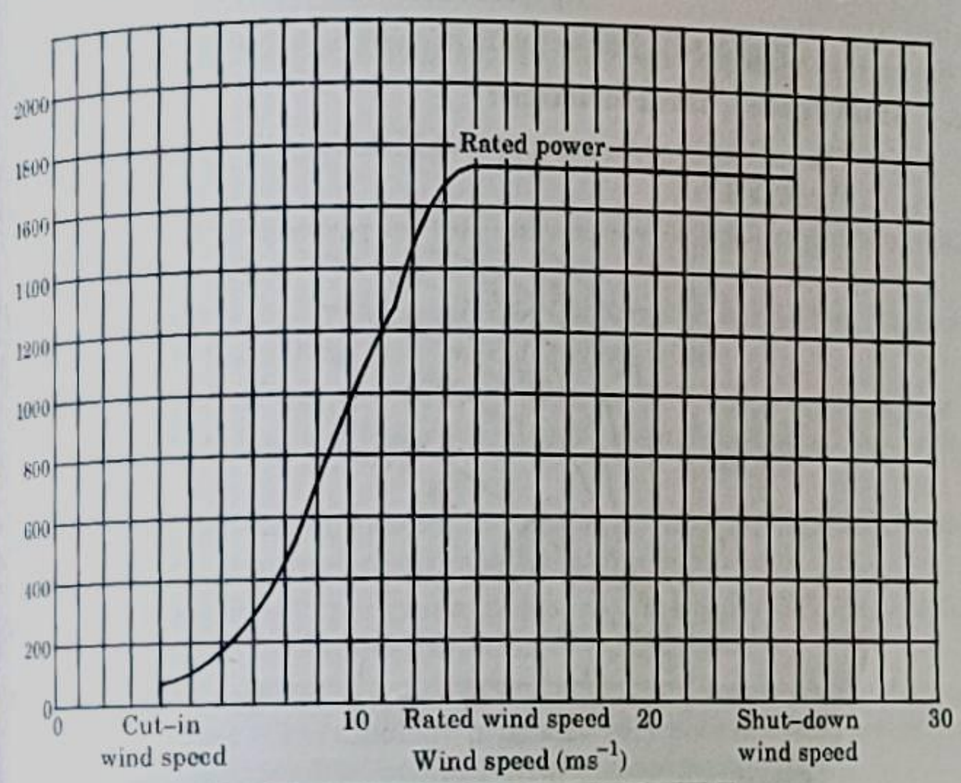

Fig. 4.16.1. Wind speed power curve.

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2. The shape of the wind speed-power curve depends upon the rotor swept area, shape of blade, number of blade, tip speed ratio, rotation of rotor, and the cut in rated speed and cut out speed of wind.
3. The area from rated speed to cut-out speed is called the constant power output area and above 25 m/s turbine output starts reducing and finally stops.

b. Dynamic Characteristics :

1. This characteristic will match the rotational frequency of the turbine for particular wind speed for optimum efficiency.
2. The power extraction will decrease for too close or too apart rotating blade.
3. The maximum value of power coefficient is achieved at particular tip speed ratio.
4. In wind turbine the two extreme conditions of operations, *i.e.*, constant turbine speed or constant tip speed are to be matched to avoid serious effect on turbine.

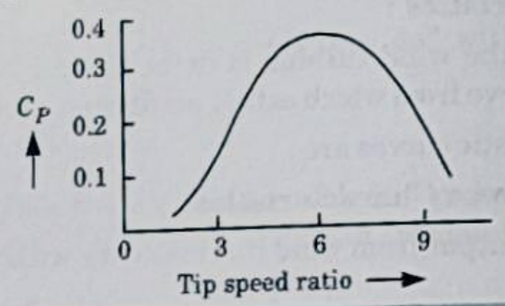

Fig. 4.16.2. Power coefficients to tip speed ratio.

c. Axial Force Characteristics :

1. The variation of thrust force coefficient with tip speed is shown in Fig. 4.13.3.

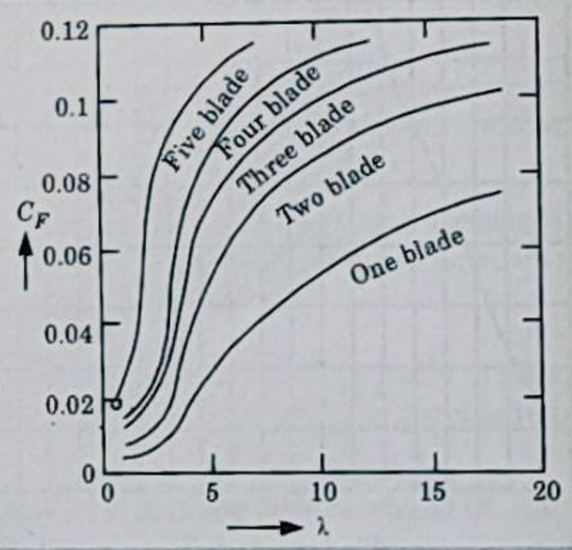

Fig. 4.16.3. Thrust force coefficients-tip speed ratio.

2. With particular speed ratio, the thrust increases with increased solidity and influences the structural design of the tower.

B. Environment Impacts of Wind Energy :

1. Wind energy creates noise pollution because of mechanical (gear box) and aerodynamic noise.
2. The wind turbine produces electromagnetic interference when placed between radio, television etc. stations, as it reflects some electromagnetic radiations.
3. It produces visual shining because of reflection and refraction which depends upon turbine size, number of turbines in wind farm, design etc.
4. Fatal collisions of birds caused by rotating turbine blades.
5. Safety consideration for life because of accidental breaking of blade.

PART-11

Limitations of Energy Conversion System.

Questions-Answers

Long Answer Type and Medium Answer Type Questions

Que 4.17. **What are the limitation of wind energy conversion system ?**

Answer

Limitations : Following are the limitations of Wind Energy Conversion System (WECS) :

1. No electricity is produced when the wind is not blowing. Thus, it cannot be used as a dependable source of base load power.
2. Operating the generator over a wide speed range may result in a considerable reduction in the overall efficiency of the energy conversion process.
3. Due to the variable wind speed, the output power of the WECS fluctuates and may create a frequency deviation of the power grid.
4. The functionality of energy management systems for WECSs is highly affected by the continuous variation of wind speed.
5. In WECS the rotational speed of the generator is controlled by current source inverter. This topology produces very poor quality current and has several issues when operating near synchronous speed.

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VERY IMPORTANT QUESTIONS

Following questions are very important. These questions may be asked in your SESSIONALS as well as UNIVERSITY EXAMINATION.

Q. 1. Explain the working of a thermoelectric generator. Derive an expression for its power output.

Ans. Refer Q. 4.4, Unit-4.

Q. 2. What is the basic difference between thermoelectric and thermionic conversion systems ? Also, explain the working of thermoelectric generators.

Ans. Refer Q. 4.7, Unit-4.

Q. 3. Derive the expression for power and efficiency of a thermionic generator.

Ans. Refer Q. 4.8, Unit-4.

Q. 4. What are the most favorable sites for installing wind turbines ? Using Betz model of a wind turbine, derive the expression for power extracted from wind. Under what condition does the maximum theoretical power can be extracted from the wind turbine ?

Ans. Refer Q. 4.12, Unit-4.

Q. 5. What do you mean by the nature of wind ? Describe the construction and working of a wind energy conversion system (WECS) with the help of a neat sketch.

Ans. Refer Q. 4.13, Unit-4.

Q. 6. What is the basic principle of wind energy conversion ? What methods are used to overcome the fluctuating power generation of a windmill ?

Ans. Refer Q. 4.15, Unit-4.

☺☺☺

5 UNIT

Biomass, OTEC and Wave and Tidal Wave Energy

CONTENTS

Part-1	:	Biomass	5–2M to 5–2M
Part-2	:	Availability	5–2M to 5–3M
Part-3	:	Conversion Theory	5–3M to 5–9M
Part-4	:	Ocean Thermal Energy Conversion, Theory and Working Principle	5–9M to 5–12M
Part-5	:	Performance	5–12M to 5–12M
Part-6	:	Limitations	5–13M to 5–14M
Part-7	:	Wave and Tidal Wave	5–14M to 5–15M
Part-8	:	Principle of Working	5–15M to 5–16M
Part-9	:	Performance and Limitations	5–16M to 5–19M
Part-10	:	Waste Recycling Plant	5–19M to 5–20M

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PART-1

Biomass.

Questions-Answers

Long Answer Type and Medium Answer Type Questions

Que 5.1. **What do you understand by "biomass" ?**

Answer

1. Biomass refers to solid carbonaceous material derived from plants and animals.
2. These include residues of agriculture and forestry, animal waste and discarded material from food processing plants.
3. Biomass being organic matter from terrestrial and marine vegetation, renews naturally in a short span of time, thus, classified as a renewable source of energy.
4. It is a derivative of solar energy as plants grow by the process of photosynthesis by absorbing CO_2 from the atmosphere to form hexose (dextrose, glucose, etc.) expressed by the reaction.

$$6CO_2 + 6H_2O \xrightarrow[\text{photosynthesis}]{\text{sunlight}} C_6H_{12}O_6 + 6O_2$$

5. Biomass does not add CO_2 to the atmosphere as it absorbs the same amount of carbon in growing the plants as it releases when consumed as fuel.
6. It is a superior fuel as the energy produced from biomass is 'carbon cycle neutral'.
7. Biomass fuel is used in over 90 % of rural households and in about 15 % urban dwellings.

PART-2

Availability.

Questions-Answers

Long Answer Type and Medium Answer Type Questions

Que 5.2. **Discuss the availability of biomass.**

Answer

1. Biomass is derived from the plant, forest residues, animal dung etc.
2. For use of biomass as energy fast growing trees, sugar, starch and oil containing plants can be cultivated.
3. The cultivated biomass also includes sweet sorghum crops, sugar beets, cereals, herbaceous crops, aquatic crops grown in fresh water, sea water, muddy water etc.
4. Algae are also the source of renewable energy. Algae contain organic matter which can be converted into methane gas. Algae can be cultivated on large scale.
5. Wood and straw come in dry biomass. The energy can be obtained by burning of this biomass.
6. Wood in the form of cut logs, chips, sawdust is commonly used for domestic applications.
7. Straw can be burn in straw burning stoves and furnaces.
8. Energy can also be obtained by converting the organic wastes to intermediate energy such as heat, biogas fuels etc.
9. The wastes can be classified as urban wastes, industrial wastes, animal wastes, forest wastes, fishery and poultry wastes, animal and human excreta and agri-wastes.
10. These wastes may be converted into heat, biogas, and biochemical by various processes such as combustion (as in case of wood, straw etc.), biochemical and biothermal reactions.
11. Biogas contains methane. Biogas plants produce methane by anaerobic digestion.
12. The human waste can also be used for production of biogas.

PART-3

Conversion Theory.

Questions-Answers

Long Answer Type and Medium Answer Type Questions

Que 5.3. **Classify biomass conversion technologies. Explain anaerobic digestion process for production of methane.**

AKTU 2017-18, Marks 10

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Answer

A. Biomass Conversion :

1. The following processes are used for the biomass conversion to energy or to biofuels :

a. Direct combustion,

b. Thermochemical conversion, and

c. Biochemical conversion.

a. Direct Combustion :

1. Combustion is the process of burning in presence of oxygen to produce heat, light and byproducts.
2. Complete combustion to ashes is called incineration.
3. Wood, dung, vegetable waste can be dried and burnt to provide heat or converted into low calorific value gas by pyrolysis.
4. In the pyrolysis process, the organic material is converted to gases, solids and liquids by heating to 500 °C to 900 °C in the absence of oxygen.
5. The combustion of biomass is more difficult than other fuels, since it contains relatively higher moisture content.
6. Biomass is free from toxic metals and their ash.
7. The technology of "fluidised bed combustion" may be used for the efficient combustion of forestry and agricultural waste material such as sawdust, wood chips, hog fuel, rice husks, straws, nutshells and chips.
8. In fluidised bed combustion of biomass, the biomass is fed into a bed of hot inert particles such as sand kept in fluidised state with air at sufficient velocity from below.
9. The operating temperature is normally controlled within the range 750-950 °C; ideally it is kept as high as possible in order to maximise the rate of combustion and heat transfer but low enough to avoid the problem of sintering of the bed particles.
10. The rapid mixing and turbulence within the fluidised bed enables efficient combustion to be achieved with high heat releases, as well as effective transfer, than in a conventional boiler. This can result in more compact boiler with less number of tubes.

b. Thermochemical Conversion :

1. Biomass is decomposed in thermochemical processes having various combinations of temperatures and pressures.
2. Thermochemical conversion takes two forms : gasification and liquefaction.

3. Gasification takes place by heating the biomass with limited oxygen to produce low heating value gas or by reacting it with steam and oxygen at high pressure and temperature to produce medium heating value gas.
4. The latter may be used as fuel directly or used in liquefaction by converting it to methanol (methyl alcohol), or ethanol (ethyl alcohol) or it may be converted to high heating value gas.

c. Biochemical Conversion :

1. In biochemical conversion there are two principal conversion processes :
   i. Anaerobic digestion, and
   ii. Fermentation

B. Anaerobic Digestion :

1. This process involves microbial digestion of biomass.
2. The process and end products depend upon the micro-organisms cultivated and culture conditions.
3. This process generates mostly methane (CH_4) and CO_2 gas with small impurities such as hydrogen sulphide.
4. The output gas obtained from anaerobic digestion can be directly burnt, or upgraded to superior fuel gas (methane) by removal of CO_2 and other impurities.
5. The residue may consist of protein-rich sludge and liquid effluents which can be used as annual feed or soil treatment after certain processing.

a. Stages of Digestion System :

i. Anaerobic digestion system consists of three stages.

ii. A flow chart showing these stages is given below :

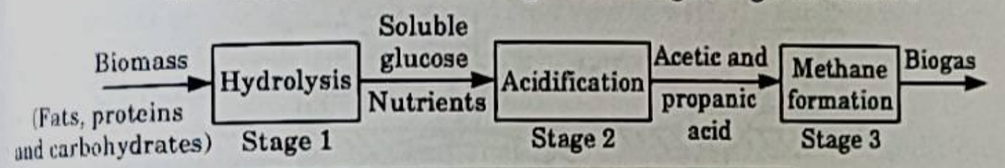

Fig. 5.3.1. Anaerobic digestion process.

Stage 1 : **Hydrolysis :** In this stage, the complex compounds (such as fats, proteins carbohydrates) are broken into small size compounds through the influence of water enzymes called hydrolysis. The reaction of hydrolysis is given below :

$$(C_6H_{10}O_5)_n + nH_2O \xrightarrow{\text{Enzymes}} n(C_6H_{12}O_6)$$

Cellulose + Nutrients → Soluble glucose + Nutrients

Stage 2 : **Acidification :** In this stage, the soluble glucose and nutrients are converted in simpler volatile fatty acid and acetic acid as by-product, that accounts for 70 % of methane byproduct with the help of acid forming bacteria.

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The reaction of acidification is given below :

$$C_6H_{12}O_6 \xrightarrow[\text{bacteria}]{\text{Acid producing}} 2CH_3CH_2OH\,(\text{Ethylalcohol}) + 2CO_2 + \text{fatty acid} + H_2 + \text{other products}$$

Stage 3 : Methane Formation : In this stage, the methane producing bacteria converts the organic acids into biogas having its main constituents as methane.

The reaction of methane formation is given below :

$$2CH_3CH_2OH + CO_2 \xrightarrow[\text{bacteria}]{\text{Methane producing}} 2CH_3COOH + CH_4$$

$$CH_3COOH \longrightarrow CH_4 + CO_2$$

Note : The biogas production depends on the environment maintained in the digester.

Que 5.4. Explain the process of production of biogas from biomass. Describe Deenbandhu biogas plant.

AKTU 2016-17, Marks 10

Answer

A. Production of Biogas from Biomass : Refer Q. 5.3, Page 5–3M, Unit-5.

B. Deenbandhu Biogas Plant :

1. Deenbandhu model was developed by the Action For Food Production (AFPRO), New Delhi.
2. Deenbandhu plants are made entirely of brick masonry work with a spherical shaped gas holder at the top and a concave bottom as shown in Fig. 5.4.1.

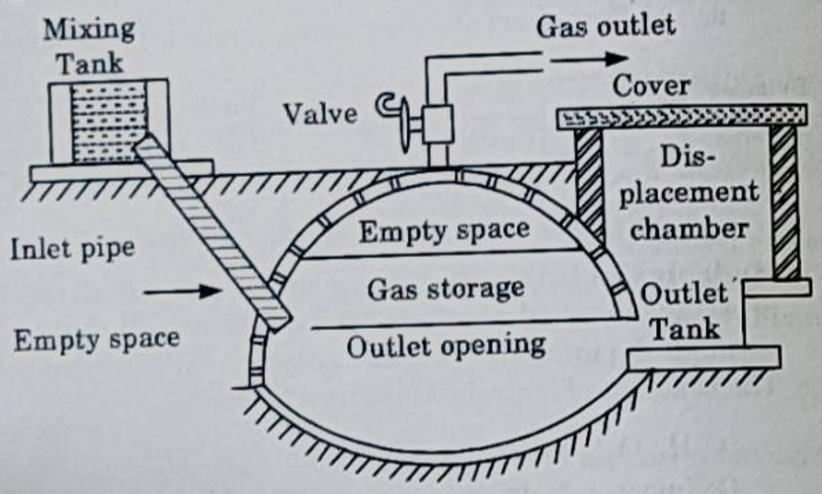

Fig. 5.4.1.

3. The inlet pipe connects the mixing tank with digester where as displaced slurry after digestion moves to outside as there is no displacement space on inlet side.

4. The gas pushes the slurry downward till the level in the digester reaches the upper end of outlet opening.
5. Recently, environmental protection and social development association (EPA), a NGO, has constructed modified Deenbandhu design plants in Bardiya district which is also approved by biogas support programme (BSP).
6. In India, this model proved 30 percent cheaper than Janata model and is known as friend to poor. It also proved to be about 45 % cheaper than a KVIC plant of comparable size without affecting the efficiency of plant.

Que 5.5. Explain the process of gasification of solid biomass. What is the general composition of the gas produced and what is its heating value. What are its applications ?

AKTU 2016-17, 2018-19; Marks 10

Answer

A. Gasification :

The process of gasification involves the following four processes :

a. **Drying :** Biomass fuels usually contain 10 % – 35 % moisture. When biomass is heated to about 100 °C, the moisture is converted into steam.

b. **Pyrolysis :** After drying as heating continues, the biomass undergoes pyrolysis. It involves burning biomass completely without supplying any oxygen. As a result, the biomass is decomposed or separated into solids, liquids and gases charcoal is the solid part, tar is the liquid part and flue gases make up the gaseous part.

c. **Oxidation :** Air is introduced into the gasifier after the decomposition process. During oxidation which takes place at about 700 – 1400 °C, charcoal or the solid carbonized fuel reacts with the oxygen in the air to produce carbon dioxide and heat.

$$C + O_2 \longrightarrow CO_2 + \text{heat}$$

d. **Reduction :** At higher temperatures and under reducing conditions that is when not enough oxygen is available, the following reactions takes place forming carbon dioxide, hydrogen and methane.

$$C + CO_2 \longrightarrow 2CO$$

$$C + H_2O \longrightarrow CO + H_2$$

$$CO + H_2O \longrightarrow CO_2 + H_2$$

$$C + 2H_2 \longrightarrow CH_4$$

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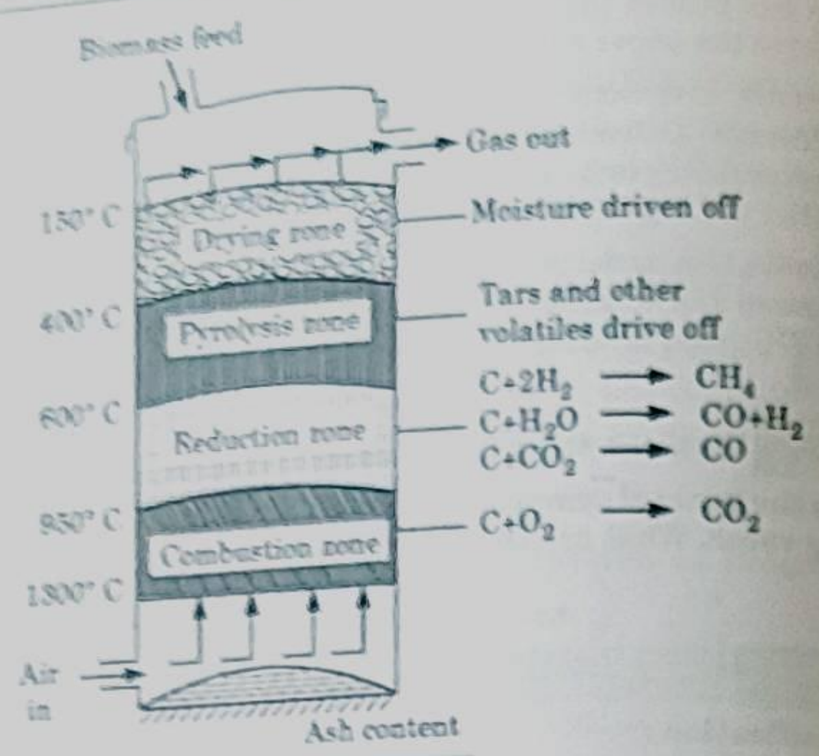

Fig. 5.5.1.

B. **Composition :**

Carbon Monoxide	– 18 – 22 %
Hydrogen	– 13 – 19 %
Methane	– 1 – 5 %
Heavier Hydrocarbons	– 0.2 – 0.4 %
Carbon dioxide	– 9 – 12 %
Nitrogen	– 45 – 55 %
Water vapour	– 4 %

C. **Heating value :**

1. The gas produced in the gasifier is a clean burning fuel having heating value of about 950 – 1200 kcal/m^3.

D. **Application :**

1. SI engines can be made to run entirely on producer gas.
2. CI engines can be made to operate with about 60 – 80 % fuel oil replacement by producer gas.
3. In an industrial oil fired boiler.
4. In gas turbine.

Que 5.6. **What is biomass ? How does biomass conversion take place ? Describe the materials used for biogas generation and factors that affect the size of a biogas plant.** **AKTU 2015-16, Marks 15**

Answer

A. **Biomass :** Refer Q. 5.1, Page 5–2M, Unit-5.

B. **Biomass Conversion :** Refer Q. 5.3, Page 5–3M, Unit-5.

C. **Material used for Biogas Generation :**

1. Biogas is produced by anaerobic decomposition of organic wastes by suitable bacteria. It contains 55- 65 % methane, 30-40 % carbon dioxide and the remainder is impurities like H_2S, N_2, H_2 gases.
2. The main source of production of biogas are crops residue, wet cow dung, vegetable wastes, water hyacinth, algae, poultry or piggery droppings, human waste, etc.
3. Any organic material of animal or plant which is easily biodegradable can be the source of biogas production.

Table 1 : Production of biogas from different types of raw materials.

Material	Amount of gas (m^3/kg)	
	Winter	Summer
Cattle dung	0.036	0.092
Pig dung	0.07	0.10
Poultry droppings	0.07	0.16

D. **Factors Affecting the Size of a Biogas Plant :**

1. The amount and type of organic waste to be disposed in the digester.
2. Demand of natural gas and consumption pattern.
3. On-site nature of the soil and the level of ground water.
4. Air temperature in the region and wind direction throughout the different seasons.
5. The training level of the staff on farm and home regarding operation of biogas units.

PART-4

Ocean Thermal Energy Conversion, Theory and Working Principle.

Questions-Answers

Long Answer Type and Medium Answer Type Questions

Que 5.7. **What is the working principle of ocean thermal energy conversion ?**

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Answer

1. The principle of ocean thermal energy conversion (OTEC) is that there is a temperature difference between water at the bottom of the sea and the water at the top.
2. This temperature difference can be used to operate a heat engine and most of the radiation is being absorbed at the surface layer of water.
3. The mixing between hot and cold water is prevented because no thermal convection occurs between hot and cold water layer. This means that the surface layer will act as a source and cold layers act as a sink.
4. Therefore, it is essential to connect the reversible heat engine between source and cold sink to produce work that can be converted into required applications.
5. The absorption of solar radiation in the water varies and can be expressed by Lambert's law :

$$-\frac{dI_y}{dy} = \mu I$$

or, $I_y = I e^{-\mu y}$

Where, I_y = Radiation intensity at depth y from water surface and falls exponentially with depth,
I = Radiation intensity at water surface, and
μ = Extinction or absorption coefficient.

Que 5.8. **Describe the basic principle of ocean thermal energy conversion (OTEC). What are the main types of OTEC power plants? Describe their working in brief.** **AKTU 2015-16, Marks 7.5**

OR

Describe the basic principle of ocean thermal energy conversion system. Describe the "Open Cycle" ocean thermal energy conversion system. **AKTU 2017-18, Marks 10**

OR

Explain with the help of the diagram, the principle of closed cycle ocean thermal energy conversion system. **AKTU 2018-19, Marks 10**

Answer

A. **Principle of OTEC :** Refer Q. 5.7, Page 5–9M, Unit-5.

B. **Types of OTEC System :**

1. There are two basic types of OTEC systems :
   a. Closed cycle system or Anderson cycle system, and
   b. Open cycle system or Claude cycle system.

a. **Closed or Anderson Cycle OTEC System :**

1. In this system, the working fluids for heat engines use the fluids like ammonia, freon 12, butane gas having low boiling point because the working temperature of sea water is small.
2. Warm water from ocean surface is circulated through a pump to a heat exchanger which acts as boiler to generate freon vapour at high pressure.
3. This vapour expands in the turbine to develop mechanical power and it is used to drive an electric generator which produces electric energy.
4. Freon vapour from turbine at low pressure is condensed in the condenser with the help of cold water drawn from the depth of ocean through a pump. The overall efficiency of such plant is very low in the range of 2 to 3 % only.

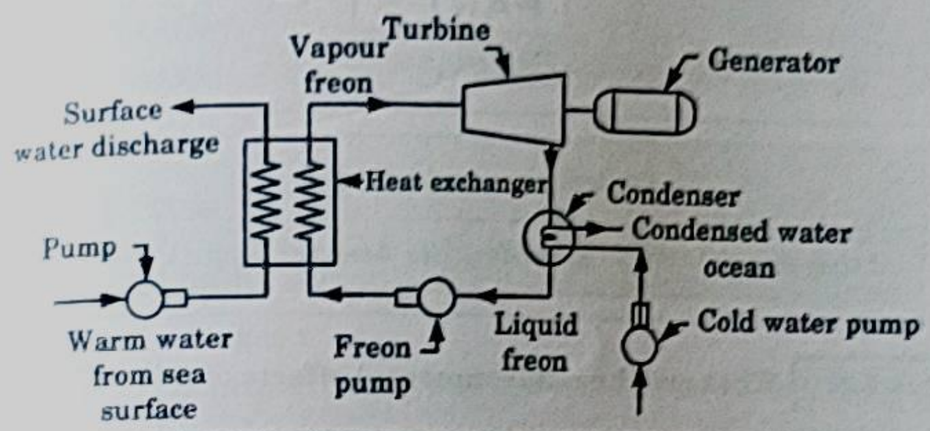

Fig. 5.8.1. Closed or Anderson cycle OTEC plant.

b. **Open Cycle or Claude Cycle OTEC System :**

1. In this system, the warm water from ocean surface is admitted through the deaerator to the flash evaporator which is maintained under high vacuum.
2. As a result, a low pressure steam is generated due to throttling effect and the remainder liquid is discharged back to the ocean at high depth.
3. The deaerator also removes the dissolved non-condensable gases from water before supplied to the evaporator.
4. This low pressure steam having very high specific volume is supplied to turbine where it expands and the mechanical power so developed is converted into electrical power by the generator.
5. The exhaust steam from turbine is discharged into a direct contact type heat exchanger and mixes with the cold water drawn from ocean at a depth of about 1 to 2 km.
6. The mixture of condensed steam and ocean cold water are discharged into the ocean.

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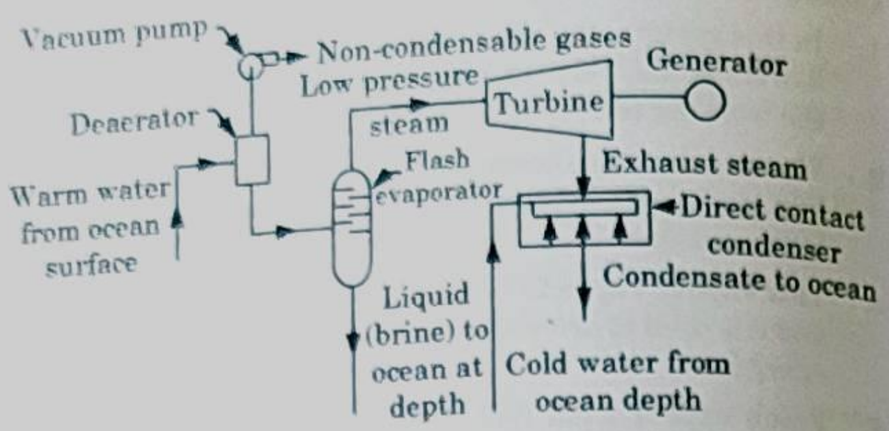

Fig. 5.8.2. Open or Claude cycle OTEC system.

PART-5

Performance.

Questions-Answers

Long Answer Type and Medium Answer Type Questions

Que 5.9. **What are the environmental effects of OTEC ?**

Answer

Following are the environmental effects of OTEC :

1. The marine environment gets affected by these plants through water heating.
2. Release of toxic chemical and entrapment of small sea organism in intake pipes is common.
3. Thermal layer of sea water near the plant gets disturbed because of discharge of low and high water at intermediate layer.
4. Affects the marine environment because of change in salinity, dissolved gases, nutrients, carbonate etc.
5. Large discharge of mixed water below the ocean surface for long time will change the environment for hatching the eggs and lower down the production rate of fishes, corals etc.
6. Toxic chemicals from the plant may leak to the environment and kill the local marine organisms.
7. The marine life gets affected because of change in pH value and dissolved oxygen.

PART-6

Limitations.

Questions-Answers

Long Answer Type and Medium Answer Type Questions

Que 5.10. **Give the advantages, disadvantages and applications of OTEC.**

Answer

A. Advantages :

1. The thermal resource of the ocean ensures that the power source is available during day or night.
2. It is eco-friendly.
3. It eliminates the need for a surface heat exchanger.
4. It produces potable water which reduces electrical generating costs up to one-third.

B. Disadvantages :

1. Due to low pressure, large size of steam turbine is used.
2. It needs very large vacuum pumps.
3. In closed cycle the working fluid is expensive.
4. Cost of plant is high.
5. Cost of electrical energy from open cycle OTEC is very high.
6. Corrosion of metal parts due to saline water.
7. Size of the plant is limited due to large size of the components.
8. Construction of floating power plant is difficult.

C. Applications :

1. A closed cycle OTEC plant can also act as a chemical treatment plant.
2. An OTEC plant can also be used to pump up the deep sea water and this cold water is used for cooling green houses, and air conditioning systems etc.
3. The enclosing area of OTEC can be used for aquaculture and mariculture.
4. The deep sea cold water is rich in nutrient and can be used for various applications.

Que 5.11. **What are the limitations of OTEC ?**

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Answer

1. Low thermal efficiency (2 – 3 %) because of low temperature difference of water available.
2. The capital cost is more.
3. Large size pump is required to handle large volume of water.
4. Plant should be capable of withstanding severe ocean storms and seasons.
5. Components life is short because of corrosion and erosion by ocean water.
6. Plant size above 100 MW is limited because it requires large size component (requires 30 m diameter pipe of 1 km long).
7. Difficult maintenance.
8. Construction of floating plant is difficult.

PART-7

Wave and Tidal Wave.

Questions-Answers

Long Answer Type and Medium Answer Type Questions

Que 5.12. **Write short notes on :**

i. **Wave Energy.**
ii. **Tidal wave Energy.**

Answer

A. Wave Energy :

1. Wave energy derives from wind energy, which derives in turn from solar energy.
2. Waves are formed on the surface of water by the frictional action of the winds resulting in the radial depression of energy from the blowing winds in all directions.

B. Tidal Wave Energy :

1. Tides are generated by the action of gravitational force of the sun and the moon on the ocean's surface, by the spinning of the earth about its axis and the relative positions of the earth, the moon and the sun.

2. Ocean tides are the periodic rise and fall of ocean water level occurring twice in each lunar day. The tidal rise and fall of water is accompanied by periodic horizontal to and fro motion of water called tidal currents.
3. The tidal currents flow in horizontal direction and have kinetic energy. This energy is called tidal current energy.

PART-8

Principle of Working.

Questions-Answers

Long Answer Type and Medium Answer Type Questions

Que 5.13. **Discuss the working principle of wave energy conversion system.**

Answer

1. Ocean waves are created by wind interaction with the ocean surface and are an indirect form of utilizing the solar energy, because the wind is created by pressure differences in the earth atmosphere, due to unequal solar heating.
2. The energy transferred to water by wind is kinetic as well as potential energy and it depends upon the wind speed, blowing time of wind, and distance of wind travel over the sea.
3. The blowing wind creates a pressure over the surface of ocean water and air pushes down each particle, which again comes up. So, it actually moves up and down in circular path.
4. Every particle passes on its motion to the next. This movement of the water particles produces a pattern, which we see as wave.
5. These waves travel a long distance as they propagate and are continuously strengthened by the new wind as they pass and retain their energy even winds die down.
6. The ocean wave energy is created because of periodic to and fro, up and down motion of water particles in the form of progressive waves.
7. It is an important to note that water does not travel with wave while the disturbance or wave travels in wind direction.

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8. The height of the wave depends on the speed of the wind.
9. These waves develop for few seconds and get superimposed on ocean water.
10. The power potential of these waves can be converted to electricity by mechanical means and harnessing this oceanic energy of waves has been developed over past 30 years using wave machines.

PART-9

Performance and Limitations.

Questions-Answers

Long Answer Type and Medium Answer Type Questions

Que 5.14. **Explain the principle and components of tidal power plants. Also, discuss the working of tidal power plants.**

OR

Explain with sketches the various methods of tidal power generation. Write the advantages and limitations of tidal power.

AKTU 2015-16, Marks 10

OR

Explain the 'single basin' and 'two basin' systems of tidal power harnessing. Discuss their advantages and limitations.

AKTU 2016-17, 2018-19; Marks 10

Answer

A. Principle : To utilize tidal energy, water must be trapped at high tide behind a dam or barrage and then made to drive turbine as it returns to sea during low tides. The available energy is proportional to the square of the amplitude.

B. Components of Tidal Power Plant :

Main components of tidal power plants are :

a. Barrage,
b. Sluice gates,
c. Turbine, and
d. Basin.

a. Barrage : It is a dam of low head and requires the following features :

1. Less sloppy towards the ocean and basin side.

2. It should be able to withstand the shock load of tides and wave.
3. Low height and shorter in length to minimize the cost of construction.
4. Steel foundation frame and channels are embedded in the ducts within the barrage for turbine and gates steel foundation.

b. Sluice Gates : These gates are opened by water pressure and no mechanical means is required.

c. Turbine : The Kaplan or bulb type turbine is used to operate with low head and the entire turbine generator unit is submerged in the water.

d. Basin : The basin can be single, pair or multiple type and have different designs.

C. Working of Tidal Power Plant : The working is described according to the following type of basins :

a. Single basin system, and
b. Double basin system.

a. Single Basin System :

1. In a tidal power plant (based on single basin system), the power house is situated at the mouth of basin.
2. The hydraulic turbine in the power house only operates during the discharge of water from the basin during ebb tide and during the high tide the basin is again filled.
3. The direction of flow through the turbine during the ebb and flood tides alternates and generation of power is accomplished, both during the emptying and filling cycle of basin.
4. Though the double cycle system has only short duration interruptions in turbine generator operation, but the continuous power generation is still not possible.
5. Further, the power generation coincides occasionally with the peak power demands. This problem is overcome in double-basin system.

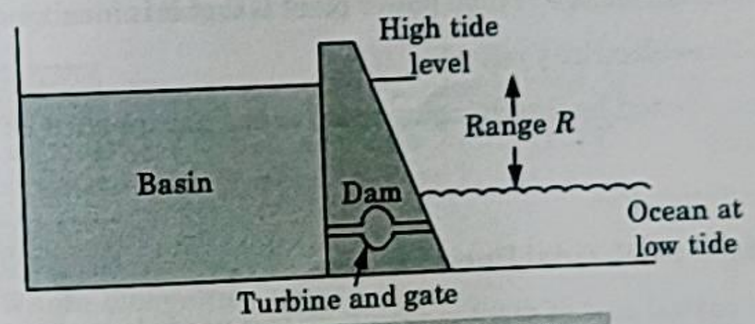

Fig. 5.14.1. Single basin system.

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b. Double Basin System :

i. Construction :

1. This system has two basins at different levels and a dam is provided in between these basins.

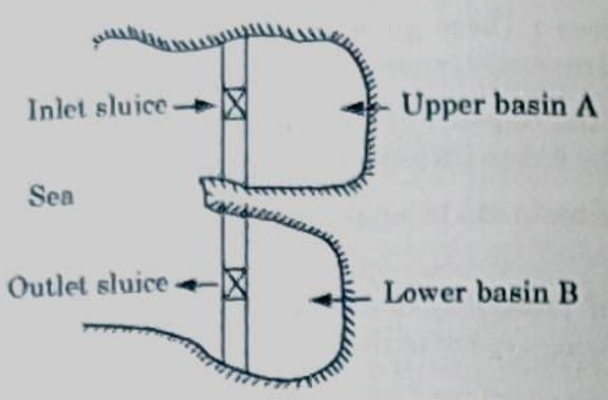

Fig. 5.14.2. Double basin system.

2. Inlet and outlet sluice gates are provided in the dam and the water level in upper basin is maintained above the level of water in the lower basin.

ii. Working :

1. When the water level in upper basin is maximum during high tide, the inlet sluice is closed and the level of water in lower basin keeps on rising due to discharge of water by the turbine.
2. When the level of water in lower basin equals during the ebb tide, the outlet sluice is opened and it is closed when the water level reaches to its minimum level which is equal to the level of water in upper basin.
3. Again the inlet sluice is opened and the cycle is repeated.

D. Advantages of Tidal Power :

1. Protection of coastline against damage from high storm tides by providing a barrage.
2. The main advantage of tidal power plant is that it is inexhaustible.
3. It produces electricity reliably.
4. It is unaffected by the changing mood of the nature such as failure of monsoon.
5. It is pollution free.

E. Limitations of Tidal Power :

1. Initial capital cost of plant is very high and needs long constructional period.

2. Output power is variable due to uneven operation.
3. Sea water is corrosive.
4. Sedimentation of basin is a serious problem.
5. Due to variable tidal range, the efficiency of plant is affected.
6. Marine life is affected.

PART-10

Waste Recycling Plant.

Questions-Answers

Long Answer Type and Medium Answer Type Questions

Que 5.15. What do you mean by recycling ?

Answer

1. Recycling involves a series of processes, which includes collection of recyclable materials and sorting out and using it as raw material after palletizing.
2. It also includes processing, manufacturing and selling of final products.
3. The collected materials are sorted and cleaned out for manufacturing into two products.
4. Some of the household materials, which can be recycled and used further, include newspapers and paper towels, aluminum, plastic and glass, soft drink containers, steel cans, and plastic laundry bottles.
5. The reuses of recycled material are fall in the field of recovered glass, in roadway asphalt or recovered plastic in carpeting, park benches, and pedestrian bridges.

Que 5.16. What do you understand by waste recycling management ? What are the basic steps involved in waste management by different sources ?

Answer

A. Waste Recycling Management :

1. Waste recycling management is the part of energy conservation.

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2. The typical route for recycling the waste material involved is collection, transport, processing and/or disposal of waste materials.
3. Recycling plays a major role in waste management. Though it is an uncommon activity, it earns good income in developing countries.

B. Steps Involved in Waste Management :

1. The various steps included in waste recycling management are :
   a. Find out the various alternate waste recycling options.
   b. Listing of steps included in the process.
   c. Economical analysis of recycling process.
   d. Organising.
   e. Execution and monitoring of program.
2. The hazardous waste is to be disposed off in a properly lined landfill or containers to prevent serious health effects.
3. The biodegradable waste goes for composting or biomethanation (biogas) process to produce energy and the remaining goes for land filling.
4. The reuse of material like glass, plastics reduces the wastes considerably after recycling into new products, plastics which can be molded to usable material.
5. The wood and agricultural residues are used to produce biomass briquettes.

VERY IMPORTANT QUESTIONS

Following questions are very important. These questions may be asked in your SESSIONALS as well as UNIVERSITY EXAMINATION.

Q. 1. Classify biomass conversion technologies. Explain anaerobic digestion process for production of methane.

Ans. Refer Q. 5.3, Unit-5.

Q. 2. Explain the process of gasification of solid biomass. What is the general composition of the gas produced and what is its heating value. What are its applications ?

Ans. Refer Q. 5.5, Unit-5.

Q. 3. What is biomass ? How does biomass conversion take place ? Describe the materials used for biogas generation and factors that affect the size of a biogas plant.

Ans. Refer Q. 5.6, Unit-5.

Q. 4. Describe the basic principle of ocean thermal energy conversion (OTEC). What are the main types of OTEC power plants ? Describe their working in brief.

Ans. Refer Q. 5.8, Unit-5.

Q. 5. Explain with sketches the various methods of tidal power generation. Write the advantages and limitations of tidal power.

Ans. Refer Q. 5.14, Unit-5.

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1 UNIT

Introduction and Solar Cells (2 Marks Questions)

1.1. What are the commercial energy sources ?

Ans. Those energy sources which are available in the market and which can be purchased from the producing agency are known as commercial energy sources. These sources are responsible for the development of industrial, agricultural and commercial sectors.
Example : Coal, Oil, Natural gas, Petroleum products, etc.

1.2. What are the non-commercial energy sources ?

Ans. Those energy sources which are not available in the commercial market for the purchase are known as non-commercial energy sources and these sources are often ignored in energy accounting.
Example : Firewood, Agro-waste, Animal dung, Solar energy, Wind energy, etc.

1.3. Give the classification of energy resources.

Ans. These energy resources can be classified as :

i. Primary energy resources :
   a. Conventional energy sources and non-conventional energy sources.
   b. Renewable and non renewable.
ii. Secondary energy resources.

1.4. What are primary energy resources ?

Ans. These resources are obtained from the environment.
Example : Fossil fuels, Solar energy, Hydro energy, and Tidal energy.

1.5. What are secondary energy resources ?

Ans. These resources do not occur in nature but are derived from primary energy resources.
Example : Electrical energy from coal burning, H_2 obtained from hydrolysis of H_2O, etc.

1.6. What are the conventional (non-renewable) energy sources ?

Ans. These are the sources of energy which are exhaustible *i.e.*, cannot be replaced if once they are used.
Example : Coal, Petroleum product, Natural gas etc.

1.7. What do you mean by non-conventional (renewable) energy sources ? **AKTU 2017-18, Marks 02**

Ans. These are the sources of energy which are inexhaustible *i.e.*, can be used to produce energy again and again.
Example : Sun, Water, Animal dung, Agro-waste etc.

1.8. How can renewable energy technologies be compared ? **AKTU 2015-16, Marks 02**

Ans. Renewable energy technologies are compared on the basis of certain parameters related to socio-economic and operational aspects that come into play while harnessing of these energy resources. Some of these parameters are as following :

1. Energy content.
2. Efficiency of conversion process.
3. Cost associated.
4. Availability and distribution.
5. Environment impact.

1.9. Are fossil fuels renewable ? **AKTU 2015-16, 2018-19; Marks 02**

Ans. No.

1.10. How many types of renewable energy are there normally said to be ? **UPTU 2015-16, Marks 02**

Ans. Sources of renewable energy :

i. Solar energy,
ii. Biomass,
iii. Biogas,
iv. Hydro energy,
v. Wind energy,
vi. Tidal energy,
vii. Ocean thermal energy,
viii. Hydrogen energy,
ix. Geothermal energy,
x. Hydrogen energy, and
xi. Photovoltaic.

1.11. Define photovoltaic.

Ans. Photovoltaics (PV) is the conversion of light into electricity using semiconducting materials that exhibit the photovoltaic effect. The photovoltaic effect is commercially utilized for electricity generation and as photosensors.

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1.12. What are the advantages of non-conventional energy resources ?

Ans. Advantages :

i. Socially relevant and useful and variety of choices available in practice to shift for renewable energy.
ii. The system has long life of 10-15 years or more.
iii. Energy cost is very low as the sources are available freely.
iv. Can be placed at any remote location.
v. Renewable energy cannot deplete unlike fossil fuel and uranium resources.

1.13. What are the disadvantages of non-conventional energy resources ?

Ans. Disadvantages :

i. Low energy density and need large size plant.
ii. Intermittent in operation as they depend on seasonal variation.
iii. Conversion efficiency is low.
iv. Large land area is required for installing the system to get same power as conventional plants.

1.14. Write a short note on solar cell.

Ans. Solar Cell : Energy conversion devices which are used to convert sunlight to electricity by the use of the photovoltaic effect are called solar cells. A single converter cell is called a solar cell or, more generally, a photovoltaic cell, and combination of such cells, designed to increase the electric power output is called a solar module or solar array.

1.15. Define solar cell materials ? **AKTU 2018-19, Marks 02**

Ans. The solar cells are made of various materials. Silicon is the most commonly used material for solar cells. A variety of compound semiconductors are to be used to manufacture thin film solar cell. These materials are $CuInSe_2$, CaS, CdTe, Cu_2S, InP, GaAs, Zinc Telluride (ZnTe), AlSb (Aluminium antimonide).

1.16. What are the various types of solar cells ?

Ans. According to type of crystal, the solar cells are of three types :

i. Monocrystalline silicon cells (band gap 1.12 eV).
ii. Polycrystalline silicon cells (band gap 1.12 eV).
iii. Amorphous silicon cells (band gap 1.75 eV).

1.17. How durable are solar panels ? **AKTU 2015-16, Marks 02**

Ans. Solar panels have a lifespan in excess of about 20 years.

1.18. Write a short note on solar photovoltaic.

OR

How can solar energy be converted into electrical energy.

AKTU 2017-18, Marks 02

Ans. Solar energy can be converted to electricity in two ways :

i. **Photovoltaic (PV devices) or "solar cells" :** Photovoltaic or solar cells change sunlight directly into electricity. PV systems are often used in remote locations that are not connected to the electric grid.
ii. **Solar Power Plants :** Solar Power Plants indirectly generate electricity when the heat from solar thermal collectors is used to heat a fluid which produces steam that is used to power generator.

1.19. Write a short note on photovoltaic effect.

Ans. Photovoltaic Effect : When a solar cell is illuminated, electron-hole pairs are generated and the electric current I is obtained. I is the difference between the solar light generated current I_L and the diode dark current I_j.

Mathematically :

$$I = I_L - I_j$$

$$= I_L - I_o\left[\exp\left(\frac{eV}{KT}\right) - 1\right]$$

where, I_o = saturation current,
e = electronic charge,
T = absolute temperature, and
K = Boltzmann's constant
$= 1.38 \times 10^{-23}$ J/K.

1.20. Define fill factor. **AKTU 2015-16, 2017-18; Marks 02**

Ans. The fill factor (FF) for a solar cell is defined as the current and voltage at maximum power unit to the light generated current and open circuit voltage.

$$FF = \frac{J_{mp} \times V_{mp}}{J_L \times V_{oc}}$$

where, J_{mp} = Current at maximum power unit,
V_{mp} = Voltage at maximum power unit,
J_L = Light generated current, and
V_{oc} = Open circuit voltage.

1.21. What do you understand by maximum efficiency of solar cells ?

Ans. Maximum efficiency of solar cell is defined as the ratio of maximum electric power output to the incident solar radiation.

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$$\eta_{max} = \frac{I_m V_m}{I_S A_C}$$

where, I_s = incident solar flux,
I_m = maximum current,
A_C = cells area, and
V_m = maximum voltage.

1.22. Explain solar cell array.

Ans. **Solar Cell Array :** Solar cells are strung in series and thus form a solar module or array. They may be tracking arrays or fixed arrays. A tracking array is defined as one which is always kept mechanically perpendicular to the sun-array line so that all times it intercepts the maximum isolation. Such arrays must be physically movable by a suitable prime mover and are considerably more complex then fixed arrays.

1.23. Explain the classification of array.

Ans. Array are classified in two type :
i. Flat plate arrays, and
ii. Concentrating arrays.

1.24. Write down the applications of solar energy.

Ans. **Applications :**
i. Solar water heating,
ii. Solar heating of buildings,
iii. Solar distillation,
iv. Solar pumping,
v. Solar drying of agricultural and animal products,
vi. Solar furnaces,
vii. Solar cooking,
viii. Solar electric power generation,
ix. Solar thermal power production, and
x. Solar green houses.

1.25. How can you conserve energy and lower your utility bills ?

AKTU 2015-16, Marks 02

Ans. Energy can be conserved by increasing the production from a given amount of energy input by reducing losses/wastage and maximizing the efficiency. By using energy wisely and adopting energy-saving measures, you can save money and cut down on energy wastage at home while meeting your needs for convenience and comfort.

1.26. Discuss the terms energy conservation and energy audit.

AKTU 2016-17, 2018-19; Marks 02

Ans.

A. **Energy Conservation :** Energy conservation means reduction in energy consumption but without making any sacrifice of quantity and quality of production. In other words, for the same energy consumption, higher production.

B. **Energy Audit :** Energy audit is a technical survey of a plant in which the machine/wise/section wise/department wise pattern of energy consumption is studied and attempts to balance the total energy input correlating with production.

1.27. What is energy poverty ? Name the five nations which consume the highest energy with suitable data.

AKTU 2017-18, Marks 02

Ans. **Energy Poverty :** It is a lack of access to modern energy services such as household access to electricity and clean cooking facilities.

Nations with Highest Energy Consumption :

i. **China :** It consumed a record 113,200 trillion British thermal units (BTU) of primary energy.
ii. **United States of America :** It consumed 89,920 trillion BTU, which accounted for over 17 % of world total.
iii. **Russia :** It consumed 27,740 trillion BTU (5.5 % of the world total).
iv. **India :** It consumed 23,610 trillion BTU (4.7 % of the world total).
v. **Japan :** It consumed 18,810 trillion BTU (3.7 %) of the world total).

1.28. Discuss the limitations of solar photovoltaic energy conversion.

AKTU 2017-18, Marks 02

Ans.

1. When photons of light energy from the sun strike the cell, some of them (30 %) are reflected (since reflectance from semiconductors is high).
2. Incident active photons produce electron-hole pairs with high quantum efficiency. Better cell design is required to ensure 95 % absorption.
3. The semiconductor with optimum band gap should be used for maximum efficiency.

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2 UNIT

Solar Thermal Energy (2 Marks Questions)

2.1. Define solar radiation.

Ans. **Solar Radiation :** The solar radiations received by the earth's surface vary with the location. However radiation received outside the earth's atmosphere is different than what we receive on the earth surface because of absorption, reflection, scattering and attenuation by particulates and clouds present in the atmosphere.

2.2. Define solar constant. What is its standard value ?

AKTU 2016-17, Marks 02

Ans.

1. This is the amount of energy received from the sun in unit time on a unit area perpendicular to sun's rays at the mean distance from the sun.
2. The standard value of the solar constant based on experimental measurements is 1367 W/m^2 with an accuracy of ±1.5 %.

2.3. What do you mean by beam radiation ?

Ans. Solar radiation that has not been absorbed or scattered and reaches the ground directly from the sun is called direct radiation or beam radiation.

2.4. What is diffuse radiation ?

Ans. Diffuse radiation is that solar radiation received from the sun after its direction has been changed by reflection and scattering by the atmosphere.

2.5. Discuss altitude angle (α) or solar altitude.

Ans. It is the vertical angle between the projection of the sun's rays on the horizontal plane and the direction of sun's rays (passing through the point).

2.6. Define irradiance and albedo.

Ans. **Irradiance :** The rate of incident energy per unit area of a surface is known as irradiance.

Albedo : The earth reflects back nearly 30% of the total solar radiant energy to the space by reflection from clouds, by scattering and by reflection at the earth's surface. This is known as albedo of earth's atmosphere system.

2.7. What do you mean by incident angle (θ) ?

Ans. It is define as the angle between the incident beam radiation and the normal to a plane surface.

2.8. What is zenith angle (θ_z) ?

Ans. It is the vertical angle between the sun's rays and line perpendicular to the horizontal plane through the point.

$$\theta_z = \frac{\pi}{2} - \alpha$$

2.9. Define latitude angle (ϕ).

Ans. The latitude of a place is the angle subtended by the radial line joining the place to the centre of earth, with the projection of the line on the equatorial plane.

2.10. What do you mean by declination angle (δ) ?

Ans. The declination is the angle made by the line joining the centres of the sun and the earth with its projection on the equatorial plane.

2.11. Define hour angle (ω).

Ans. It is the angle through which the earth must be rotated to bring the meridian of a point directly in line with the sun's ray. In other words, it is the angular displacement of the sun, east or west of the local meridian due to the rotation of the earth on its axis at an angle of 15° per hour.

$$\omega = 15\,(LST - 12)$$

where, LST = local solar time.

2.12. What is surface azimuth angle (γ) ?

AKTU 2018-19, Marks 02

Ans. It is the angle in the horizontal plane, between the line due south and the horizontal projection of the normal to the inclined plane surface. By convention, the angle will be taken negative for northern hemisphere (India) and vice-versa for southern hemisphere (Australia).

2.13. Discuss solar azimuth angle (γ_s).

Ans. It is the angle in a horizontal plane, between the line due to south and the projection of beam radiation on the horizontal plane. Thus, it gives the direction of the shadow cast in the horizontal plane by a vertical rod.

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2.14. What is day length ?

Ans. **Day Length :** The time of sunrise, sunset and the duration of the day length depend upon the latitude of the location and the month in the year. At sunrise and sunset, the sunlight is parallel to the ground surface with a zenith angle of 90°. The hour angle pertaining to sunrise or sunset (ω_s) is given below :

At solar noon, $\theta_z = 0$ and $\omega = \omega_s$

$$\cos \omega_s = -\tan \phi \tan \delta$$

On tilted surface,

$$\omega_s = \cos^{-1} [(-\tan (\phi - \beta) \tan \delta)]$$

and the corresponding day length (in hours) is given by

$$N = \frac{2}{15} \cos^{-1} [-\tan (\phi - \beta) \tan \delta]$$

2.15. What do you mean by local apparent time (LAT) ?

Ans. **Local Apparent Time (LAT) :** The time used for calculating the hour angle ω is the local apparent time. It is given by,

LAT = Standard time + Equation of time correction + 4 (Standard time longitude – longitude of location)

In India, standard time is based on 82.5° E longitude.

2.16. Write down the advantages of solar energy.

Ans. **Advantages :**

i. It is reusable source of energy.
ii. It is easily and abundantly available.
iii. It is eco-friendly *i.e.*, pollution free.
iv. It decreases greenhouse gas emissions.

2.17. Write down the disadvantages of solar energy.

Ans. **Disadvantages :**

i. High capital cost due to requirement of large area.
ii. Limited to sun shine hours.
iii. Need of tracking due to change in position of sun.
iv. There is a need of storage.

2.18. Define solar collectors.

Ans. **Solar Collectors :** Solar collectors are used to collect the solar energy and convert this energy into the thermal energy by absorbing them. This thermal energy is further used for heating a collector fluid such as water, oil or air.

2.19. What are the different types of solar collectors ?

Ans. Solar collectors are classified in two type :

i. Non-concentrating collector or flat plate type solar collector, and
ii. Concentrating collector or focusing type solar collector.

2.20. Define flat plate collector.

Ans. **Flat Plate Collector :** It is simplest in design and it is most important part of any solar thermal energy system. In this collector, both direct and diffuse radiations are absorbed and converted into useful heat.

2.21. What do you mean by fin efficiency factor (F_e) of flat plate collector ?

Ans. It is defined as the ratio of actual rate of heat transferred to the heat that would be transferred, if entire fins (plate area) are at base temperature.

Mathematically :

$$F_e = \frac{Q_{actual}}{A_c[\alpha_o \tau_o I_t - U_L(T_p - T_a)]}$$

where, Q_{actual} = actual rate of heat,
α_o = absorptivity,
A_c = collector area,
I_t = incident total radiations,
τ_o = transmitivity,
U_L = overall heat loss coefficient,
T_p = plate temperature, and
T_a = ambient temperature.

2.22. What is collector efficiency factor (F_c) for flat plate collector ?

Ans. It is defined as the ratio of useful heat removed by flowing fluid in the tubes to the rate of heat transferred to the fluid, if the fin is at local fluid temperature.

Mathematically :

$$F_c = \frac{Q_u}{A_c[\alpha_o \tau_o I_t - U_L(T_f - T_a)]}$$

Q_u = useful heat removed by flowing fluid in the tubes, and
T_f = local fluid temperature.

2.23. Define collector heat removal factor (F_H) for flat plate collector.

Ans. It is defined as the ratio of actual useful energy gain by fluid to the rate of heat transferred to the fluid, if the fin is at inlet fluid temperature.

Mathematically :

$$F_H = \frac{Q_u}{A_c[\alpha_o \tau_o I_t - U_L(T_{fi} - T_a)]}$$

or

$$F_H = \frac{mC_f (T_{fo} - T_{fi})}{A_c[\alpha_o \tau_o I_t - U_L(T_{fi} - T_a)]}$$

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where, m = mass flow rate of fluid per unit area of collector,
C_f = specific heat of fluid,
T_{fo} = outlet fluid temperature, and
T_{fi} = inlet fluid temperature.

2.24. Discuss collector efficiency (η_c) for flat plate collector.

Ans. It is defined as the ratio of useful energy absorbed by collector to the incident solar energy over it.

$$\eta_c = \frac{Q_u}{A_c I_t}$$

2.25. Define concentrating collector or focusing collector.

Ans. **Concentrating Collector :** It is a device to collect solar energy with high intensity of solar radiation on the absorbing surface by help of reflector or refractor.

2.26. Differentiate between concentrating collector and flat plate type collectors.

Ans.

S. No.	Concentrating collector	Flat plate type collector
i.	It requires less absorber area.	It requires more absorber area.
ii.	It has high collection efficiency.	It has low collection efficiency.
iii.	Heat storage cost is lower.	Heat storage cost is higher.
iv.	Suitable for large power generation.	Suitable for small power generation.

2.27. Discuss the various materials used for solar flat plate collector.

Ans.
i. Transparent cover is made of glass or radiation transmitting plastic sheet or transparent glass fibre reinforced composite material.
ii. Tubes, passages or channels are integral with collector plate are made by brass, copper or aluminium metals.
iii. Absorber plate normally metallic is blackened with black paint.
iv. Casing or container which encloses all components is made by G.I. sheets.

2.28. On what factors does at the collector efficiency of a solars flat plate collector depend ? **AKTU 2016-17, Marks 02**

Ans. The different factors which affects the performance of flat plate collector :
i. Incident solar radiation,
ii. Number of cover plate,
iii. Spacing between absorber plates and glass cover,
iv. Collector tilt,
v. Selective surface,
vi. Fluid inlet temperature, and
vii. Dust on cover plate, etc.

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3 UNIT

Geothermal Energy, MHD and Fuel Cells (2 Marks Questions)

3.1. Define geothermal energy.

Ans. **Geothermal Energy :** The enormous amount of energy available inside the earth in the form of heat is known as geothermal energy. Geothermal energy is a form of renewable energy and independent of sun, having the source of natural heat inside the earth.

3.2. Why is geothermal considered a renewable energy resource ?

AKTU 2015-16, Marks 02

Ans. The geothermal resource of the earth is estimated to be more than 2.11×10^{25} J, which is equivalent to 10^9 MTOE (million tonnes of oil equivalent). This is a huge amount of energy, enough to supply our energy needs at current rates for 3,50,000 years. Thus, it is considered an inexhaustible and renewable source.

3.3. Describe various geothermal energy resources.

AKTU 2016-17, Marks 02

Ans. Geothermal resources are of five types :

i. Hydrothermal energy resources.
   a. Hot water, and
   b. Wet steam.
ii. Vapour dominated resources.
iii. Hot dry rock resources.
iv. Geo pressured resources.
v. Magma resources.

3.4. Define thermal gradient.

Ans. Thermal gradient is defined as the ratio of heat flux to thermal conductivity.

Mathematically :

$$\text{Thermal gradient} = \frac{\text{Heat flux}}{\text{Thermal conductivity}}$$

3.5. Write down the types of geothermal power plants.

Ans. Electric power from geothermal resources can be developed in the following manner :

i. Vapour dominated power plant, and
ii. Liquid dominated power plant.

There are two types of liquid dominated power plants :

a. Flashed steam system.
b. Binary cycle system.

3.6. What are the sources of heat for hot springs ?

AKTU 2015-16, Marks 02

Ans. Steam, Hot water, Matter Rock.

3.7. Define MHD (Magneto Hydro Dynamics).

Ans. **Magneto Hydro Dynamics :** The magneto hydro dynamics deals with generation of electrical field, when ionized fluid at high temperature passes through the applied magnetic field. The direct current is generated from the system with the expense of thermal energy.

3.8. What are the different types of magneto hydro dynamic system ?

Ans. MHD systems may be classified as :

i. Open cycle systems, and
ii. Closed cycle systems.

3.9. What are the advantages of MHD generator ?

Ans. **Advantages :**

i. Less wear and tear because of no moving parts and needs less maintenance.
ii. Instant operation and is suitable as peak load plant.
iii. Less operating cost.
iv. Less costly than coal fired steam and gas power plant.

3.10. What are the disadvantages of MHD generator ?

Ans. **Disadvantages :**

i. High thermal and frictional (12%) losses are involved.
ii. The reverse flow of electrons occurs at the end of magnetic field, through the conducting gases.
iii. High temperature around 2500 °C is required to ionize the gases.
iv. There are problems for manufacturing duct, heat exchanger and reactor to withstand high temperature.

3.11. Define fuel cell.

Ans. **Fuel cell :** A fuel cell is an electrochemical device that converts chemical energy into electricity and heat without combustion, whereas, the conversion of chemical energy into electrical energy in case of fuel cell is an isothermal process.

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3.12. What are the various type of fuel cell ?

Ans. Based on electrolyte, the fuel cell can be classified as :

i. Alkaline fuel cells (AFC).
ii. Direct methanol fuel cells (DMFC).
iii. Phosphoric acid fuel cells (PAFC).
iv. Proton or polymer exchange membrane fuel cells (PEMFC).
v. Molten carbonate fuel cells (MCFC).
vi. Solid oxide fuel cells (SOFC).
vii. Zinc air fuel cells (ZAFC).
viii. Regenerative fuel cells (RFC).

3.13. Describe regenerative fuel cell.

Ans. **Regenerative fuel cell :** It operates in a closed loop. Fuel cell generates the electricity, heat and water from hydrogen and oxygen. The hydrogen would be generated from the electrolysis of water *i.e.*, water can split into hydrogen and oxygen by the use of renewable energy sources such as sun, wind etc. Oxygen can be used as oxidant and water produced is recirculated for electrolysis.

3.14. Discuss the advantages and disadvantages of PEMFC.

Ans. **Advantages :**

i. It has high power density.
ii. It can be start rapidly.
iii. Less expensive.
iv. It has fewer problems with corrosion.

Disadvantages : The main disadvantage of this type of fuel cell is that due to low operating temperature, these are not enough to perform useful cogeneration.

3.15. What do you mean by efficiency of fuel cell ?

Ans. The efficiency (η_F) of energy conversion of a fuel cell is defined as the ratio of the useful work to the heat of combustion of the fuel.

$$\eta_F = \frac{\Delta W_{max}}{(-\Delta H)} = \frac{-(\Delta G)}{-(\Delta H)}$$

3.16. What are the various types of characteristics of fuel cells ?

Ans. The various characteristics of fuel cell are :

i. Current-voltage characteristics (V-I), and
ii. Power-current characteristics (P-I).

3.17. What is the difference between fuel cell and a battery ?

Ans.

S. No.	Fuel Cell	Battery
i.	The fuel cell is a primary cell and cannot be recharged but can be refueled.	The battery is rechargeable.
ii.	The fuel and oxidizer do not mix together.	Fuel and oxidizer are not used separately.
iii.	It produces electricity continuously as long as fuel oxidizer is supplied.	Battery stores energy.

3.18. What are the environmental effects of geothermal energy sources ?

Ans. Geothermal energy is not completely pollution free energy. The main adverse environmental effects are air pollution (waste steam is sometimes vented directly to the atmosphere), thermal pollution (pumping more thermal energy to atmosphere), surface disturbance, physical effect (land subsidence) caused by fluid withdrawal.

3.19. How does geothermal heat get up to the earth's surface ?

AKTU 2015-16, Marks 02

Ans. The heat from the earth's core continuously flows outward. It transfers (conducts) to the surrounding layer of rock, the mantle. When temperatures and pressures become high enough, some mantle rock melts, becoming magma. Then, because it is lighter (less dense) than the surrounding rock, the magma rises (convects), moving slowly up toward the earth's crust, carrying the heat from below.

3.20. Write the chemical reaction takes place in alkaline fuel cell.

AKTU 2016-17, 2018-19; Marks 02

Ans. The chemical reaction in the alkaline fuel cell expressed as :

Anode : $2H_2 + 4OH^- \rightarrow 4H_2O + 4e^-$

Cathode : $O_2 + 2H_2O + 4e^- \rightarrow 4OH^-$

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4 UNIT

Thermoelectrical and Thermionic Conversion and Wind Energy (2 Marks Questions)

4.1. Define joule effect.

Ans. **Joule Effect :** In a closed electrical circuit, if the current I flows through a resistance R, the heat generated by the resistance is equal to I^2R. It is known as Joule effect.

$$Q = I^2R$$

where, I = Current flow,
R = Resistance.

4.2. What is Seebeck thermoelectric effect ?

Ans. **Seebeck Effect :** It states that if a closed circuit is made of two dissimilar metals and the two junctions are maintained at different temperatures, than an emf is setup in circuit. The magnitude of current depends on both the metals and the temperature difference between the junctions.

The emf produced is the function of the difference in temperature between hot and cold junction, and is given by,

Mathematically :

$$E = \alpha_s \, \Delta T$$

where, ΔT = Difference between hot and cold junction, and
α_s = Seebeck coefficient.

4.3. What is Peltier effect ?

Ans. **Peltier Effect :** When an electric current flows across an isothermal junction of two dissimilar materials, there is either an evolution or absorption of heat at the junction. This effect is called the Peltier effect.

Mathematically :

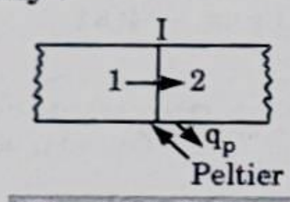

Fig. 4.3.1. Peltier effect.

$$\alpha_{p1-2} = \alpha_{p1} - \alpha_{p2} = \frac{q_p}{I}$$

where, I = Peltier heat per unit time.

4.4. Define Peltier coefficient.

Ans. **Peltier Coefficient :** The Peltier coefficient α_p is defined as the heat evolved or absorbed at the junction per unit current flow, per unit time.

Mathematically :

$$\alpha_{p1-2} = \frac{q_p}{I}$$

$$q_p = \alpha_{1-2} I$$

where, I = Peltier heat per unit time.

4.5. What is Thomson effect ?

Ans. **Thomson Effect :** It states that "Any current carrying conductor with a temperature difference between two points will either absorb or emit heat, depending upon the material".

4.6. Define Thomson coefficient.

Ans. **Thomson Coefficient :** The Thomson coefficient (σ) is defined as the heat absorbed or evolved per unit time per unit electric current and per unit temperature gradient.

Mathematically :

$$\sigma = I \frac{dq_T / dx}{dT / dx}$$

where, $\frac{dq_T}{dx}$ = heat interchanger per unit time per unit length.

$\frac{dT}{dx}$ = temperature gradient.

4.7. Define Seebeck coefficient.

Ans. **Seebeck Coefficient :** It is defined as the ratio of the resulting voltage and temperature difference.

$$E \propto \Delta T$$

$$E = \alpha_s \, . \, \Delta T$$

$$\alpha_s = \frac{E}{\Delta T}$$

α_s = Seebeck coefficient.

4.8. What is thermoelectrical conversion ?

Ans. **Thermoelectrical Conversion :** In thermoelectrical conversion, heat energy (thermal energy) is converted into electrical energy through semiconductors or conductors. This process is based on the Seebeck effect.

4.9. Define thermionic conversion.

Ans. **Thermionic Conversion :** In thermionic conversion, heat energy is directly converted into electrical energy by utilizing thermionic

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emission effect. In such a device, electrons are emitted from the surface of heated metal. The energy required to extract one electron from the metal is known as work function and express in electron volt.

4.10. What are the different types of semiconductors used in thermoelectric power generation ?

Ans. Semiconductors used in thermoelectric power generation are :

i. Bismuth telluride (Bi_2Te_3).
ii. Lead telluride (PbTe).
iii. Germanium telluride (GeTe).
iv. Cesium sulphide (CeS).
v. Zinc antimonide (ZnSb).

4.11. Differentiate between thermoelectric and thermionic conversion system.

Ans.

S. No.	Thermoelectric Conversion System	Thermionic Conversion System
i.	This process is based on Seebeck effect.	This process is based on thermionic emission effect.
ii.	In thermoelectric conversion, heat energy (thermal energy) is converted into electric energy.	In thermionic conversion, heat energy is directly converted into electrical energy.

4.12. What do you understand by wind energy ?

Ans. The movement of air is called wind which is created by the pressure difference between two uneven heated places. The energy developed by the wind is called wind energy. It is a clean, eco-friendly, safe and renewable source of energy.

4.13. What is the basic principle of wind energy conversion ?

Ans. **Principle :** The basic principle of wind energy is to convert the kinetic energy of wind into more useful forms that are mechanical power and electrical power.

4.14. Give the classification of wind mills.

Ans.

i. **Based on orientation of the axis of rotor :**
   a. Horizontal axis, and
   b. Vertical axis.

ii. **Based on type of rotor :**
   a. Propeller type (horizontal axis),
   b. Multiple blade type (horizontal axis),
   c. Savonius type (vertical axis), and
   d. Darrieus type (vertical axis).

4.15. Define the term solidity.

Ans. **Solidity :** It is the ratio of the projected area of the rotor to the swept area of the rotor.

OR

Solidity is defined as the percentage of the circumference of the rotor which is filled by the rotor blades.

$$\text{Solidity } (s) = \frac{N.b}{\pi d}$$

where, N = number of blades,
b = average breadth of blade, and
d = diameter of the circle described by a blade.

4.16. What are the main components of wind energy conversion system (WECS) ?

Ans. The main components are as follows :

i. Rotor with blade,
ii. Electromagnetic brakes,
iii. Controller,
iv. Mechanical brakes,
v. Gear box,
vi. Generator,
vii. Shaft,
viii. Flap or tail vane, and
ix. Tower top.

4.17. What is the maximum energy conversion efficiency of a wind turbine for a given swept area ?

AKTU 2016-17, Marks 02

Ans.

$$P_{max} = 0.593 \times \left(\frac{1}{2} \times \frac{\rho A v_i^3}{g_c}\right)$$

$$P_{max} = 0.593 \times P_{total}$$

4.18. What do you understand by "coefficient of performance of a wind mill rotor" ?

Ans. **Coefficient of Performance (Power Coefficient) :** The coefficient of performance of a wind mill is defined as the ratio of the power of the wind rotor to the total maximum power available in the wind.

Mathematically :

$$C_p = \frac{\text{Power of wind rotor}}{\text{Total maximum power available in the wind}}$$

The maximum theoretical power coefficient is 0.593.

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4.19. Discuss pitch angle and pitch control.

Ans. **Pitch Angle :** It is the angle between the direction of wind and direction perpendicular to the plane of blades.
Pitch Control : It is the control of pitch by turning the blades or blade tips.

4.20. What do you mean by tip speed ratio ?

Ans. **Tip speed ratio :** The ratio of the speed of the rotor blade tips to the speed of the wind is known as tip speed ratio.

4.21. Define nacelle and yaw control.

Ans. **Nacelle :** Assembly consists of wind turbine, gears, bearings, generator etc., mounted in a housing.
Yaw Control : Control of orienting the axis of wind turbine in the direction of wind.

4.22. What is wind turbine (wind machine) ?

Ans. **Wind Turbine :** A turbine which converts wind power into rotary mechanical power. A wind turbine has aerofoil blades mounted on the rotor. The wind drives the rotor and produces rotary mechanical energy.

4.23. Discuss the main criteria's for site selection for setting up a wind firm.

Ans. The main criteria's are :
i. The area should be open and away from cities.
ii. Minimum wind speed is available throughout the area.
iii. Ground surface should be stable and high soil strength.
iv. The proposed altitude is to be selected by taking average wind speed data.
v. Approach road up to site.

4.24. How much of our daily CO_2 emissions can wind avoid ?

AKTU 2015-16, Marks 02

Ans. An estimate said that wind energy avoided the emission of 140 million tonnes of CO_2 in 2011. In 2020, it will avoid 342 million tones and in 2030, it is projected to avoid 646 million tonnes.

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5 UNIT Biomass, OTEC and Wave and Tidal Wave Energy (2 Marks Questions)

5.1. What do you understand by biomass ?

Ans. **Biomass :** It is an organic matter from plants, animals and micro-organism grow on land and water and their derivatives. The energy obtained from biomass is called biomass energy.

5.2. What are the various processes used for biomass conversion ?

Ans. The following processes are used for biomass conversion to energy or to biofuels :
i. Direct combustion,
ii. Thermochemical conversion, and
iii. Biochemical conversion.

5.3. What is OTEC ? Discuss in brief. **AKTU 2016-17, Marks 02**

Ans. It is a means of converting ocean thermal energy into useful energy. OTEC is a technology that converts solar radiation into electric energy. It uses the ocean's temperature gradient to generate power, based on second law of thermodynamics.

5.4. What is the efficiency of OTEC ?

Ans. The maximum possible efficiency of heat engine working between two temperature limits cannot be more than that of a Carnot cycle operating between the same temperature limit.

5.5. What are the types of OTEC systems ?

Ans. There are two basic types of OTEC system :
i. Closed cycle system or Anderson cycle system, and
ii. Open cycle system or Claude cycle system.

5.6. Give the advantages of OTEC.

Ans. **Advantages :**
i. It is eco-friendly.
ii. The thermal resources of the ocean ensure that the power source is available during day or night.
iii. It can provide employment.
iv. It eliminates the need for a surface heat exchanger.

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5.7. Explain the disadvantages of OTEC.

Ans. **Disadvantages :**

i. It needs very large vacuum pumps.
ii. Due to low pressure, large size of steam turbine is used.
iii. Cost of plant is high.
iv. Cost of electrical energy from open cycle OTEC is very high.

5.8. Discuss the resources of biomass.

Ans.
i. Biomass obtained from cultivated fields, crops, forests and algae.
ii. Biomass derived from waste like municipal waste, animal and human excreta, etc.
iii. Biomass converted from liquid fuels.

5.9. Define fermentation in biomass energy.

AKTU 2018-19, Marks 02

Ans. Fermentation is the process of chemical change in organic matter brought about by living organism like bacteria. There are two basic types of organic decomposition that can occur in all organic matter. Both animal and vegetable can be broken down by these two processes :

i. Aerobic, and
ii. Anaerobic.

5.10. What is an aerobic digestion ?

AKTU 2016-17, 2018-19; Marks 02

Ans. Aerobic decomposition is carried out in the presence of oxygen and will produce carbon dioxide, ammonia and some other gases in small quantities and large quantities of heat. The final product can be used as a fertilizer.

5.11. Define anaerobic decomposition.

Ans. Anaerobic decomposition carried out in the presence of oxygen and will produce carbon dioxide, ammonia and some other gases in traces, very little heat. The final product with higher nitrogen content is produced through the process of aerobic fermentation and the product is used as a fertilizer.

5.12. Define wave energy.

Ans. Ocean and sea waves are caused directly by solar energy like the wind and OTEC. Wave energy derives from wind energy, which derives in turn from solar energy. Waves are formed on the surface of water by the frictional action of the winds resulting in the radial depression of energy from the blowing wind in all directions.

5.13. What are the advantages of wave energy ?

Ans. **Advantages :**

i. It is a concentrated form of energy and can naturally accumulate over time.
ii. It is an eco-friendly renewable source of energy.
iii. Large concentrated power carried in wave's motion.
iv. No space coverage on land as required by wind and solar device.

5.14. What are the disadvantages of wave energy ?

Ans. **Disadvantages :**

i. Capital cost of system is high.
ii. Problems in maintenance occur.
iii. The device operates in ocean and needs consideration for construction, maintenance and reliability.

5.15. How tides are generated ?

AKTU 2015-16, 2017-18; Marks 02

Ans. Tides are generated by the action of gravitational forces of the sun and the moon in the ocean, by the spinning of the earth about its axis and the relative positions of the earth, moon and the sun. The highest level of tidal water is known as high tide and the lowest level of tidal water is known as low tide.

5.16. What is the principle of tidal power plant ?

Ans. **Principle :** To utilize tidal energy, water must be trapped at high tides behind a dam or barrage and then made to drive turbine as it returns to sea during low tides. The available energy is proportional to the square of the amplitude.

5.17. What do you mean by recycling ?

Ans. Recycling involves a series of processes, which includes collection of recyclable materials and sorting out and using it as raw material after palletizing. It also includes processing, manufacturing and selling of final products. The collected materials are sorted and cleaned out for manufacturing into two products.

5.18. What are the advantages of recycling of waste ?

Ans. **Advantages :**

i. The amount of waste which goes for disposal reduces.
ii. The amount of raw material to produce a new product reduces.

5.19. What are the disadvantages of recycling of waste ?

Ans. **Disadvantages :**

i. The reprocessing of recovered materials is not always pollution free.

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ii. The effects of de-inking and re-pulping waste paper are relatively important in assessing the environmental effects as recycling of paper produces salt in the environment.
iii. The costs of collection, transport and reprocessing are higher.

5.20. What are the factors affecting the size of biomass plant ?

Ans. To following factors are affecting the size of biomass plant :
i. Volume of waste to be digested daily.
ii. The type and amount of waste available for digestion consistently.
iii. Period of digestion.
iv. Method of stirring.
v. Climate condition of the region.
vi. Efficiency of the collection of the raw waste.

5.21. Write down the name of popular biogas plant developed in India.

Ans. Following biogas plant are developed in India :
i. Khadi village industries commission (KVIC) model.
ii. Janta biogas plant.
iii. Deen Bandhu biogas plant.
iv. Pragati design biogas plant.
v. Ganesh biogas plant.

5.22. What are the advantages of tidal power ?

AKTU 2018-19, Marks 02

Ans. **Advantages :**
i. Protection of coastline against damage from high storm tides by providing a barrage.
ii. It produces electricity reliably.
iii. It is pollution free.
iv. The main advantage of tidal power plant is that it is inexhaustible.

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